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United States Patent	12416001
Kind Code	B2
Date of Patent	September 16, 2025
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Nucleobase editors and methods of use thereof

Abstract

The present disclosure relates to nucleobase editors and methods of use thereof. Disclosed herein are fusion proteins, systems, and compositions for editing disease-associated mutations and methods of use thereof. In some aspects, disclosed herein is a fusion protein comprising a Cas9 nickase and a nucleotide deaminase, wherein the Cas9 nickase comprises a first amino acid substitution at a position selected from the group consisting of 262, 324, 409, 480, 543, 694, and 1219 when compared to SEQ ID NO: 11, and wherein the Cas9 nickase comprises a second amino acid substitution at a position selected from the group consisting of 1111, 1135, 1218, 1219, 1322, 1335, and 1337 when compared to SEQ ID NO: 11.

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Appl. No.:	17/799440
Filed (or PCT Filed):	February 12, 2021
PCT No.:	PCT/US2021/017868
PCT Pub. No.:	WO2021/163492
PCT Pub. Date:	August 19, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20230116627 A1	Apr. 13, 2023

Related U.S. Application Data

us-provisional-application US 62976590 20200214

Publication Classification

Int. Cl.: **C12N15/00** (20060101); **C12N9/22** (20060101); **C12N9/78** (20060101); **C12N15/10** (20060101); **C12N15/11** (20060101); **C12N15/86** (20060101)

U.S. Cl.:

CPC **C12N15/102** (20130101); **C12N9/22** (20130101); **C12N9/78** (20130101); **C12N15/11** (20130101); **C12N15/86** (20130101); C12N2310/20 (20170501); C12N2750/14143 (20130101); C12N2800/40 (20130101); C12Y305/04002 (20130101); C12Y305/04005 (20130101)

Field of Classification Search

USPC: None

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a national stage application filed under 35 U.S.C. § 371 of PCT/US2021/017868, filed Feb. 12, 2021, which claims the benefit of U.S. Provisional Application No. 62/976,590, filed Feb. 14, 2020, which are expressly incorporated herein by reference.

FIELD

(1) The present disclosure relates to nucleobase editors and methods of use thereof.

BACKGROUND

(2) Many genetic diseases arise from single nucleotide alterations. Duchenne muscular dystrophy (DMD) is a fatal genetic muscle disease affecting approximately 1 in ~5000 male births worldwide, which is caused by mutations in the DMD gene. Most of the DMD mutations are due to deletions or duplications with over 500 point mutations accounting for ~10% of the cases. DMD codes for the dystrophin protein, a cytoskeletal protein that functions in the muscle force transmission and sarcolemmal stability of muscle fibers. Loss of dystrophin leads to progressive muscle weakness and wasting, loss of ambulation, respiratory impairment, cardiomyopathy, and eventual death. Previous studies showed that exon deletion through CRISPR genome editing can restore dystrophin expression and function. Although promising, this strategy has safety concerns as it relies on repair of the double strand DNA break (DSB) created by CRISPR/Cas9, which can cause unwanted large deletion and even chromosomal rearrangement.

(3) Therefore, what is needed are compositions for precise correction of disease-associated mutations.

SUMMARY

(4) Disclosed herein are fusion proteins, systems, and compositions for editing disease-associated mutations and methods of use thereof.

(5) In some aspects, disclosed herein is a fusion protein comprising a Cas9 nickase and a nucleotide deaminase, wherein the Cas9 nickase comprises a first amino acid substitution at a position selected from the group consisting of 262, 324, 409, 480, 543, 694, and 1219 when compared to SEQ ID NO: 11, and wherein the Cas9 nickase comprises a second amino acid substitution at a position selected from the group consisting of 1111, 1135, 1218, 1219, 1322, 1335, and 1337 when compared to SEQ ID NO: 11.

(6) In some embodiments, the first amino acid substitution is selected from the group consisting of A262T, R324L, S409I, E480K, E543D, M694I, and E1219V when compared to SEQ ID NO: 11. In some embodiments, the second amino acid substitution is selected from the group consisting of L1111R, D1135V, G1218R, E1219F, A1322R, R1335V, R1335Q, R1335E, and T1337R when compared to SEQ ID NO: 11.

(7) In some embodiments, the Cas9 nickase comprises the amino acid sequence selected from the

group consisting of SEQ ID NOs: 12-20.

(8) In some embodiments, the Cas9 nickase when in conjunction with a bound guide RNA (gRNA) specifically binds to a target nucleic acid sequence.

(9) In some embodiments, the Cas9 nickase recognizes a NG protospacer adjacent motif (PAM) sequence.

(10) In some embodiments, the nucleotide deaminase is a cytidine deaminase or an adenine deaminase. In some embodiments, the adenine deaminase is a dimeric adenine deaminase or a monomeric adenine deaminase domain. In some embodiments, the monomeric adenine deaminase comprises amino acid substitutions A56G and V82G when compared to SEQ ID NO: 47. In some embodiments, the adenine deaminase comprises the amino acid sequence set forth in SEQ ID NO: 22 or 23.

(11) In some aspects, disclosed herein is a system for base editing comprising: a first nucleotide sequence encoding a N-terminal portion of a Cas9 nickase fused at its C-terminus to an intein-N; and a second nucleotide sequence encoding an intein-C fused to the N-terminus of a C-terminal portion of the Cas9 nickase; wherein the first nucleotide sequence further comprises a nucleotide sequence encoding a nucleotide deaminase fused to the N-terminus of the N-terminal portion of the Cas9 nickase, wherein the Cas9 nickase comprises a first amino acid substitution at a position selected from the group consisting of 262, 324, 409, 480, 543, 694, and 1219 when compared to SEQ ID NO: 11, and wherein the Cas9 nickase comprises a second amino acid substitution at a position selected from the group consisting of 1111, 1135, 1218, 1219, 1322, 1335, and 1337 when compared to SEQ ID NO: 11.

(12) In some embodiments, the N-terminal portion of the Cas9 nickase comprises the amino acid positions 2-573 of the amino acid sequence selected from the group consisting of SEQ ID NOs: 12-20 and the C-terminal portion of the Cas9 nickase comprises the amino acid positions 574-1368 of the amino acid sequence selected from the group consisting of SEQ ID NOs: 12-20.

(13) In some embodiments, the first nucleotide sequence or the second nucleotide sequence further comprises a nucleotide encoding a guide RNA (gRNA).

(14) In some aspects, disclosed herein is a composition comprising: a first recombinant adeno-associated virus (AAV) particle comprising a first nucleotide sequence encoding a N-terminal portion of a Cas9 nickase fused at its C-terminus to an intein-N; and a second recombinant AAV particle comprising a second nucleotide sequence encoding an intein-C fused to the N-terminus of a C-terminal portion of the Cas9 nickase; wherein the first nucleotide sequence further comprises a nucleotide sequence encoding a nucleotide deaminase fused to the N-terminus of the N-terminal portion of the Cas9 nickase, wherein the Cas9 nickase comprises a first amino acid substitution at a position selected from the group consisting of 262, 324, 409, 480, 543, 694, and 1219 when compared to SEQ ID NO: 11, and wherein the Cas9 nickase comprises a second amino acid substitution at a position selected from the group consisting of 1111, 1135, 1218, 1219, 1322, 1335, and 1337 when compared to SEQ. ID NO: 11.

(15) In some embodiments, the first nucleotide sequence comprises the sequence as set forth in SEQ ID NO: 3 or SEQ ID NO: 5. In some embodiments, the second nucleotide sequence comprises the sequence as set forth in SEQ ID NO: 4 or SEQ ID NO: 6.

(16) In some aspects, disclosed herein is a method of treating a genetic disease or disorder in a subject, comprising administering a therapeutically effective amount of the composition of any preceding aspect. In some embodiments, the genetic disease or disorder is Duchenne muscular dystrophy, dysferlinopathy, or spinal muscular atrophy. In some embodiments, the genetic disease or disorder is due to a point mutation in a causative gene from a G:C pair to an A:T pair.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying figures, which are incorporated in and constitute a part of this specification, illustrate several aspects described below.

(2) FIGS. **1a-1e** show in vitro studies of mdx.sup.4cv mutation correction using ABE-NG. FIG. **1a** shows genomic DNA, encoded amino acids and guide RNA with PAM (highlighted in blue) sequences at the stop codon mutation site (red). FIG. **1b** shows the reporter construct containing a puromycin resistance cassette fused with E2A peptide, mdx.sup.4cv target sequence and ATG-removed. EGFP. Correction of the stop codon within the target sequence would allow EGFP expression. FIG. **1c** shows fluorescence microscopy images of HEK293 cells transfected with reporter alone, or reporter, gRNA and one of the base editors (ABEmax, ABE-x and ABE-NG). Scale bar: 500 μ m. FIGS. **1d-1e** shows flow cytometry analysis of EGFP expression in HEK293 cells transfected as described in FIG. **1c**. * $p < 0.05$; **** $p < 0.0001$ (one-way ANOVA). The sequences in FIG. **1** are GAACAGCTGCAGAACAGGAGATAACAG (SEQ ID NO: 620), GTTASTCTCCTGTTCTG CAGCTGT (SEQ ID NO: 621).

(3) FIGS. **2a-2f** show optimization of the PAM-interacting domain to improve the editing efficiency of ABE-NG at the NGN sites. FIGS. **2a-2f** show measurements of the base editing efficiencies of different ABE variants on a NGG-PAM site and five different sites with NGH or GAT PAM. The gRNA sequences are listed on the top of each graph with the PAM sequences in red and the target adenines in blue ns, not significant; ** $p < 0.01$; *** $p < 0.001$; **** $p < 0.0001$ (one-way ANOVA). The sequences in FIG. **2** are ATGACAGGCAGGGGCACCGCGG (SEQ ID NO: 622), GAGCGAGCAGCGTCTTCGAGAGT (SEQ ID NO: 623), GCAGACGGCA GTCAGTAGGGGGC (SEQ ID NO: 624), GTCGCAGGACAGCTTTTCCTAGA (SEQ ID NO: 625), GGGAAAGCTGGGTGAATGGAGCGA (SEQ ID NO: 626), GATCCAGGT GCTGCAGAAGGGAT (SEQ ID NO: 627).

(4) FIGS. **3a-3d** show re-engineering of the adenine deaminase domain to improve the efficiency and specificity of ABE-NG. FIG. **3a** shows schematics of the adenine deaminase domain used in ABE-NG editors. FIG. **3b** shows quantification of the editing efficiency of different ABE-NG variants with modified TadA* domain at the mdx.sup.4cv target site. **** $p < 0.0001$ (one-way ANOVA). FIG. **3c** shows the number of off-target RNA editing events in Neuro-2a cells transfected with different ABE-NG variants. *** $p < 0.05$; *** $p < 0.001$ (one-way ANOVA). FIG. **3d** shows quantification of the off-target RNA editing (A-to-I) activities on four RNA adenines previously identified as being efficiently modified by ABEmax in HEK293 cells. **** $p < 0.0001$ (one-way ANOVA).

(5) FIGS. **4a-4b** show comparison of the editing efficiency of different ABEs at the mdx.sup.4cv target site (FIG. **4a**) and human dysferlin Q605X site (FIG. **4b**). All base editors carry the SpCas9-NG and differ at the adenine deaminase domain. ns, not statistically significant; ** $p < 0.01$; **** $p < 0.0001$ (one-way ANOVA for FIG. **4a**; two-way ANOVA for FIG. **4b**). The sequence in FIG. **4** is GTTATCTCCTGTTCTGCAGC (SEQ ID NO: 570) and ATCCTACAGCATGGTGGCTG (SEQ ID NO: 590).

(6) FIGS. **5a-5b** show the intein split of ABEmax had relatively low editing activity. FIG. **5a** shows schematics showing the two halves of intein-split ABEmax. The TadA-TadA* was fused with Npu intein N-terminal fragment and SpCas9 nickase (nSpCas9) was fused with Npu intein C-terminal fragment. FIG. **5b** shows genomic DNA PCR analysis of HEK293 cells at 5 days after transfection with S2-gRNA and different versions of ABEs.

(7) FIGS. **6a-6h** show intein-mediated assembly of full-length ABE-NG. FIG. **6a** shows schematics of the intein split ABE-NG. The N-terminal and C-terminal intein sequences reconstitute the full-length protein when co-expressed within cells. FIG. **6b** shows Western blot analysis of HEK293 cell lysates transfected with different versions of ABEs. FIG. **6c** shows fluorescence microscopy images of HEK293 cells transfected with reporter alone, or reporter, gRNA and one of the base

editors (ABE-NG, split_v1 N+C or Split_v2 N+C). Scale bar: 500 μ m. FIG. 6d shows flow cytometry analysis of EGFP expression in HEK293 cells transfected as described in FIG. 6c. FIG. 6e shows Western blot analysis of HEK293 cell lysates transfected with full-length iABE-NGA, Gp41-1 or Npu split of iABE-NGA. FL, the full-length iABE-NGA band; N, the N-terminal fragment of the iABE-NGA. FIG. 6f shows densitometry quantification of the Western blot data shown in FIG. 6e, FIG. 6g shows the assembly efficiency of the Gp41-1 and Npu split of iABE-NGA (defined as the percentage of the full-length iABE-NGA bands). FIG. 6h shows quantification of the editing efficiency of full-length iABE-NGA, Gp41-1 split and Npu split of iABE-NGA at the mdx.sup.4cv target site, Npu Split_1g is same as Npu Split except that only the C-terminal construct carries the gRNA. ns, not significant; * $p < 0.05$; *** $p < 0.001$; **** $p < 0.0001$; (one-way ANOVA).

(8) FIG. 7. Immunofluorescence staining of heart and gastrocnemius muscle sections with anti-dystrophin antibody and DAPI. The mdx.sup.4cv mice at 5 weeks of age received either 5×10^{13} , 1×10^{14} AAV9-iNG/mdx.sup.4cv-gRNA or 1×10^{14} vg/kg AAV9-iNG/non-targeting-gRNA (NT-gRNA) through tail vein injection, and sacrificed at 10 weeks of age for immunofluorescence staining analysis. Scale bar: 100 μ m.

(9) FIGS. 8a-8k show widespread restoration of dystrophin expression in mdx.sup.4cv mice following systemic administration of AAV9-iNG. FIG. 8a shows dystrophin and laminin- α 2 co-immunostaining of heart sections from WT and mdx.sup.4cv mice (10 weeks or 10 months of age) with or without tail vein injection of AAV9-iNG (a total of 1×10^{14} vg/kg, 1:1 of the N and C-terminal half). Scale bar: 100 μ m. FIGS. 8b-8d show quantification of dystrophin-positive fibers in the heart (FIG. 8b), gastrocnemius (GA, FIG. 8c) and diaphragm (Diaph, FIG. 8d) muscles. 10 w, 10 weeks old; 10 m, 10 months old. ** $p < 0.01$; **** $p < 0.0001$ (Student's t-test). FIG. 8e shows Western blot analysis of heart homogenates from 10-week-old mice with anti-dystrophin, Cas9 and Gapdh antibodies. The WT muscle lysates were loaded at 5 μ g/lane while the mdx.sup.4cv muscle lysates were loaded at 25 μ g/lane. FIG. 8f shows densitometry quantification of Western blot data shown in FIG. 8e. * $p < 0.05$ (one-way ANOVA). FIG. 8g shows Western blot analysis of heart homogenates from 10-month-old mice with anti-dystrophin and Gapdh antibodies. FIG. 8h shows densitometry quantification of Western blot data in FIG. 8g, ** $p < 0.01$; **** $p < 0.0001$ (one-way ANOVA). FIG. 8i shows representative sequencing trace of dystrophin transcripts of WT and mdx.sup.4cv mouse hearts (10 weeks or 10 months old) with or without AAV9-iNG treatment. FIGS. 8j and 8k show quantification of the targeted T-to-C editing efficiency in the mdx.sup.4cv mouse hearts (FIG. 8j, 10 weeks; FIG. 8k, 10 months) as assayed by sequencing of dystrophin transcripts. **** $p < 0.0001$ (Student's t-test).

(10) FIG. 9 shows stitched large images showing dystrophin and laminin- α 2 immunostaining of the entire heart sections of a WT mouse at 10 weeks of age. Scale bars: 0.5 mm.

(11) FIG. 10 shows stitched large images showing dystrophin and laminin- α 2 immunostaining of the entire heart sections of a control mdx.sup.4cv mouse at 10 weeks of age. Scale bars: 0.5 mm.

(12) FIG. 11 shows stitched large images showing dystrophin and laminin- α 2 immunostaining of the entire heart sections mdx.sup.4cv mouse #142 five weeks after intravenous injection of AAV9-iNG at 5 weeks of age. Mouse number is shown in yellow. Scale bars: 0.5 mm.

(13) FIG. 12 shows stitched large images showing dystrophin and laminin- α 2 immunostaining of the entire heart sections mdx.sup.4cv mouse #147 five weeks after intravenous injection of AAV9-iNG at 5 weeks of age. Mouse number is shown in yellow. Scale bars: 0.5 mm.

(14) FIG. 13 shows stitched large images showing dystrophin and laminin- α 2 immunostaining of the entire heart sections mdx.sup.4cv mouse #148 five weeks after intravenous injection of AAV9-iNG at 5 weeks of age. Mouse number is shown in yellow. Scale bars: 0.5 mm.

(15) FIG. 14 shows stitched large images showing dystrophin and laminin- α 2 immunostaining of the entire heart sections mdx.sup.4cv mouse #149 five weeks after intravenous injection of AAV9-iNG at 5 weeks of age. Mouse number is shown in yellow. Scale bars: 0.5 mm.

(16) FIG. 15 shows stitched large images showing dystrophin and laminin- α 2 immunostaining of the entire heart sections mdx.sup.4cv mouse #150 five weeks after intravenous injection of AAV9-iNG at 5 weeks of age. Mouse number is shown in yellow. Scale bars: 0.5 mm.

(17) FIGS. 16a-16b show immunofluorescence staining of dystrophin and laminin α 2 in the gastrocnemius (FIG. 16a) and diaphragm (FIG. 16b) muscles from WT and mdx.sup.4cv (10 weeks of age) treated with or without tail vein injection of AAV9-iNG. Scale bar: 100 μ m.

(18) FIG. 17 shows Western blot analysis of gastrocnemius muscles from WT and mdx.sup.4cv (10 weeks of age) treated with or without tail vein injection of AAV9-iNG.

(19) FIG. 18 shows stitched large images showing dystrophin and laminin- α 2 immunostaining of the entire heart sections of a WT mouse at 10 months of age. Scale bars: 0.5 mm.

(20) FIG. 19 shows stitched large images showing dystrophin and laminin- α 2 immunostaining of the entire heart sections of a control mdx.sup.4cv mouse at 10 months of age. Scale bars: 0.5 mm.

(21) FIG. 20 shows stitched large images showing dystrophin and laminin- α 2 immunostaining of the entire heart sections mdx.sup.4cv mouse #1976 9-10 months after intravenous injection of AAV9-iNG at 5 weeks of age. Mouse number is shown in yellow, Scale bars: 0.5 mm.

(22) FIG. 21 shows stitched large images showing dystrophin and laminin- α 2 immunostaining of the entire heart sections mdx.sup.4cv mouse #1982 9-10 months after intravenous injection of AAV9-iNG at 5 weeks of age. Mouse number is shown in yellow. Scale bars: 0.5 mm.

(23) FIG. 22 shows stitched large images showing dystrophin and laminin- α 2 immunostaining of the entire heart sections mdx.sup.4cv mouse #1985 9-10 months after intravenous injection of AAV9-iNG at 5 weeks of age. Mouse number is shown in yellow. Scale bars: 0.5 mm.

(24) FIG. 23 shows stitched large images showing dystrophin and laminin- α 2 immunostaining of the entire heart sections mdx.sup.4cv mouse #1984 9-10 months after intravenous injection of AAV9-iNG at 5 weeks of age. Mouse number is shown in yellow, Scale bars: 0.5 mm.

(25) FIGS. 24a-24c show restoration of dystrophin expression in the skeletal muscles of 10-month-old mdx.sup.4cv mice after tail vein injection of AAV9-iNG at 5 weeks of age. FIG. 24a shows immunofluorescence staining of dystrophin in diaphragm and gastrocnemius muscles of WT and mdx.sup.4cv mice with or without systemic AAV9-iNG delivery. FIG. 24b shows Western blot of dystrophin expression in gastrocnemius muscles. FIG. 24c shows quantification of Western blot data.

(26) FIGS. 25a-25m show systemic delivery of AAV9-iNG improved histopathology and contractility in mdx.sup.4cv mice. FIG. 25a shows trichrome staining of muscle and heart sections showing the extensive fibrosis in diaphragm and gastrocnemius (GA) muscles of mdx.sup.4cv mice (10 months of age), which were substantially reduced following systemic AAV9-iNG delivery. The mdx.sup.4cv mouse heart had little fibrosis at 10 months of age. Scale bar: 200 μ m. FIGS. 25b-25d show quantification of fibrotic area of the diaphragm, gastrocnemius and heart muscles. ns, not significant; * p <0.05; ** p <0.01; *** p <0.001; **** p <0.0001 (one-way ANOVA). FIGS. 25e-25h show measurement of CNF in the diaphragm (FIGS. 25e and 25f) and gastrocnemius (FIGS. 25g and 25h) muscles of mdx.sup.4cv mice with or without AAV9-iNG treatment at 10 weeks (FIGS. 25e and 25g) or 10 months (FIGS. 25f and 25h) of age (two-tailed, unpaired Student's t-test). FIGS. 25i-25l show muscle fiber size measurement and distribution in diaphragm and gastrocnemius muscles of the mice (WT and mdx.sup.4cv with or without AAV9-iNG) at 10 weeks of age. ns, not significant; ** p <0.01 (one-way ANOVA). FIG. 25m shows tetanic torque measurements of the posterior compartment muscles. * p <0.05; **** p <0.0001 (one-way ANOVA).

(27) FIGS. 26a-26d show measurement of muscle fiber size in diaphragm and gastrocnemius muscles at 10 months of age. FIGS. 26a-26b show average CSA of diaphragm (FIG. 26a) and gastrocnemius (FIG. 26b) muscle fibers. c, d, Distribution of diaphragm (FIG. 26c) and gastrocnemius (FIG. 26d) muscle fiber areas. ns, not significant; * p <0.05 (one-way ANOVA).

(28) FIGS. 27a-27j show host immune response, toxicity and off-target studies of AAV9-iNG therapy in mdx.sup.4cv mice. FIGS. 27a and 27b show host immune response to AAV9 capsid and

the base editor transgene (anti-Cas9). IM, intramuscular injection; IV, intravenous injection. **** $p < 0.0001$ (one-way ANOVA) compared to WT or mdx.sup.4cv. FIGS. 27c-27e show measurements of serum AST (FIG. 27c), ALT (FIG. 27d) and BUN (FIG. 27e) of mice treated with or without AAV9-iNG. ns, not statistically significant (one-way ANOVA). FIG. 27f shows predicted off-target sites with different number of mismatches from the mdx.sup.4cv target gRNA, and the sequences of two most similar off-target sites located on chromosome 16 and 1, respectively. FIGS. 27g and 27h show quantification of deep sequencing reads of the genomic DNA PCR amplicons of the chromosome 16 off-target site (FIG. 27g) or the chromosome 1 off-target site (FIG. 27h) from Neuro-2a cells transfected with ABE-NG, iABE-NGA or control plus the gRNA. ns, not significant (one-way ANOVA). FIG. 27i shows quantification of the A.sup.4-to-G editing in mdx.sup.4cv mice treated with or without AAV9-iNG. **** $p < 0.0001$ (student's t-test). FIG. 27j shows quantification of the bystander C.sup.6-to-T editing in mdx.sup.4cv mice treated with or without AAV9-iNG. *** $p < 0.001$ (student's t-test). The sequences in FIG. 27 are GTTATCTCCTGTTCTGCAGCTGT (SEQ ID NO: 621), GTTATCTCCTGCTCTGCAGCAGA (SEQ ID NO: 628), GATATCTCCTGT TCTGCAGGAGA (SEQ ID NO: 629).

(29) FIG. 28 shows off-target activities of AAV9-iNG. The nucleotide frequency at the on-target site of the four mdx.sup.4cv mice at 10 months after treatment with AAV9-iNG. The desired edit at A.sup.4 is highlighted in green and the bystander C.sup.6 edit in red. The sequence in FIG. 28 is GTTATCTCCTGTTCTGCAGCTGT (SEQ ID NO: 621),

(30) FIGS. 29a-29c show off-target RNA editing activities in the mdx.sup.4cv mouse hearts (10 weeks and 10 months of age) induced by systemic delivery of AAV9-iNG. FIG. 29a shows the number of RNA SNVs in three AAV9-iNG treated mouse hearts after filtering the RNA SNVs in three control mouse hearts. FIG. 29b shows the A-to-I editing rate of all RNA SNVs. FIG. 29c shows the A-to-I editing rate of the common RNA SNVs in the three AAV9-iNG treated mouse hearts.

(31) FIGS. 30a-30b show iABE-NGA editing-mediated exon skipping of human DMD exon 55 in DMD-hiPSC-derived cardiomyocytes, FIG. 30a shows Sanger sequencing showing that iABE-NGA mediated efficient conversion of A to G at the splice acceptor site of human DMD exon 55. FIG. 30b shows RT-PCR analysis showing that iABE-NGA editing led to skipping of exon 55 in Δ E48-54 DMD-hiPSC-derived cardiomyocytes. The WT, Δ E48-54 (green arrow) and E55 skipped transcript (red arrow) were clearly discernable.

(32) FIGS. 31a-31c show iABE-NGA editing-mediated exon 7 inclusion of human SMN2. FIG. 31a shows sequences of SMN1 and SMN2 exon 7 with the different nucleotides at position 6 highlighted in red. Two gRNAs were designed to edit A36 to G with the PAM highlighted in purple. FIG. 31b shows RT-PCR analysis of Neuro-2a cells transfected with a SMN2 reporter gene plus or minus the base editors and gRNAs. The exon 7 inclusion (FL) or exclusion (Δ 7) transcripts were measured. FIG. 31c shows quantification of the RT-PCR products. ** $p < 0.01$; **** $p < 0.0001$; ns, not significant when compared with the control group, using one-way ANOVA with Turkey post test. The sequences in FIG. 31a are GGTTCAGACAAAATCA AAAAGAAGGAAGGTGCTCACATTCCTTAAATTAAGGA (SEQ ID NO: 632), GGTTCAGACAAAATCAAAAAGAAGGAAGGTGCTCACATTCCTTAAATTAAGGA (SEQ ID NO: 633), GTGCTCACATTCCTTAAATTAAGG (SEQ ID NO: 634), gGCTCACATTCCTTA AATTAAGGA (SEQ ID NO: 635),

(33) FIGS. 32a-32c show the iABE-MGA editing of mouse Angptl3 (FIG. 32a), Apoc3 (FIG. 32b) and Asgr1 (FIG. 32c) in Neuro-2a cells. ** $p < 0.01$; *** $p < 0.001$; **** $p < 0.0001$

DETAILED DESCRIPTION

(34) Fusing the CRISPR-Cas9 nickase with nucleobase deaminases (e.g. cytidine or adenine deaminase), a new paradigm-shifting class of genome editing technology, termed “base editors”, have recently been developed. DNA base editors, via catalyzing the conversion of one base to another, directly and precisely install point mutations into chromosomal DNA without making

DSBs. Therefore, base editing can be developed as promising therapeutics to correct the genetic diseases without DNA cleavage. In particular, the adenine base editors (ABEs) show remarkable fidelity in mouse embryos and rice as compared to cytosine base editors (CBEs), making them highly attractive in therapeutic development. Moreover, nearly half of the point mutations causing human diseases are G-to-A or C-to-T, highlighting the potential of ABEs in correcting a large number of human diseases. In particular, 174 out of 508 pathogenic point mutations for DMD are due to G:C to A:T conversion (Table 5), which can be targeted by ABE editing.

(35) In vivo base editing can correct a custom-made mouse model of Duchenne muscular dystrophy (DMD), which carries a nonsense mutation in exon 20 with a classical 5'-TGG protospacer adjacent motif (PAM) sequence in the noncoding strand for recognition by the Cas9 from *Streptococcus pyogenes* (SpCas9). In silico analysis of the ClinVar database showed that about 42.8% of the 53469 human disease-causing mutations can be potential targets for base editing correction; however, the majority (~72.4%) of these potential targets cannot be suitable for SpCas9 base editing due to the lack of the 5'-NGG PAM sequence within the suitable distance from the mutations. Several variants of SpCas9 have recently been engineered with relaxed PAM (such as xCas9-3.7, SpCas9-NG and ScCas9) and non-G PAM. These enzymes greatly increase the target scope for correcting human mutations. However, their performance to correct genetic mutations in preclinical animal models remains to be determined. Here, the efficacy of correcting a commonly used mouse model of DMD, mdx.sup.4cv mice were explored, using NG-targeting base editors.

(36) The present disclosure provides fusion proteins, systems, and compositions for editing disease-associated mutations and uses thereof for treating a disease (for example, Duchenne muscular dystrophy).

(37) Reference will now be made in detail to the embodiments of the invention, examples of which are illustrated in the drawings and the examples. This invention may, however, be embodied in many different forms and should not be construed as limited to the embodiments set forth herein.

(38) Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood to one of ordinary skill in the art to which this disclosure belongs.

Terminology

(39) Terms used throughout this application are to be construed with ordinary and typical meaning to those of ordinary skill in the art. However, Applicant desires that the following terms be given the particular definition as defined below.

(40) As used herein, the article “a,” “an,” and “the” means “at least one,” unless the context in which the article is used clearly indicates otherwise.

(41) The term “comprising” and variations thereof as used herein is used synonymously with the term “including” and variations thereof and are open, non-limiting terms. Although the terms “comprising” and “including” have been used herein to describe various embodiments, the terms “consisting essentially of” and “consisting of” can be used in place of “comprising” and “including” to provide for more specific embodiments and are also disclosed.

(42) As used herein, the terms “may,” “optionally,” and “may optionally” are used interchangeably and are meant to include cases in which the condition occurs as well as cases in which the condition does not occur. Thus, for example, the statement that a formulation “may include an excipient” is meant to include cases in which the formulation includes an excipient as well as cases in which the formulation does not include an excipient.

(43) The terms “about” and “approximately” are defined as being “close to” as understood by one of ordinary skill in the art. In one non-limiting embodiment, the terms are defined to be within 10%. In another non-limiting embodiment, the terms are defined to be within 5%. In still another non-limiting embodiment, the terms are defined to be within 1%.

(44) The term “Cas9” or “Cas9 nuclease” refers to an RNA-guided nuclease comprising a Cas9 protein, or a fragment thereof (e.g., a protein comprising an active, inactive, or partially active DNA cleavage domain of Cas9, and/or the gRNA binding domain of Cas9). A Cas9 nuclease is also

referred to sometimes as a cas1 nuclease or a CRISPR (clustered regularly interspaced short palindromic repeat)-associated nuclease. CRISPR is an adaptive immune system that provides protection against mobile genetic elements (viruses, transposable elements and conjugative plasmids). CRISPR clusters contain spacers, sequences complementary to antecedent mobile elements, and target invading nucleic acids. CRISPR clusters are transcribed and processed into CRISPR RNA (crRNA). In type II CRISPR systems correct processing of pre-crRNA requires a trans-encoded small RNA (tracrRNA), endogenous ribonuclease 3 (rnc) and a Cas9 protein. The tracrRNA serves as a guide for ribonuclease 3-aided processing of pre-crRNA. Subsequently, Cas9/crRNA/tracrRNA endonucleolytically cleaves linear or circular dsDNA target complementary to the spacer. The target strand not complementary to crRNA is first cut endonucleolytically, then trimmed 3'-5' exonucleolytically. In nature, DNA-binding and cleavage typically requires protein and both RNAs. However, single guide RNAs ("sgRNA", or simply "gRNA") can be engineered so as to incorporate aspects of both the crRNA and tracrRNA into a single RNA species. See, e.g., Jinek M., Chylinski K., Fonfara I., Hauer M., Doudna J. A., Charpentier E. Science 337:816-821(2012), the entire contents of which is hereby incorporated by reference. Cas9 recognizes a short motif in the CRISPR repeat sequences (the PAM or protospacer adjacent motif) to help distinguish self versus non-self. Cas9 nuclease sequences and structures are well known to those of skill in the art (see, e.g., "Complete genome sequence of an M1 strain of *Streptococcus pyogenes*." Ferretti et al., J. J., McShan W. M., Ajdic D. J., Savic D. J., Savic G., Lyon K., Primeaux C., Sezate S., Suvorov A. N., Kenton S., Lai H. S., Lin S. P., Qian Y., Jia H. G., Najjar F. Z., Ren Q., Zhu H., Song L., White J., Yuan X., Clifton S. W., Roe B. A., McLaughlin R. E., Proc. Natl. Acad. Sci. U.S.A. 98:4658-4663(2001); "CRISPR RNA maturation by trans-encoded small RNA and host factor RNase III." Deltcheva E., Chylinski K., Sharma C. M., Gonzales K., Chao Y., Pirzada Z. A., Eckert M. R., Vogel J., Charpentier E., Nature 471:602-607(2011); and "A programmable dual-RNA-guided DNA endonuclease in adaptive bacterial immunity." Jinek M., Chylinski K., Fonfara I., Hauer M., Doudna J. A., Charpentier E. Science 337:816-821(2012), the entire contents of each of which are incorporated herein by reference). Cas9 orthologs have been described in various species, including, but not limited to, *S. pyogenes* and *S. thermophiles*. Additional suitable Cas9 nucleases and sequences will be apparent to those of skill in the art based on this disclosure, and such Cas9 nucleases and sequences include Cas9 sequences from the organisms and loci disclosed in Chylinski, Rhun, and Charpentier, "The tracrRNA and Cas9 families of type II CRISPR-Cas immunity systems" (2013) RNA Biology 10:5, 726-737; the entire contents of which are incorporated herein by reference. In some embodiments, a Cas9 nuclease has an inactive (e.g., an inactivated) DNA cleavage domain, that is, the Cas9 is a nickase.

(45) A "composition" is intended to include a combination of active agent and another compound or composition, inert (for example, a fusion protein, nucleic acid, or virus) or active, such as an adjuvant.

(46) Dystrophinopathies are a group of muscular dystrophies resulting from mutations in the dystrophin gene, located on the short arm of the X chromosome in the Xp21 region (Kunkel et al. 1985; Monaco et al. 1985; Ray et al. 1985). Of these, "Duchenne muscular dystrophy" or "DMD" is the most common dystrophinopathy resulting from complete absence of the dystrophin gene product, the subsarcolemmal protein dystrophin (Hoffman et al. 1987a; Koenig et al. 1987; Hoffman et al. 1988). While dystrophin deficiency can be a primary cause of DMD, multiple secondary pathways are responsible for the progression of muscle necrosis, abnormal fibrosis and failure of regeneration that results in a progressively worsening clinical status. There is evidence supporting oxidative radical damage to myofibers (Rando 2002), inflammation (Spencer and Tidball 2001; Porter et al. 2002), abnormal calcium homeostasis (Allen 2010; Millay 2009), myonuclear apoptosis (Rando 2001b; Sandri et al. 2001; Tews 2002), abnormal fibrosis and failure of regeneration (Rando 2001b; Bernasconi 1995); (Melone 2000; Morrison 2000; Luz 2002). This body of literature has been validated by cross sectional genome-wide approaches that allow an

overall analysis of multiple defective mechanisms in DMD (Chen et al. 2000; Porter 2003). The main symptom of DMD is muscle weakness associated with muscle wasting first with the voluntary muscles, e.g., the hips, pelvic area, thighs, shoulders, and calf muscles.

(47) As used herein, the term “effective amount” refers to an amount of a composition necessary or sufficient to realize a desired biologic effect. An effective amount of the composition would be the amount that achieves a selected result, and such an amount could be determined as a matter of routine experimentation by a person skilled in the art. For example, an effective amount of the composition could be that amount necessary for preventing, treating and/or ameliorating Duchenne muscular dystrophy in a subject. The term is also synonymous with “sufficient amount.”

(48) The term “subject” is defined herein to include animals such as mammals, including, but not limited to, primates (e.g., humans), cows, sheep, goats, horses, dogs, cats, rabbits, rats, mice and the like. In some embodiments, the subject is a human.

(49) “Pharmaceutically acceptable carrier” (sometimes referred to as a “carrier”) means a carrier or excipient that is useful in preparing a pharmaceutical or therapeutic composition that is generally safe and non-toxic, and includes a carrier that is acceptable for veterinary and/or human pharmaceutical or therapeutic use. The terms “carrier” or “pharmaceutically acceptable carrier” can include, but are not limited to, phosphate buffered saline solution, water, emulsions (such as an oil/water or water/oil emulsion) and/or various types of wetting agents.

(50) As used herein, the term “carrier” encompasses any excipient, diluent, filler, salt, buffer, stabilizer, solubilizer, lipid, stabilizer, or other material well known in the art for use in pharmaceutical formulations. The choice of a carrier for use in a composition will depend upon the intended route of administration for the composition. The preparation of pharmaceutically acceptable carriers and formulations containing these materials is described in, e.g., *Remington's Pharmaceutical Sciences*, 21st Edition, ed. University of the Sciences in Philadelphia, Lippincott, Williams R. Wilkins, Philadelphia, PA, 2005. Examples of physiologically acceptable carriers include saline, glycerol, DMSO, buffers such as phosphate buffers, citrate buffer, and buffers with other organic acids; antioxidants including ascorbic acid; low molecular weight (less than about 10 residues) polypeptides; proteins, such as serum albumin, gelatin, or immunoglobulins; hydrophilic polymers such as polyvinylpyrrolidone; amino acids such as glycine, glutamine, asparagine, arginine or lysine; monosaccharides, disaccharides, and other carbohydrates including glucose, mannose, or dextrans; chelating agents such as EDTA; sugar alcohols such as mannitol or sorbitol; salt-forming counterions such as sodium; and/or nonionic surfactants such as TWEEN™ (ICI, Inc.; Bridgewater, NJ), polyethylene glycol (PEG), and PLURONICS™ (BASF; Florham Park, NJ). To provide for the administration of such dosages for the desired therapeutic treatment, compositions disclosed herein can advantageously comprise between about 0.1% and 99% by weight of the total of one or more of the subject compounds based on the weight of the total composition including carrier or diluent.

(51) A gRNA is a component of the CRISPR/Cas system. A “gRNA” (guide ribonucleic acid) herein refers to a fusion of a CRISPR-targeting RNA (crRNA) and a trans-activation crRNA (tracrRNA), providing both targeting specificity and scaffolding/binding ability for Cas9 nuclease. A “crRNA” is a bacterial RNA that confers target specificity and requires tracrRNA to bind to Cas9. A “tracrRNA” is a bacterial RNA that links the crRNA to the Cas9 nuclease and typically can bind any crRNA. The sequence specificity of a Cas DNA-binding protein is determined by gRNAs, which have nucleotide base-pairing complementarity to target DNA sequences. The native gRNA comprises a Specificity Determining Sequence (SDS), which specifies the DNA sequence to be targeted. At least a portion of the target DNA sequence is complementary to the SDS of the gRNA. For Cas9 to successfully bind to the DNA target sequence, a region of the target sequence is complementary to the SDS of the gRNA sequence and is immediately followed by the correct protospacer adjacent motif (PAM) sequence (e.g., NGG or NG for Cas9 used herein). In some embodiments, an SDS is 100% complementary to its target sequence. In some embodiments, the

SDS sequence is less than 100% complementary to its target sequence and is, thus, considered to be partially complementary to its target sequence. For example, a targeting sequence may be 99%, 98%, 97%, 96%, 95%, 94%, 93%, 92%, 91%, or 90% complementary to its target sequence.

(52) The term “linker,” as used herein, refers to a chemical group or a molecule linking two molecules or moieties, e.g., two domains of a fusion protein, such as, for example, a Cas9 domain and a nucleic acid editing domain (e.g., a deaminase domain). In some embodiments, a linker joins a gRNA binding domain of an RNA-programmable nuclease, including a Cas9 nuclease domain, and the catalytic domain of a nucleic acid editing protein. In some embodiments, a linker joins a Cas9 and a nucleic acid editing protein. Typically, the linker is positioned between, or flanked by, two groups, molecules, or other moieties and connected to each one via a covalent bond, thus connecting the two. In some embodiments, the linker is an amino acid or a plurality of amino acids (e.g., a peptide or protein). In some embodiments, the linker is an organic molecule, group, polymer, or chemical moiety. In some embodiments, the linker is 5-100 amino acids in length, for example, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 30-35, 35-40, 40-45, 45-50, 50-60, 60-70, 70-80, 80-90, 90-100, 100-150, or 150-200 amino acids in length. Longer or shorter linkers are also contemplated.

(53) The term “mutation,” as used herein, refers to a substitution of a residue within a sequence, e.g., a nucleic acid or amino acid sequence, with another residue, or a deletion or insertion of one or more residues within a sequence. Mutations are typically described herein by identifying the original residue followed by the position of the residue within the sequence and by the identity of the newly substituted residue. Various methods for making the amino acid substitutions (mutations) provided herein are well known in the art, and are provided by, for example, Green and Sambrook, *Molecular Cloning: A Laboratory Manual* (4th ed., Cold Spring Harbor Laboratory Press, Cold Spring Harbor, N.Y. (2012)).

(54) The term “nickase” as used herein, refers to a nuclease that cleaves only a single DNA strand, either due to its natural function or because it has been engineered to cleave only a single DNA strand, Jinek et al., “A programmable dual-RNA-guided DNA endonuclease in adaptive bacterial immunity” *Science* 337(6096):816-821 (2012) and Cong et al. *Multiplex genome engineering using CRISPR/Cas systems* *Science* 339(6121):819-823 (2013).

(55) The term “nucleic acid editing domain,” as used herein refers to a protein or enzyme capable of making one or more modifications (e.g., deamination of a cytidine residue) to a nucleic acid (e.g., DNA or RNA). Exemplary nucleic acid editing domains include, but are not limited to a deaminase, a nuclease, a nickase, a recombinase, a methyltransferase, a methylase, an acetylase, an acetyltransferase, a transcriptional activator, or a transcriptional repressor domain. In some embodiments the nucleic acid editing domain comprises a deaminase (e.g., a cytidine deaminase or an adenine deaminase).

(56) An “adenine deaminase” is an enzyme involved in purine metabolism. It is needed for the breakdown of adenosine from food and for the turnover of nucleic acids in tissues. Its primary function in humans is the development and maintenance of the immune system. An adenine deaminase catalyzes hydrolytic deamination of adenosine (forming inosine, which base pairs as G) in the context of DNA.

(57) The term “recombinant” as used herein in the context of proteins or nucleic acids refers to proteins or nucleic acids that do not occur in nature, but are the product of human engineering. For example, in some embodiments, a recombinant protein or nucleic acid molecule comprises an amino acid or nucleotide sequence that comprises at least one, at least two, at least three, at least four, at least five, at least six, or at least seven mutations as compared to any naturally occurring sequence.

(58) As used herein, the terms “treating” or “treatment” of a subject includes the administration of a drug to a subject with the purpose of curing, healing, alleviating, relieving, altering, remedying, ameliorating, improving, stabilizing or affecting a disease or disorder, or a symptom of a disease or

disorder. The terms “treating” and “treatment” can also refer to reduction in severity and/or frequency of symptoms, elimination of symptoms and/or underlying cause, and improvement or remediation of damage. For example, Duchenne muscular dystrophy, may result in e.g., a slowing of muscle degeneration, decreased fatigue, increased muscle strength, reduced blood levels of creatine kinase (CK), decreased difficulty with motor skills, decreased muscle fiber deformities, decreased inflammation or fibrotic tissue infiltration in the muscle, stabilization of the progression of the disease (e.g., by halting progressive muscle weakness) etc.

(59) “Therapeutically effective amount” or “therapeutically effective dose” of a composition (e.g. a fusion protein, a nucleic acid, or virus) refers to an amount that is effective to achieve a desired therapeutic result. In some embodiments, a desired therapeutic result is the prevention of Duchenne muscular dystrophy. In some embodiments, a desired therapeutic result is the treatment of Duchenne muscular dystrophy. Therapeutically effective amounts of a given therapeutic agent will typically vary with respect to factors such as the type and severity of the disorder or disease being treated and the age, gender, and weight of the subject. The term can also refer to an amount of a therapeutic agent, or a rate of delivery of a therapeutic agent (e.g., amount over time), effective to facilitate a desired therapeutic effect, such as coughing relief. The precise desired therapeutic effect will vary according to the condition to be treated, the tolerance of the subject, the agent and/or agent formulation to be administered (e.g., the potency of the therapeutic agent, the concentration of agent in the formulation, and the like), and a variety of other factors that are appreciated by those of ordinary skill in the art. In some instances, a desired biological or medical response is achieved following administration of multiple dosages of the composition to the subject over a period of days, weeks, or years.

(60) A “vector” is a composition of matter which comprises an isolated nucleic acid and which can be used to deliver the isolated nucleic acid to the interior of a cell. Numerous vectors are known in the art including, but not limited to, linear polynucleotides, polynucleotides associated with ionic or amphiphilic compounds, plasmids, and viruses. Thus, the term “vector” includes an autonomously replicating plasmid or a virus. The term should also be construed to include non-plasmid and non-viral compounds which facilitate transfer of nucleic acid into cells, such as, for example, polylysine compounds, liposomes, and the like. Examples of viral vectors include, but are not limited to, lentiviral vectors, adenoviral vectors, adeno-associated virus vectors, retroviral vectors, and the like.

(61) An “adeno-associated virus” or “AAV” is a virus which infects humans and some other primate species. The wild-type AAV genome is a single-stranded deoxyribonucleic acid (ssDNA), either positive- or negative-sensed. The genome comprises two inverted terminal repeats (ITRs), one at each end of the DNA strand, and two open reading frames (ORFs): rep and cap between the ITRs. The rep ORF comprises four overlapping genes encoding Rep proteins required for the AAV life cycle. The cap ORF comprises overlapping genes encoding capsid proteins: VP1, VP2 and VP3, which interact together to form the viral capsid. VP1, VP2 and VP3 are translated from one mRNA transcript, which can be spliced in two different manners: either a longer or shorter intron can be excised resulting in the formation of two isoforms of mRNAs: a ~2.3 kb- and a ~2.6 kb-long mRNA isoform. The capsid forms a supranuclear assembly of approximately 60 individual capsid protein subunits into a non-enveloped, T-1 icosahedral lattice capable of protecting the AAV genome. The mature capsid is composed of VP1, VP2, and VP3 (molecular masses of approximately 87, 73, and 62 kDa respectively) in a ratio of about 1:1:10.

(62) The term “nucleic acid” as used herein means a polymer composed of nucleotides, deoxyribonucleotides or ribonucleotides.

(63) The terms “ribonucleic acid” and “RNA” as used herein mean a polymer composed of ribonucleotides.

(64) The terms “deoxyribonucleic acid” and “DNA” as used herein mean a polymer composed of deoxyribonucleotides.

(65) The term “oligonucleotide” denotes single- or double-stranded nucleotide multimers of from about 2 to up to about 100 nucleotides in length. Suitable oligonucleotides may be prepared by the phosphoramidite method described by Beaucage and Carruthers, *Tetrahedron Lett.*, 22: 1859-1862 (1981), or by the triester method according to Matteucci, et al., *J. Am. Chem. Soc.*, 103:3185 (1981), both incorporated herein by reference, or by other chemical methods using either a commercial automated oligonucleotide synthesizer or VLSIPS™ technology. When oligonucleotides are referred to as “double-stranded,” it is understood by those of skill in the art that a pair of oligonucleotides exist in a hydrogen-bonded, helical array typically associated with, for example, DNA. In addition to the 100% complementary form of double-stranded oligonucleotides, the term “double-stranded,” as used herein is also meant to refer to those forms which include such structural features as bulges and loops, described more fully in such biochemistry texts as Stryer, *Biochemistry*, Third Ed., (1988), incorporated herein by reference for all purposes.

(66) The term “polynucleotide” refers to a single or double stranded polymer composed of nucleotide monomers.

(67) The term “polypeptide” refers to a compound made up of a single chain of D- or L-amino acids or a mixture of D- and L-amino acids joined by peptide bonds.

(68) The terms “identical” or percent “identity,” in the context of two or more nucleic acids or polypeptide sequences, refer to two or more sequences or subsequences that are the same or have a specified percentage of amino acid residues or nucleotides that are the same (i.e., about 60% identity, preferably 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or higher identity over a specified region when compared and aligned for maximum correspondence over a comparison window or designated region) as measured using a BLAST or BLAST 2.0 sequence comparison algorithms with default parameters described below, or by manual alignment and visual inspection (see, e.g. NCBI web site or the like). Such sequences are then said to be “substantially identical.” This definition also refers to, or may be applied to, the complement of a test sequence. The definition also includes sequences that have deletions and/or additions, as well as those that have substitutions. As described below, the preferred algorithms can account for gaps and the like. Preferably, identity exists over a region that is at least about 10 amino acids or 20 nucleotides in length, or more preferably over a region that is 10-50 amino acids or 20-50 nucleotides in length. As used herein, percent (%) nucleotide sequence identity is defined as the percentage of amino acids in a candidate sequence that are identical to the nucleotides in a reference sequence, after aligning the sequences and introducing gaps, if necessary, to achieve the maximum percent sequence identity. Alignment for purposes of determining percent sequence identity can be achieved in various ways that are within the skill in the art, for instance, using publicly available computer software such as BLAST, BLAST-2, ALIGN, ALIGN-2 or Megalign (DNASTAR) software. Appropriate parameters for measuring alignment, including any algorithms needed to achieve maximal alignment over the full-length of the sequences being compared can be determined by known methods.

(69) For sequence comparisons, typically one sequence acts as a reference sequence, to which test sequences are compared. When using a sequence comparison algorithm, test and reference sequences are entered into a computer, subsequence coordinates are designated, if necessary, and sequence algorithm program parameters are designated. Preferably, default program parameters can be used, or alternative parameters can be designated. The sequence comparison algorithm then calculates the percent sequence identities for the test sequences relative to the reference sequence, based on the program parameters.

(70) One example of an algorithm that is suitable for determining percent sequence identity and sequence similarity are the BLAST and BLAST 2.0 algorithms, which are described in Altschul et al. (1977) *Nuc. Acids Res.* 25:3389-3402, and Altschul et al. (1990) *J. Mol. Biol.* 215:403-410,

respectively. Software for performing BLAST analyses is publicly available through the National Center for Biotechnology information (www.ncbi.nlm.nih.gov). This algorithm involves first identifying high scoring sequence pairs (HSPs) by identifying short words of length W in the query sequence, which either match or satisfy some positive-valued threshold score T when aligned with a word of the same length in a database sequence. T is referred to as the neighborhood word score threshold (Altschul et al. (1990) *J. Mol. Biol.* 215:403-410). These initial neighborhood word hits act as seeds for initiating searches to find longer HSPs containing them. The word hits are extended in both directions along each sequence for as far as the cumulative alignment score can be increased. Cumulative scores are calculated using, for nucleotide sequences, the parameters M (reward score for a pair of matching residues; always >0) and N (penalty score for mismatching residues; always <0). For amino acid sequences, a scoring matrix is used to calculate the cumulative score. Extension of the word hits in each direction are halted when: the cumulative alignment score falls off by the quantity X from its maximum achieved value; the cumulative score goes to zero or below, due to the accumulation of one or more negative-scoring residue alignments; or the end of either sequence is reached. The BLAST algorithm parameters W, T, and X determine the sensitivity and speed of the alignment. The BLASTN program (for nucleotide sequences) uses as defaults a wordlength (W) of 11, an expectation (E) of 10, M=5, N=-4 and a comparison of both strands. For amino acid sequences, the BLASTP program uses as defaults a wordlength of 3, and expectation (E) of 10, and the BLOSUM62 scoring matrix (see Henikoff and Henikoff (1989) *Proc. Natl. Acad. Sci. USA* 89:10915) alignments (B) of 50, expectation (E) of 10, M=5, N=-4, and a comparison of both strands.

(71) The BLAST algorithm also performs a statistical analysis of the similarity between two sequences (see, e.g., Karlin and Altschul (1993) *Proc. Sci. USA* 90:5873-5787). One measure of similarity provided by the BLAST algorithm is the smallest sum probability (P(N)), which provides an indication of the probability by which a match between two nucleotide or amino acid sequences would occur by chance. For example, a nucleic acid is considered similar to a reference sequence if the smallest sum probability in a comparison of the test nucleic acid to the reference nucleic acid is less than about 0.2, more preferably less than about 0.01.

(72) The term “increased” or “increase” as used herein generally means an increase by a statically significant amount; for the avoidance of any doubt, “increased” means an increase of at least 10% as compared to a reference level, for example an increase of at least about 20%, or at least about 30%, or at least about 40%, or at least about 50%, or at least about 60%, or at least about 70%, or at least about 80%, or at least about 90% or up to and including a 100% increase or any increase between 10-100% as compared to a reference level, or at least about a 2-fold, or at least about a 3-fold, or at least about a 4-fold, or at least about a 5-fold or at least about a 10-fold increase, or any increase between 2-fold and 10-fold or greater as compared to a reference level.

(73) The term “reduced”, “reduce”, “reduction”, or “decrease” as used herein generally means a decrease by a statistically significant amount. However, for avoidance of doubt, “reduced” means a decrease by at least 10% as compared to a reference level, for example a decrease by at least about 20%, or at least about 30%, or at least about 40%, or at least about 50%, or at least about 60%, or at least about 70%, or at least about 80%, or at least about 90% or up to and including a 100% decrease (i.e. absent level as compared to a reference sample), or any decrease between 10-100% as compared to a reference level.

(74) Throughout this application, various publications are referenced. The disclosures of these publications in their entireties are hereby incorporated by reference into this application in order to more fully describe the state of the art to which this pertains. The references disclosed are also individually and specifically incorporated by reference herein for the material contained in them that is discussed in the sentence in which the reference is relied upon.

(75) Compositions

(76) In some aspects, disclosed herein is a fusion protein comprising a Cas9 nickase and a

nucleotide deaminase, wherein the Cas9 nickase comprises a first amino acid substitution at a position selected from the group consisting of 262, 324, 409, 480, 543, 694, and 1219 when compared to SEQ ID NO: 11, and wherein the Cas9 nickase comprises a second amino acid substitution at a position selected from the group consisting of 1111, 1135, 1218, 1219, 1322, 1335, and 1337 when compared to SEQ ID NO: 11. In some embodiments, the Cas9 nickase and the nucleotide deaminase are operably linked.

(77) In some embodiments, the first amino acid substitution is selected from the group consisting of A262T, R324L, S409I, E480K, E543D, M694I, and E1219V when compared to SEQ ID NO: 11. In some embodiments, the second amino acid substitution is selected from the group consisting of L1111R, D1135V, G1218R, E1219F, A1322R, R1335V, R1335Q, R1335E, and T1337R when compared to SEQ ID NO: 11.

(78) In some embodiments, the fusion protein comprises a Cas9 nickase fused to a deaminase and further fused to a Uracil DNA glycosylase inhibitor (UGI) domain.

(79) In some embodiments, the Cas9 nickase comprises an amino acid sequence at least 80% identity (at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or at least 99.5%) to the amino acid sequence selected from the group consisting of SEQ ID NOs: 12-20. In some embodiments, the Cas9 nickase comprises the amino acid sequence selected from the group consisting of SEQ ID NOs: 12-20.

(80) In some embodiments, the Cas9 nickase when in conjunction with a bound guide RNA (gRNA) specifically binds to a target nucleic acid sequence.

(81) In some embodiments, the Cas9 nickase recognizes a NG protospacer adjacent motif (PAM) sequence.

(82) In some embodiments, the nucleotide deaminase is a cytidine deaminase or an adenine deaminase. In some embodiments, the nucleotide deaminase is a cytidine deaminase. In some embodiments, the nucleotide deaminase is an adenine deaminase. In some embodiments, the adenine deaminase is a dimeric adenine deaminase or a monomeric adenine deaminase domain. In some embodiments, the dimeric adenine deaminase comprises an amino acid sequence at least 80% identity (at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or at least 99.5%) to SEQ ID NOs: 22. In some embodiments, the monomeric adenine deaminase comprises an amino acid sequence at least 80% identity (at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or at least 99.5%) to SEQ NOs: 23. In some embodiments, the monomeric adenine deaminase comprises amino acid substitutions A56G and V82G when compared to SEQ ID NO: 47. In some embodiments, the adenine deaminase comprises the amino acid sequence set forth in SEQ ID NO: 22 or 23.

(83) In some embodiments, the adenine deaminase is encoded by a nucleotide sequence at least 80% identity (at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or at least 99.5%) to SEQ ID NO: 41 or 42. In some embodiments, the adenine deaminase is encoded by the nucleotide sequence as set forth in SEQ ID NO: 41 or 42.

(84) In some embodiments, the Cas9 nickase comprising the monomeric adenine deaminase domain has a lower off-target RNA editing activity than a Cas9 nickase comprising a dimeric adenine deaminase domain (e.g., at least 10% lower, at least 20% lower, at least 30% lower, at least 40% lower, at least 50% lower, at least 60% lower, at least 70% lower, at least 80% lower, at least 90% lower, at least 95% lower, at least 99% lower, or at least 2 times lower, at least 3 times lower, at least 4 times lower, at least 5 times lower, at least 6 times lower, at least 7 times lower, at least 8 times lower, at least 9 times lower, at least 10 times lower, at least 20 times lower, at least 50 times lower, at least 100 times lower, at least 150 times lower, at time 200 times lower, at least 500 times lower, or at least 1000 times lower).

(85) In some aspects, disclosed herein is a construct comprising a nucleotide sequence encoding the fusion protein of any preceding aspect. In some embodiments, the construct further comprises a

promoter operably linked to the nucleotide sequence, wherein the promoter is a CMV promoter. Accordingly, in some embodiments, the nucleotide sequence of any preceding aspect is at least 80% identity (at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or at least 99.5%) to SEQ ID NO: 1 or SEQ ID NO: 2.

(86) In some aspects, disclosed herein is a system for base editing comprising: a first nucleotide sequence encoding a N-terminal portion of a Cas9 nickase fused at its C-terminus to an intein-N; and a second nucleotide sequence encoding an intein-C fused to the N-terminus of a C-terminal portion of the Cas9 nickase; wherein the first nucleotide sequence further comprises a nucleotide sequence encoding a nucleotide deaminase fused to the N-terminus of the N-terminal portion of the Cas9 nickase, wherein the Cas9 nickase comprises a first amino acid substitution at a position selected from the group consisting of 262, 324, 409, 480, 543, 694, and 1219 when compared to SEQ ID NO: 11, and wherein the Cas9 nickase comprises a second amino acid substitution at a position selected from the group consisting of 1111, 1135, 1218, 1219, 1322, 1335, and 1337 when compared to SEQ ID NO: 11.

(87) In some embodiments, the Cas9 nickase comprises an amino acid sequence at least 80% identity (at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or at least 99.5%) to the amino acid sequence selected from the group consisting of SEQ ID NOs: 12-20. In some embodiments, the Cas9 nickase comprises the amino acid sequence selected from the group consisting of SEQ ID NOs: 12-20.

(88) Accordingly, in some embodiments, the Cas9 nickase is encoded by a nucleotide sequence at least 80% identity (at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or at least 99.5%) to the nucleotide sequence selected from the group consisting of SEQ ID NOs: 31-39. In some embodiments, the Cas9 nickase is encoded by a nucleotide sequence selected from the group consisting of SEQ ID NOs:

(89) In some embodiments, the N-terminal portion of the Cas9 nickase comprises the amino acid positions 2-560, 2-561, 2-562, 2-563, 2-564, 2-565, 2-566, 2-567, 2-568, 2-569, 2-570, 2-571, 2-572, 2-573, 2-574, 2-575, 2-576, 2-577, 2-578, 2-579, or 2-580 of the amino acid sequence selected from the group consisting of SEQ ID NOs: 12-20. In some embodiments, the C-terminal portion of the Cas9 nickase comprises the amino acid positions 574-1368 of the amino acid sequence selected from the group consisting of SEQ ID NOs: 12-20.

(90) An “intein” is a segment of a protein that is able to excise itself and join the remaining portions (the exteins) with a peptide bond in a process known as protein splicing. Inteins are also referred to as “protein introns.” The process of an intein excising itself and joining the remaining portions of the protein is herein termed “protein splicing” or “intein-mediated protein splicing.” In some embodiments, an intein of a precursor protein (an intein containing protein prior to intein-mediated protein splicing) comes from two genes. Such intein is referred to herein as a split intein. The inteins used herein can be, for example, Npu DnaE intein, Cfa, DnaE intein or gp41-1 intein. The intein encoded by the DnaE-n gene is herein referred to as “intein-N.” The intein encoded by the DnaE-c gene is herein referred to as “intein-C.”

(91) Other intein systems may also be used. For example, a synthetic intein based on the dnaE intein, the Cfa-N and Cfa-C intein pair, has been described (e.g., in Stevens et al., J Am Chem Soc. 2016 Feb. 24; 138(7):2162-5, incorporated herein by reference). Non-limiting examples of intein pairs that may be used in accordance with the present disclosure include: Ssp GyrB intein, Ssp DnaX intein, Ter DnaE3 intein, Ter ThyX intein, Rma DnaB intein and Cne Prp8 intein (e.g., as described in U.S. Pat. No. 8,394,604, incorporated herein by reference).

(92) In some embodiments, the intein-N is a Cfa intein-N or a gp41-1 intein-N. Accordingly, in some embodiments, the intein-N comprises the amino acid sequence of SEQ ID NO: 24 or 26. In some embodiments, the intein-N is encoded by the nucleotide sequence of SEQ ID NO: 43 or 45. Accordingly, the first nucleotide sequence of any preceding aspects comprises SEQ ID NO: 43 or 45.

- (93) In some embodiments, the intein-C is a Cfa intein-C or a gp41-1 intein-C. Accordingly, in some embodiments, the intein-C comprises the amino acid sequence of SEQ ID NO: 25 or 27. In some embodiments, the intein-C is encoded by the nucleotide sequence of SEQ ID NO: 44 or 46. Accordingly, the second nucleotide sequence of any preceding aspects comprises SEQ ID NO: 44 or 46.
- (94) In some embodiments, the first nucleotide sequence or the second nucleotide sequence further comprises a nucleotide encoding a guide RNA (gRNA).
- (95) In some embodiments, the first nucleotide sequence comprises an N terminal portion of SEQ ID NO: 1 or 2. In some embodiments, the first nucleotide comprises a portion of SEQ ID NO: 1 or 2.
- (96) In some embodiments, the second nucleotide sequence comprises a C terminal portion of SEQ ID NO: 1 or 2. In some embodiments, the second nucleotide comprises a portion of SEQ ID NO: 1 or 2.
- (97) In some aspects, disclosed herein is a composition comprising: a first recombinant viral particle comprising a first nucleotide sequence encoding a N-terminal portion of a Cas9 nickase fused at its C-terminus to an intein-N; and a second recombinant viral particle comprising a second nucleotide sequence encoding an intein-C fused to the N-terminus of a C-terminal portion of the Cas9 nickase; wherein the first nucleotide sequence further comprises a nucleotide sequence encoding a nucleotide deaminase fused to the N-terminus of the N-terminal portion of the Cas9 nickase, wherein the Cas9 nickase comprises a first amino acid substitution at a position selected from the group consisting of 262, 324, 409, 480, 543, 694, and 1219 when compared to SEQ ID NO: 11, and wherein the Cas9 nickase comprises a second amino acid substitution at a position selected from the group consisting of 1111, 1135, 1218, 1219, 1322, 1335, and 1337 when compared to SEQ ID NO: 11.
- (98) In some embodiments, the Cas9 nickase comprises an amino acid sequence at least 80% identity (at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or at least 99.5%) to the amino acid sequence selected from the group consisting of SEQ ID NOs: 12-20. In some embodiments, the Cas9 nickase comprises the amino acid sequence selected from the group consisting of SEQ ID NOs: 12-20.
- (99) Accordingly, in some embodiments, the Cas9 nickase is encoded by a nucleotide sequence at least 80% identity (at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or at least 99.5%) to the nucleotide sequence selected from the group consisting of SEQ ID NOs: 31-39. In some embodiments, the Cas9 nickase is encoded by a nucleotide sequence selected from the group consisting of SEQ ID NOs: 31-39.
- (100) In some embodiments, the N-terminal portion of the Cas9 nickase comprises the amino acid positions 2-560, 2-561, 2-562, 2-563, 2-564, 2-565, 2-566, 2-567, 2-568, 2-569, 2-570, 2-571, 2-572, 2-573, 2-574, 2-575, 2-576, 2-577, 2-578, 2-579, or 2-580 of the amino acid sequence selected from the group consisting of SEQ ID NOs: 12-20. In some embodiments, the C-terminal portion of the Cas9 nickase comprises the amino acid positions 574-1368 of the amino acid sequence selected from the group consisting of SEQ ID NOs: 12-20.
- (101) In some embodiments, the intein-N is a Cfa intein-N or a gp41-1 intein-N. Accordingly, in some embodiments, the intein-N comprises the amino acid sequence of SEQ ID NO: 24 or 26. In some embodiments, the intein-N is encoded by the nucleotide sequence of SEQ ID NO: 43 or 45. Accordingly, the first nucleotide sequence of any preceding aspects comprises SEQ ID NO: 43 or 45.
- (102) In some embodiments, the intein-C is a Cfa intein-C or a gp41-1 intein-C. Accordingly, in some embodiments, the intein-C comprises the amino acid sequence of SEQ ID NO: 25 or 27. In some embodiments, the intein-C is encoded by the nucleotide sequence of SEQ ID NO: 44 or 46. Accordingly, the second nucleotide sequence of any preceding aspects comprises SEQ ID NO: 44 or 46.

(103) A number of viral based systems have been developed for gene transfer into mammalian cells. For example, retroviruses provide a convenient platform for gene delivery systems. A selected gene can be inserted into a vector and packaged in retroviral particles using techniques known in the art. The recombinant virus can then be isolated and delivered to cells of the subject either in vivo or ex vivo. A number of retroviral systems are known in the art. In some embodiments, adenovirus vectors are used. A number of adenovirus vectors are known in the art. In one embodiment, lentivirus vectors are used

(104) AAV is a preferred vector because it can infect many cell types and is nonpathogenic to humans. AAV type vectors can transport about 4 to 5 kb and wild type AAV is known to stably insert into chromosome 19. Vectors which contain this site-specific integration property are preferred. In some embodiments, the AAV vector of any preceding aspect further a herpes simplex virus thymidine kinase gene, HSV-tk, and/or a marker gene, such as the gene encoding the green fluorescent protein, GFP.

(105) Accordingly, in some aspects, disclosed herein is a composition comprising: a first recombinant adeno-associated virus (AAV) particle comprising a first nucleotide sequence encoding a N-terminal portion of a Cas9 nickase fused at its C-terminus to an intein-N; and a second recombinant AAV particle comprising a second nucleotide sequence encoding an intein-C fused to the N-terminus of a C-terminal portion of the Cas9 nickase; wherein the first nucleotide sequence further comprises a nucleotide sequence encoding a nucleotide deaminase fused to the N-terminus of the N-terminal portion of the Cas9 nickase, wherein the Cas9 nickase comprises a first amino acid substitution at a position selected from the group consisting of 262, 324, 409, 480, 543, 694, and 1219 when compared to SEQ ID NO: 11, and wherein the Cas9 nickase comprises a second amino acid substitution at a position selected from the group consisting of 1111, 1135, 1218, 1219, 1322, 1335, and 1337 when compared to SEQ ID NO: 11.

(106) In some embodiments, the AAV contains a pair of inverted terminal repeats (ITRs) which flank at least one cassette containing a promoter which directs cell-specific expression operably linked to a heterologous gene. Heterologous in this context refers to any nucleotide sequence or gene which is not native to the AAV or B19 parvovirus. Typically the AAV and B19 coding regions have been deleted, resulting in a safe, noncytotoxic vector. The AAV ITRs, or modifications thereof, confer infectivity and site-specific integration. U.S. Pat. No. 6,261,834 is herein incorporated by reference for material related to the AAV vector.

(107) The inserted genes in viral and retroviral usually contain promoters, and/or enhancers to help control the expression of the desired gene product. A promoter is generally a sequence or sequences of DNA that function when in a relatively fixed location in regard to the transcription start site. A promoter contains core elements required for basic interaction of RNA polymerase and transcription factors, and may contain upstream elements and response elements. The spacing between promoter elements frequently is flexible, so that promoter function is preserved when elements are inverted or moved relative to one another. Depending on the promoter, it appears that individual elements can function either cooperatively or independently to activate transcription.

(108) The term “promoter” as used herein is defined as a DNA sequence recognized by the synthetic machinery of the cell, or introduced synthetic machinery, required to initiate the specific transcription of a polynucleotide sequence. One example of a suitable promoter is the immediate early cytomegalovirus (CMV) promoter sequence. This promoter sequence is a strong constitutive promoter sequence capable of driving high levels of expression of any polynucleotide sequence operatively linked thereto. Another example of a suitable promoter is Elongation Growth Factor-1 alpha (EF-1 α). However, other promoter sequences may also be used, including, but not limited to the simian virus 40 (SV40), early promoter, mouse mammary tumor virus (MMTV), human immunodeficiency virus (HIV) long terminal repeat (LTR) promoter, promoter, PGK-1 promoter, an avian leukemia virus promoter, an Epstein-Barr virus immediate early promoter, a Rous sarcoma virus promoter, as well as human gene promoters such as, but not limited to, the actin promoter, the

myosin promoter, the hemoglobin promoter, and the creatine kinase promoter as well as synthetic protein, such as a CAG promoter. Further, the invention should not be limited to the use of constitutive promoters, inducible promoters are also contemplated as part of the invention. The use of an inducible promoter provides a molecular switch capable of turning on expression of the polynucleotide sequence which it is operatively linked when such expression is desired, or turning off the expression when expression is not desired. In some embodiments, the AAV of any preceding aspect comprises a constitutive promoter or a muscle tissue specific promoter, e.g., a muscle-specific MHP1 promoter.

(109) In some embodiments, the promoter is a CMV promoter. In some embodiments, the CMV promoter comprises the nucleotide sequence of SEQ ID NO: 28. Accordingly, in some embodiments, the first recombinant adeno-associated virus (AAV) particle of any preceding aspect comprises a first nucleotide sequence that is at least 80% identity (at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or at least 99.5%) to SEQ ID NO: 3 or SEQ ID NO: 5, wherein the second recombinant adeno-associated virus (AAV) particle of any preceding aspect comprises a first nucleotide sequence that is at least 80% identity (at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or at least 99.5%) to SEQ ID NO: 4 or SEQ ID NO: 6.

(110) In some embodiments, the promoter is a MHP1 promoter. In some embodiments, the MHP1 promoter comprises the nucleotide sequence of SEQ ID NO: 29. Accordingly, in some embodiments, the first recombinant adeno-associated virus (AAV) particle of any preceding aspect comprises a first nucleotide sequence that is at least 80% identity (at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or at least 99.5%) to SEQ ID NO: 7, wherein the second recombinant adeno-associated virus (AAV) particle of any preceding aspect comprises a first nucleotide sequence that is at least 80% identity (at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or at least 99.5%) to SEQ ID NO: 8.

(111) In some embodiments, viral vector is a lentivirus vector. Accordingly, disclosed herein is a composition comprising: a first recombinant lentivirus particle comprising a first nucleotide sequence encoding a N-terminal portion of a Cas9 nickase fused at its C-terminus to an intein-N; and a second recombinant lentivirus particle comprising a second nucleotide sequence encoding an intein-C fused to the N-terminus of a C-terminal portion of the Cas9 nickase; wherein first nucleotide sequence further comprises a nucleotide sequence encoding a nucleotide deaminase fused to the N-terminus of the N-terminal portion of the Cas9 nickase, wherein the Cas9 nickase comprises a first amino acid substitution at a position selected from the group consisting of 262, 324, 409, 480, 543, 694, and 1219 when compared to SEQ ID NO: 11, and wherein the Cas9 nickase comprises a second amino acid substitution at a position selected from the group consisting of 1111, 1135, 1218, 1219, 1322, 1335, and 1337 when compared to SEQ ID NO: 11.

(112) Methods of introducing and expressing genes into a cell are known in the art. In the context of an expression vector, the vector can be readily introduced into a host cell, e.g., mammalian, bacterial, yeast, or insect cell by any method in the art. For example, the expression vector can be transferred into a host cell by physical, chemical, or biological means. See, e.g., WO2012079000A1, incorporated by reference herein in their entirety.

(113) Reporter genes are used for identifying potentially transfected cells and for evaluating the functionality of regulatory sequences. In general, a reporter gene is a gene that is not present in or expressed by the recipient organism or tissue and that encodes a polypeptide whose expression is manifested by some easily detectable property, e.g., enzymatic activity. Expression of the reporter gene is assayed at a suitable time after the DNA has been introduced into the recipient cells. Suitable reporter genes may include genes encoding luciferase, beta-galactosidase, chloramphenicol acetyl transferase, secreted alkaline phosphatase, or the green fluorescent protein gene (e.g., Ui-Tei et al., 2000 FEBS Letters 479: 79-82). Suitable expression systems are well

known and may be prepared using known techniques or obtained commercially. In general, the construct with the minimal 5' flanking region showing the highest level of expression of reporter gene is identified as the promoter. Such promoter regions may be linked to a reporter gene and used to evaluate agents for the ability to modulate promoter-driven transcription. In some embodiments, the recombinant nucleic acid of any preceding aspect further comprises a reporter gene. In some embodiments, the reporter gene in invention is GFP.

(114) In some embodiments, the fusion protein, nucleotide, system, or composition of any preceding aspect can be further formulated in a pharmaceutically acceptable carrier.

(115) Genetic Diseases and Methods of Treatment

(116) It is estimated that over 10,000 human diseases are caused by genetic disorders, which are abnormalities in genes or chromosomes, See, e.g., McClellan, J. and M. C. King, Genetic heterogeneity in human disease. *Cell*. 141(2): p. 210-7; Leachman, S. A., et al., Therapeutic siRNAs for dominant genetic skin disorders including pachyonychia congenita. *J Dermatol Sci*, 2008. 51(3): p. 151-7. The compositions disclosed herein can be used to treat a number of these genetic disorders.

(117) In some aspects, disclosed herein is a method of treating a genetic disease in a subject, comprising administering to the subject a therapeutically effective amount of the fusion protein, system, or composition of any preceding aspect.

(118) In some embodiments, the genetic disease is a muscular dystrophy. Muscular dystrophies are a group of muscle diseases caused by mutations in a person's genes. Over time, muscle weakness decreases mobility, making everyday tasks difficult. The methods and compositions disclosed herein can be used for treating a muscular dystrophy, including, for example, Duchenne muscular dystrophy, Becker muscular dystrophy, myotonic dystrophy, limb-girdle muscular dystrophy, facioscapulohumeral dystrophy, congenital muscular dystrophy, distal muscular dystrophy, oculopharyngeal muscular dystrophy, or Emery-Dreifuss muscular dystrophy.

(119) In some embodiments, the genetic disease or disorder is Duchenne muscular dystrophy, dysferlinopathy, or spinal muscular atrophy. In some embodiments, the methods and compositions disclosed herein can be used for treating Duchenne muscular dystrophy. In some embodiments, the genetic disease is spinal muscular atrophy.

(120) In some embodiments, the genetic disease or disorder is hypercholesterolemia. For treating hypercholesterolemia, iABE-NGA is used to create a protective silent mutation in functional genes, for example ANGPTL3, APOC3, PCSK9, or ASGR1, to lower plasma cholesterol levels.

(121) In some embodiments, the genetic disease or disorder is due to a point mutation in a causative gene from a G:C pair to an A:T pair.

(122) In some aspects, disclosed herein is a method of treating a genetic disease in a subject, comprising administering to the subject a therapeutically effective amount of the fusion protein, system, and composition of any preceding aspect.

(123) The disclosed methods can be performed any time prior to the onset of a genetic disease, even prior to the appearance of any symptom. In one aspect, the disclosed methods can be employed 60, 59, 58, 57, 56, 55, 54, 53, 52, 51, 50, 49, 48, 47, 46, 45, 44, 43, 42, 41, 40, 39, 38, 37, 36, 35, 34, 33, 32, 31, 30, 29, 28, 27, 26, 25, 24, 23, 22, 21, 20, 19, 18, 17, 16, 15, 14, 13, 12, 11, 10, 9, 8, 7, 6, 5, 4, 3, 2, or 1 years; 12, 11, 10, 9, 8, 7, 6, 5, 4, 3, 2, or 1 months; 30, 29, 28, 27, 26, 25, 24, 23, 22, 21, 20, 19, 18, 17, 16, 15, 14, 13, 12, 11, 10, 9, 8, 7, 6, 5, 4, or 3 days; 60, 48, 36, 30, 24, 18, 15, 12, 10, 9, 8, 7, 6, 5, 4, 3, or 2 hours prior to the onset of the genetic disease or any symptom thereof; or 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 15, 18, 24, 30, 36, 48, 60 hours; 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 45, 60, 90 or more days; 4, 5, 6, 7, 8, 9, 10, 11, 12 or more months; 1, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 45, or 60 years after the onset of the genetic disease or any symptom thereof.

(124) Dosing frequency for the composition of any preceding aspects, includes, but is not limited

to, at least once every year, once every two years, once every three years, once every four years, once every five years, once every six years, once every seven years, once every eight years, once every nine years, once every ten years, at least once every two months, once every three months, once every four months, once every five months, once every six months, once every seven months, once every eight months, once every nine months, once every ten months, once every eleven months, at least once every month, once every three weeks, once every two weeks, once a week, twice a week, three times a week, four times a week, five times a week, six times a week, daily, two times per day, three times per day, four times per day, five times per day, six times per day, eight times per day, nine times per day, ten times per day, eleven times per day, twelve times per day, once every 12 hours, once every 10 hours, once every 8 hours, once every 6 hours, once every 5 hours, once every 4 hours, once every 3 hours, once every 2 hours, once every hour, once every 40 min, once every 30 min, once every 20 min, once every 10 min, once every 5 min, or once per min. Administration can also be continuous and adjusted to maintaining a level of the compound within any desired and specified range.

(125) The compositions of the present invention can be administered to the appropriate subject in any manner known in the art, e.g., orally, intramuscularly, intravenously, sublingual mucosal, intraarterially, intrathecally, intradermally, intraperitoneally, intranasally, intrapulmonarily, intraocularly, intravaginally, intrarectally, subcutaneously, or through by inhalation. Parenteral administration, if used, is generally characterized by injection. Injectables can be prepared in conventional forms, either as liquid solutions or suspensions, solid forms suitable for solution or suspension in liquid prior to injection, or as emulsions.

EXAMPLES

(126) The following examples are set forth below to illustrate the compounds, systems, methods, and results according to the disclosed subject matter. These examples are not intended to be inclusive of all aspects of the subject matter disclosed herein, but rather to illustrate representative methods and results. These examples are not intended to exclude equivalents and variations of the present invention which are apparent to one skilled in the art.

Example 1. In Vitro Reporter Assay Demonstrates the Feasibility to Correct the mdx.SUP.4cv .Mutation Using ABE-NG

(127) The mdx.sup.4cv mouse carries a premature stop codon (CAA-to-TAA) in the exon 53 of Dmd gene, which disrupts the expression of dystrophin and leads to the development of muscular dystrophy. Targeting the noncoding strand with ABEs can correct this nonsense mutation. However, in the noncoding strand, there is a lack of 5'-NGG sequence at the downstream of this mutation within the suitable editing window, but a 5'-TGT PAM is present with the mutated A located at position 4 in the guide RNA (gRNA) (FIG. 1a), making it feasible to correct the stop codon with the NG-targeting base editors in this widely used mouse model of DMD. A reporter plasmid with the targeting sequence from the mdx.sup.4cv mice was first constructed (FIG. 1b). The nonsense mutation in the mdx.sup.4cv targeting sequence disrupts the expression of downstream EGFP and successful editing of the nonsense mutation is indicated by the restoration of EGFP expression. As shown in FIG. 1c, transfection with the reporter alone resulted in minimal background fluorescence. Similarly, co-transfection with the reporter, mdx.sup.4cv-gRNA and ABEmax failed to restore EGFP expression. However, ABE-NG (based on SpCas9-NG) successfully restored EGFP expression in this reporter assay. In contrast, ABE-x (based on xCas9-3.7) was found to be less efficient in restoring EGFP expression even though xCas9-3.7 was also engineered to target 5'-NG PAM, consistent with previous reports that xCas9-3.7 is generally less efficient than SpCas9-NG. FACS analysis showed that ABE-x and ABE-NG restored EGFP expression in 10% and 20% cells, respectively (FIGS. 1d, 1e). These in vitro studies showed that ABE-NG corrects the nonsense mdx.sup.4cv mutation.

Example 2. Improvements in the Editing Efficiency and Specificity of ABE-NG

(128) The relative low efficiency of ABE-NG, together with the recently reported off-target RNA

editing activity, prompted the re-design of ABE-NG in order to improve the editing efficiency and specificity. First, the targeting efficiency of ABE-NG at the sites with 5'-NG PAM can be improved by optimizing the PAM-interacting domain. The targeting property of ABE-NG can be modified by combining the mutations in SpCas9-NG (R1335V/L1111R/D1135V/G1218R/E1219F/A1322R/T1337R) with other mutations designed to target different PAM sequences such as those in xCas9(3.7) (A262T/R324L/S409I/E480K/E543D/M694I/E1219V), VQR (D1135V/R1335Q/T1337R), VRER (D1135V/G1218R/R1335E/T1337R) and the loop sequence in ScCas9 (amino acids 367-376). Seven new ABE variants were generated with different combinations of the aforementioned variants (Table 1) and compared their base editing activities at six different loci with those of ABE-NG and ABEmaxSC. While all variants except ABE-NGC (containing all NG mutations plus R1335E) performed similarly at the NGG site (FIG. 2a), it was observed that ABE-NGA (carrying all NG mutations plus R1335Q) had a small improvement in editing the NGH sites as compared to ABE-NG (FIGS. 2b-2e). ABE-NGA and ABE-NGX-NGC (carrying the xCas9(3.7) mutations, NG mutations and R1335E) worked equally well at the NGC site (FIG. 2c). The ABE-NG and ABE-NGA also edited the site containing a 5'-GAT PAM with high efficiency (FIG. 2f), with similar efficiency as compared to the ABE-NGX variant carrying both the xCas9(3.7) mutations and ABE-NG mutations, which was previously reported to have the broadest targeting scope active in plant. However, the efficiency of ABE-NGX at the NGH sites was lower than ABE-NGA (FIG. 2b). Since ABE-NGA is generally superior to other variants tested at NGH sites, ABE-NGA was chosen for further in vitro and in vivo studies.

(129) Previous studies showed that the deaminase domain in the ABEs can elicit transcriptome-wide RNA off-target editing activity, and that the off-target RNA editing activity can be substantially reduced by removing the WT ecTadA domain and mutating the evolved ecTadA domain. The dimeric adenine deaminase domain (ecTadA-ecTadA*) in ABE-NG was replaced with the originally evolved ecTadA* monomer or its high-fidelity version (ecTadA*-V82G) (FIG. 3a) in order to minimize the off-target RNA editing activity. The miniABE-NG (the monomeric TadA* fused with SpCas9-NG nickase) performed slightly better at the mdx.sup.4cv target site as compared to ABE7.10-NG (FIG. 3b). However, the on-target DNA editing activity of miniABE(V82G)-NG was remarkably reduced by over 50% when compared to ABE7.10-NG (FIG. 3b). Then the next experiment attempted to improve the on-target DNA editing efficiency of the high-fidelity mini ABE(V82G)-NG without compromising its low off-target RNA editing activity. The V82G is one of the 26 amino acid residue positions in ecTadA that reside near the enzymatic pocket around the substrate tRNA, inferred from the *S. aureus* TadA-tRNA co-crystal structure. The V82G mutation does not only affect the non-specific affinity to RNA substrates, but can also reduce its affinity to the DNA substrates. It was noticed that the A56G mutation (which is also inferred to lie near the enzymatic pocket around the tRNA substrate) had higher on-target DNA editing activity without affecting the off-target RNA editing activity as compared to miniABEmax. Installing the A56G mutation into miniABE(V82G)-NG can improve its on-target DNA editing activity without compromising its off-target RNA editing profile. Indeed, it was observed that adding the A56G mutation into miniABE(V82G)-NG (named miniABE(GG)-NG) completely restored its on-target DNA editing activity (FIG. 3b).

(130) RNA-seq was used to compare the transcriptome-wide off-target RNA editing activities of miniABE(GG)-NG to other ABE variants in mouse Neuro2a cells. These studies were performed in triplicate. Edited RNA adenines were identified from RNA-seq experiments by filtering out background editing observed with read-count-matched controls. MiniABE-NG and miniABE(A56G)-NG induced much higher numbers of adenine editing as compared to miniABE(V82G)-NG or miniABE(GG)-NG (FIG. 3c). There was no statistical difference between miniABE(V82G)-NG and miniABE(GG)-NG. To further verify the off-target RNA editing activity of mini ABE(GG)-NG, four RT-PCR amplicons were amplified and sequenced, which were shown

to be highly modified by ABEmax in human cells. Transfection of HEK293 cells with mini ABE-NG induced high levels of A-to-I RNA editing in all these transcripts (FIG. 3d); however, such A-to-I RNA editing was essentially eliminated in cells transfected with miniABE(GG)-NG (FIG. 3d). Taken together, these results showed that miniABE(GG)-NG does not only have increased DNA editing activity, but also inherits the high fidelity of miniABE(V82G)-NG. Hereafter, the improved ABE-NG carrying the miniABE(GG) domain and Cas9-NGA nickase was referred to as iABE-NGA.

(131) A new generation of ABEs were developed through directed evolution, namely, ABE8s (such as ABE8.17 and ABE8.20) and ABE8e. To directly compare miniABE(GG) with ABE8.17, ABE8.20 and ABE8e, each of them was fused with SpCas9-NG and tested their activities for editing the mdx.sup.4cv target site using the reporter assay in Neuro2A cells. All these editors showed above 60% editing efficiency with the ABE8e-NG exhibiting the highest activity (FIG. 4a). ABE8e can have also increased bystander activity than miniABE(GG). Since the mdx.sup.4cv target site has no extra adenine within the editing window, the bystander editing activity of ABE8e and miniABE(GG) was compared by testing their performance to edit a nonsense mutation in human DYSF gene (encoding dysferlin) that causes limb girdle muscular dystrophy. The target A.sup.6 was edited with ~58% and 88% efficiency by iABE-NG and ABE8e-NG, respectively (FIG. 4b). The two bystander adenines at positions 8 and 11 were also edited at substantially higher (quantitative numbers) rates by ABE8e-NG than miniABE(GG)-NG. Even the ABE8e with V106W mutation still displayed very high bystander editing activity at position 8. Thus, for in vivo applications that require not only efficiency but also precision, the use of miniABE(GG) was recommended.

Example 3. Intein-Split Allows Efficient Assembly of Full-Length ABE-NG and Editing

(132) The large size of the ABE-NG and other base editors poses a major challenge for viral packaging and in vivo delivery. A dual trans-splicing adeno-associated virus (AAV) approach was used to deliver ABE and a dual protein trans-splicing (PTS) approach using the split-intein moiety from *Nostoc punctiforme* (Npu) was used to deliver CBE. The PTS approach was adopted to deliver ABE. The ABE was split between the ecTad-ecTadA* and the Cas9 nickase with Npu intein moieties, and this split renders low editing efficiency (FIG. 5). To improve the editing efficiency of the split ABE, the amino acid position 573 and 574 of the Cas9 nickase were chose as the splitting site because previous studies showed that 573/574 split Cas9 exhibited near the full-length Cas9 activity. Moreover, split at this site would produce a roughly equal size of the two halves for AAV packaging (FIG. 4a). the split ABE can be further improved by using inteins with fast rate of PTS. Two inteins were selected with the remarkably fast rate of PTS: Cfa ($t_{1/2}=20$ s at 30° C.) and Gp41-1 ($t_{sub.1/2}=5$ s at 37° C.), which are ~2.5-fold and ~10-fold faster than the rate reported for the Npu DnaE intein ($t_{sub.1/2}=50$ s at 37° C.), respectively. Transfection of both split versions into HEK293 cells resulted in robust expression of full-length ABEs as detected by the anti-Cas9 antibody (FIG. 6b), although the expression level was generally lower than the ABEmax but higher than the original ABE7.10. Co-transfection with the split ABH-NG, mdx.sup.4cv-gRNA and the mdx.sup.4cv reporter restored EGFP expression to a similar level as the full-length ABE-NG (FIG. 6c). There was no significant difference between the Cfa and Gp41-1 intein splits (FIG. 6d). The Gp41-1 version was chosen for further studies.

(133) It was shown that the Npu intein split of ABE worked well in vivo and in vivo. The Gp41-1 split and Npu split was directly compared. While both the Gp41-1 split and Npu split allowed the assembly of full-length iABE-NGA, Western blotting analysis showed that the Gp41-1 split rendered significantly more full-length iABE-NGA protein as compared to the Npu split (FIGS. 6e, 6f). The assembly efficiency (as measured by the percentage of the full-length band) of the Gp41-1 split was about 70%, while the Npu split resulted in only 21% (FIG. 6g). To further compare the editing efficiency of the Gp41-1 split and Npu split, the T-to-C conversion of the mdx.sup.4cv stop codon was quantified in Neuro2A cells using the reporter assay. As compared to the full-

length iABE-NGA, the Gp41-1 split and Npu split retained about 85.5% and 78.8% of its activity, respectively (FIG. 6h). The difference in the editing efficiency between the Gp41-1 split and Npu split was small but statistically significant ($p=0.009$) (FIG. 6h). In the Gp41-1 split and Npu split shown herein, each half carries a U6-gRNA expression cassette, while only the C-terminal half of the Npu split reported by the Liu group carries the U6-gRNA expression cassette. To test if the double U6-gRNA cassette has higher editing activity than a single U6-gRNA cassette, the U6-gRNA cassette was removed from the N-terminal construct of the Npu split used herein, which resulted in greatly reduced editing (FIG. 6h), indicating that the gRNA dosage is a limiting factor for ancient Cas9-mediated editing. Moreover, to test the specificity of intein-mediated assembly of iABE-NGA, the N and C-terminal fragments of the Gp41-1 and Npu splits were swapped, and observed no editing (FIG. 6h), indicating that the intein-mediated protein splicing and assembly of full-length iABE-NGA are required for efficient editing.

Example 4. Systematic Delivery of AAV9-iNG Leads to Widespread Dystrophin Restoration
(134) The two Gp41 intein split halves of the iABE-NGA were packaged into AAV9 (hereafter referred to as AAV9-iNG) and tested if in vivo delivery of iABE-NG-A could correct the mutation in mdx.sup.4cv mice. A truncated MHCK7 promoter was used to drive the expression of two halves of iABE-NGA. A preliminary testing of two dosages (a total of 5×10^{13} or 1×10^{14} vg/kg, 1:1 of the N and C-terminal half) showed that the higher dose appeared to increase the dystrophin-positive myocytes in the mdx.sup.4cv mouse heart (FIG. 7). The higher dose (1×10^{14} vg/kg, 1:1 of the N and C-terminal half) was chosen for the rest of the study. In addition, the study also showed that injection of AAV9-iNG carrying a non-targeting gRNA failed to induce dystrophin rescue (FIG. 7).

(135) A cohort of nine mdx.sup.4cv mice were treated with AAV9-iNG (a total of 1×10^{14} vg/kg, 1:1 of the N and C-terminal half) through a single tail vein injection at 5 weeks of age. A subset of the mice was sacrificed at 5 weeks after AAV9-iNG administration. Dystrophin was found to be widely rescued in mdx.sup.4cv heart (FIG. 8a and FIGS. 9-15). Quantification of the entire heart sections showed that $41.9 \pm 10.5\%$ cardiomyocytes of mdx.sup.4cv mice became dystrophin positive at 10 weeks of age after systematic AAV9-iNG treatment ($N=5$) while the control mdx.sup.4cv hearts were essentially dystrophin negative ($0.03 \pm 0.02\%$, $N=4$) (FIG. 8b). Dystrophin was also rescued in skeletal muscles (diaphragm and gastrocnemius) of mdx.sup.4cv mice treated with AAV9-iNG, albeit the recovery was less efficient as compared to that in the heart (FIGS. 8c, 8d, and FIG. 16). Western blot analysis showed that dystrophin was rescued in mdx.sup.4cv mouse heart to $45.9 \pm 11.7\%$ of the WT level following systemic AAV9-iNG treatment (FIGS. 8e, 8f). Consistent with the immunofluorescence data, Western blot showed dystrophin was restored to about $8.0 \pm 2.6\%$ of the WT level in the gastrocnemius muscle of mdx.sup.4cv mice (FIG. 17).

(136) A group of mdx.sup.4cv mice treated with intravenous administration of AAV9-iNG at 5 weeks of age were kept for 10 months to study the long-term impact of systemic ABE editing therapy. A near complete dystrophin restoration was observed in the hearts of all four treated mdx.sup.4cv mice at 10 months of age (FIGS. 8a, 8b and FIGS. 18-23). Dystrophin was also rescued in the skeletal muscles of these older animals with a similar percentage of dystrophin-positive muscle fibers as analyzed at the 10 weeks of age (FIGS. 5c, 5d and FIG. 24). Western blot analysis showed near WT levels of dystrophin expression in the hearts of the 10-month-old mdx.sup.4cv mice treated with AAV9-iNG (FIGS. 8g, 8h).

(137) The heart and muscle tissues contain many different types of cells, which makes it challenging to precisely determine the DNA editing efficiency in myocytes. To estimate the editing efficiency of the Dmd gene, the total RNA was extracted from the heart tissues treated with or without AAV9-iNG, amplified the target region by RT-PCR, and analyzed the resulting amplicons by Sanger sequencing and BEAT program. The AAV9-iNG treated mdx.sup.4cv hearts showed an average $32.6 \pm 2.0\%$ T-to-C editing at 10 weeks of age (FIGS. 8i, 8j) and $84.6 \pm 2.6\%$ at 10 months of

age (FIGS. 8i, 8k).

(138) Repeated cycles of muscle degeneration and regeneration in muscular dystrophy result in muscle fibrosis. To examine if systemic AAV9-iNG delivery can improve the histopathology of mdx.sup.4cv mice, Trichrome staining was performed in 10-month-old mice. As compared to WT mice, the mdx.sup.4cv mice showed significantly elevated fibrosis in both diaphragm and gastrocnemius muscles and the fibrotic areas in these muscles were significantly reduced in the mdx.sup.4cv mice treated with AAV9-iNG (FIGS. 25a-25c). Consistent with previous studies that the mdx mice do not develop overt cardiomyopathy before one year old, there were no significant changes in cardiac fibrosis in mdx.sup.4cv mice with or without AAV9-iNG treatment at 10 months of age as compared to the WT controls (FIG. 25d). AAV9-iNG treatment also significantly reduced the percentage of centrally nucleated fibers (CNF) in both diaphragm and gastrocnemius muscles at 10 weeks of age (FIGS. 25e, 25g). By 10 months of age, the effects of AAV9-iNG treatments on CNF were blunted (FIGS. 25f, 25h). Although a significant difference in cross-sectional area (CSA) of muscle fibers was not observed following AAV9-iNG treatment (FIGS. 25i, 25j, and FIGS. 26a, 26b), the AAV9-iNG treatment appeared to shift the fiber size distribution towards those of the WT muscles (FIGS. 25k, 25l, and FIGS. 26c, 26d), particularly in gastrocnemius muscles at 10 weeks of age.

(139) To test if systemic AAV9-iNG treatment can improve the muscle function, the muscle contractility was measured using an in vivo muscle test system. Maximum plantarflexion tetanic torque was measured during supramaximal electric stimulation of the tibial nerve at 150 Hz. While the mdx.sup.4cv mice produced significantly reduced torque as compared to the WT controls, systemic delivery of AAV9-iNG significantly increased the tetanic torque in mdx.sup.4cv mice (FIG. 25m).

Example 5. The Safety Profile and Off-Target Activity of AAV9-iNG Treatment

(140) Previous studies showed that AAV-mediated delivery of CRISPR/Cas9 into neonatal mice resulted in humoral immune responses to AAV capsid but not Cas9. In contrast, AAV-mediated delivery of CRISPR/Cas9 into adult mice evoked robust anti-Cas9 immunity. Serum samples were collected to analyze the host immune responses to the AAV9 capsid and the base editor iABE-NGA. Intramuscular injection of AAV9-iNG into 5-6 weeks old mdx.sup.4cv mice produced robust anti-AAV9 capsid (FIG. 27a) and anti-Cas9 antibodies (FIG. 27b) at 2 weeks after injection. The anti-AAV9 titers were similar at different time points from 2 to 7 weeks post intramuscular injection and from 7 to 9 weeks post intravenous injection (FIG. 27a). The anti-Cas9 antibody titers showed a large variation among mice at 2 weeks after intramuscular injection, but all increased to peak by 4 weeks (FIG. 27b).

(141) The liver toxicity of AAV9-iNG treatment was examined by measuring serum aspartate aminotransferase (AST) and alanine aminotransferase (ALT), and kidney toxicity by measuring blood urine nitrogen (BUN). As compared to WT mice, the mdx.sup.4cv mice showed elevated AST (FIG. 27c) and ALT (FIG. 27d). However, treatment of mdx.sup.4cv mice with AAV9-iNG did not further increase the serum levels of AST and ALT at either 8 weeks or 10 months of age. Measurement of BUN did not find significant changes in the treated or untreated mdx.sup.4cv mice (FIG. 27e).

(142) One concern with ABE-mediated gene correction is the off-target activities such as gRNA mismatch tolerance, bystander editing, and off-target RNA editing. Previous studies showed that ABE can tolerate 1-2 mismatches between the gRNA and its target sites. Prediction by Cas-OFFinder showed that one site on chromosome 16 (Chr16_OT) has only one mismatch, two other sites have two mismatches and 55 sites have three mismatches (FIG. 27f). The Chr16_OT differs from the mdx.sup.4cv target sequence by only one C at position 12. Neuro-2a cells were transfected with ABE-NG or iABE-NG plus the gRNA, amplified the Chr16_OT by PCR and subjected the amplicon to next generation sequencing (NGS). As shown in FIG. 27g, no significant editing of the A4 in either ABE-NG or iABE-NG transfected cells was observed. Similarly, the off-

target site on chromosome 1 (Chr1_OT) was analyzed, which differs from the mdx.sup.4cv target sequence by an A at position 2 and a G at position 20. Again, it was found that ABE-NG or iABE-NG did not edit the A4 at Chr1_OT (FIG. 27h).

(143) Next, the bystander editing at the on-target mdx.sup.4cv locus was analyzed in the mice treated with AAV9-iNG. Since the 10-month treated mouse hearts showed a high level of dystrophin rescue, the on-target editing efficiency was first determined in these mouse hearts by NGS. As mouse hearts contain multiple different cell types, analysis of the genomic DNA PCR products can significantly underestimate the editing efficiency. To verify this, NGS of the genomic DNA PCR products was performed from two mouse hearts receiving AAV9-iNG and exhibiting high dystrophin rescue, and an up to 11% edits at A4 was detected. Thus, the RT-PCR products were sequenced to estimate the editing efficiency at the on-target mdx.sup.4cv locus. The A at position 4 (corresponding to the T within the premature stop codon in the coding strand) was converted to G with high efficiency from all four mouse hearts (FIG. 28). On average, $86.2 \pm 2.4\%$ A-to-G conversion was measured (FIG. 27i). At the mdx.sup.4cv target site, there was only one A within the editing window of 4-8, disallowing us to analyze the bystander A-to-G editing at this particular site. Another type of undesired ABE-mediated genome edits at an on-target locus is ABE-dependent cytosine-to-uracil conversion resulting in C.Math.G to T.Math.A mutation at that site. It was found that C6 at the mdx.sup.4cv target site was edited above background with an average efficiency of $1.6 \pm 0.1\%$ (FIG. 27j).

(144) Finally, RNA-seq was performed to characterize the transcriptome-wide RNA off-target RNA editing induced by AAV9-iNG in the mdx.sup.4cv mouse heart samples. After filtering the confident variants from control mdx.sup.4cv heart samples, a few hundred RNA editing events were found in the three AAV9-iNG treated heart samples with only 32 shared by all of them (FIG. 29).

(145) The present study has improved the split ABE-NG for AAV-mediated in vivo delivery by engineering a new NG PAM-interacting domain variant, a new adenine deaminase domain with higher on-target DNA editing efficiency without compromising the high fidelity of ABE-V82G, and a Gp41-1 intein split that mediates higher efficiency of protein splicing and editing. Together, these improvements allowed to achieve widespread dystrophin rescue and functional improvement in dystrophic mice. The editing efficiency in the heart was extraordinarily high in mdx.sup.4cv mice following systemic delivery of AAV9-iNG and over 90% of cardiomyocytes were corrected to express dystrophin in mdx.sup.4cv hearts at 10 months of age after a single intravenous administration of AAV9-iNG at 5 weeks old. There was no obvious toxicity detected following AAV9-iNG treatment, despite the host immune response to the AAV9 capsid and ABE. This has tremendous implication for base correction of genetic cardiomyopathies.

(146) It was shown herein that the ecTadA* domain with the V82G mutation had significantly reduced on-target DNA editing activity as compared to the original ecTadA*. By adding the A56G mutation into the V82G variant of ecTadA*, the on-target DNA editing activity was dramatically improved without compromising the high fidelity of the V82G variant in terms of its low off-target RNA editing activity. Although the A56G_V82G variant was less efficient than the recently reported ABE8e, ABE8e had also significantly higher bystander editing activity than the A56G_V82G variant. It is essential to consider both the on-target DNA editing efficiency and the bystander DNA editing as well as off-target RNA editing activity for in vivo applications. The editors have high on-target DNA editing but induce minimal bystander DNA editing and off-target RNA editing events. The A56G_V82G variant offers a balance between the editing efficiency and the editing precision.

(147) The iABE-NG and iABE-NGA can be broadly applied to correct DMD mutations and many other disease-causing mutations. Analysis of the ClinVar database showed that over 100 of the 174 total G>A or T>C point mutations for DMD can be targeted for repair by at least one of the ABEs (iABE-NGA). Describe all other cases including targeting the splicing sites for exon skipping and

other diseases.

(148) The recent advances in engineering Cas9 variants with non-G PAM further increases targeting capacity. Moreover, the ABE editing can be designed to induce skipping of mutant exons via targeting the canonical splicing donor or acceptor, thus further broadening the applicability of ABE editing therapy for a larger population of DMD.

(149) The mice at ten months after AAV9-iNG delivery showed significantly higher dystrophin rescue than the mice at 10 weeks after the treatment. One explanation is that the DMD cardiomyocytes with restored dystrophin expression can gain advantage for selective survival and regeneration during the development stages after delivery of AAV9-iNG. Additionally, transduced cardiomyocyte-derived extracellular vesicles can deliver genetic materials such as transcripts encoding iABE-NG into proximal un-transduced cardiomyocytes and confer base editing in those cells.

(150) This study has also shown that systemic delivery of AAV9-iNG resulted in dystrophin restoration in skeletal muscles and functional improvement. As compared to cardiomyocytes, the editing efficiency in skeletal muscles was substantially lower. This can be attributed to the observation that AAV9 has higher tropism towards cardiomyocytes than skeletal muscles. However, other mechanisms can also be responsible for the lower editing efficiency in skeletal muscles. For example, the dystrophic and inflammatory microenvironment in skeletal muscles can pose further constraints on AAV9 delivery and base editing. In addition, targeting muscle satellite cells can be required to improve the overall editing outcomes in skeletal muscle as they are constantly activated to replace injured skeletal muscle in DMD. Although AAV9 has been shown to transduce muscle satellite cells, the efficiency is relatively low. Moreover, the use of a muscle-specific promoter can further reduce the base editing in muscle satellite cells in the present study.

(151) Improvements in these examples show exceptionally high editing efficiency in comparison to other approaches. First, the intein used in the current study (Gp41-1) has superfast kinetics, which allows more efficient assembly of full-length ABE (FIG. 6). Second, each half of the AAVs carries a gRNA-expressing cassette in the present study, while in the other study, gRNA is present in only the C-terminal half of the Npu intein split constructs. The data (FIG. 6h) showed that the gRNA dosage affects the editing efficiency. Third, the promoters used in these studies were also different, which can drive different expression levels of ABE in heart tissues. Finally, the intrinsic difference in the gRNAs and ABE variants can have impacts on the overall editing outcomes. Thus, the exceptionally high editing efficiency achieved in adult dystrophic mice indicates that the AAV9-iNG vectors disclosed herein are useful for clinical applications.

Example 6. Methods

(152) Mice. Mice (C57BL/6J and B6Ros.Cg-Dmd.sup.mdx-4Cv/J) were purchased from the Jackson Laboratory and maintained at The Ohio State University Laboratory Animal Resources in accordance with animal use guidelines. All the experimental procedures were approved by the Animal Care, Use, and Review Committee of the Ohio State University.

(153) Plasmid construction. The pCMV-ABE7.10, pCMV-ABE-xCas9(3.7) and pCMV-ABEmax were obtained from Addgene. The NG mutations were introduced by fusion PCR of pCMV-ABEmax and subcloned into pCMV-ABEmax to make pCMV-ABEmaxNG. The A56G and V82G mutations were introduced into TadA* domain by fusion PCR and cloned into pCMV-ABEmaxNG to generate pCMV-iABEmaxNG. The CfaN minigene was synthesized by IDTdna and fused at the amino acid 573 of SpCas9-max through PCR amplification. The TadA-TadA*-SpCas9max(2-573)-CfaN fragment was PCR amplified and subcloned into pAAV under the control of meCMV promoter to generate pAAV-ABEmaxN-temp. The hU6 promoter with mdx.sup.4cv-targeting gRNA was PCR amplified and cloned into pAAV-ABEmaxN-temp to make pAAV-ABEmaxN. The CfaC fused with SpCas9max(574-end) was generated by PCR and cloned into pAAV-ABEmaxN-temp to make pAAV-ABEmaxC. Similarly pAAV-ABEmaxN2 and pAAV-ABEmaxC2NG with the Gp41-1 intein, and pAAV-ABEmaxN3 and pAAV-ABEmaxC3NG with the Npu intein were

constructed. The mdx.sup.4cv gRNA and other gRNA oligos (listed in Table 2) were annealed and ligated into pLenti-ogRNA. The mdx.sup.4cv reporter oligos were annealed and ligated into pLKO-puro-2A-mdx.sup.4cv-EGFP. All plasmids used in this study are listed in Table 3.

(154) Generation of AAV particles. AAV vectors were produced at the viral vector core of the Nationwide Children's Hospital as previously described. The Gp41-1 intein split of iABE-NGA and the gRNA targeting mdx4cv mutation (GTTaTCTCCTGTTCTGCAGC TGT (SEQ ID NO: 621); note: the underlined PAM sequences were not included in the gRNA) or a non-targeting gRNA (GTTTaTGTCACCAGAGTAAC (SEQ ID NO: 571), the different nucleotides are highlighted in blue) expression cassettes were packaged into AAV9 capsid using the standard triple transfection protocol. A quantitative PCR-based titration method was used to determine an encapsulated vector genome titer utilizing a Prism 7500 Fast Taqman detector system (PE Applied Biosystems Grand Island, NY USA). The following primers/probes were used: 5'-GGATTTC CAAGTCTCCACCC-3' (SEQ ID NO: 630) and 5'-TCCCACCGTACACGCCTAC-3' (SEQ ID NO: 631) for titrating AAV9-NG, and AAV9-iNG was titered using digital droplet PCR. Titers are expressed as DNase resistant particles per ml (DRP/ml) and rAAV titers used for injection in mice were 8.9×10^{12} DRP/ml (AAV9-NG) and 3.0×10^{13} DRP/ml (AAV9-iNG).

(155) Cell culture and transfection. HEK293 cells were cultured in Dulbecco's modified eagle's medium (DMEM) (Corning, Manassas, VA) containing 10% fetal bovine serum (FBS) and 1% 100× penicillin-streptomycin (10,000 U/ml, invitrogen). Cells were plated in 6-well plates and transfected with the 2 µg plasmids (0.5 µg reporter, 0.75 µg gRNA and 0.75 µg ABE) per well unless specified otherwise by polyethylenimine (PEI) as previously described.

(156) Flow cytometry. At 72 hour post transfection, HEK293 cells transfected with ABE plasmids were collected from 6-well plate and analyzed on Becton Dickinson LSR II (BD Biosciences) to determine GFP-positive cells. A total of 100,000 cell events were collected and data analysis was performed using the FlowJo software (Tree Star, Ashland, OR, USA).

(157) Intramuscular and intravenous administration of AAV/9 particles. AAV9-iNG viral particles (2×10^{11} vg, 25 µl) were injected into the right gastrocnemius compartment of the male mdx.sup.4cv mice at 5-6 weeks of age or day 3. For systematic delivery, the male mdx.sup.4cv mice at 5-6 weeks of age were administered with AAV9-NG, AAV9-iNG or AAV9-GFP viral particles (1×10^{14} vg/kg) via tail vein injection.

(158) Serological analysis. Blood samples were collected at various time points after intramuscular or intravenous injection. The blood samples were allowed to clot for 15 min to 30 min and centrifuged at 5000 rpm for 10 min in room temperature. The supernatant was collected as serum and stored at -80° C. for the biochemical assays. Measurement of ALT (BioVision Incorporated), AST (BioVision Incorporated), BUN (Arbor Assays, Michigan, USA) and cardiac Troponin I (Life Diagnostics, Inc) were performed according to the manufacturer's protocols.

(159) Antibody ELISA. Antibodies against AAV9 and SpCas9 were detected by adapting previously published protocols. In brief, recombinant AAV9 (2×10^9 vg/well) and SpCas9 protein (0.27 µg/well) were diluted in 1× Coating Buffer A (BioLegend) and used to coat a 96-well Nunc MaxiSorp plate. Proteins were incubated overnight at 4° C. to adsorb to the plate. Plates were washed four times 5 min each with PBS plus 0.05% Tween-20 and then blocked with 1× Assay Diluent A (BioLegend) for 1 h at room temperature. The anti-AAV2 (A20, cat. # 03-65155, American Research Products, Inc) and anti-SpCas9 antibody (Diagenode C15310258) was used as positive control for detection of anti-AAV9 and anti-SpCas9 antibodies, respectively. Serum samples were added in 1:50 dilution and plates were incubated for 2 h at room temperature with shaking. Plates were washed four times 5 min each and 100 µl of blocking solution containing goat anti-mouse IgG (Sigma 1:3,000) was added to each well and incubated at 1 h at room temperature. Plates were washed four times 5 min each, 100 µl of freshly mixed TMB Substrate Solution (BioLegend) was added to each well, and incubated in the dark for 20 min. The reaction was stopped by adding 100 µl 2N H.sub.2SO.sub.4 Stop Solution. Optical density at 450 nm was

measured with a plate reader.

(160) Muscle contractility measurements. At 5 weeks after intramuscular AAV9-NG or intravenous AAV9-iNG injection, muscle contractility was measured using an in vivo muscle test system (AuroraScientific Inc). Mice were anesthetized with 3% (w/v) isoflurane and anesthesia was maintained by 1.5% isoflurane (w/v) during muscle contractility measurement. Maximum plantarflexion tetanic torque was measured during a train of supramaximal electric stimulations of the tibial nerve (pulse frequency 150 Hz, pulse duration 0.2 ms).

(161) Histopathological assessment of tissues. Mice were sacrificed at various time points, and tissues (heart, lung, diaphragm, spleen, kidney, liver, quadriceps and gastrocnemius) were harvested for histological, histochemical, biochemical and molecular analyses. For immunohistological examinations, tissues were embedded in optimal cutting temperature (OCT, Sakura Finetek, Netherlands) compound and snap-frozen in cold isopentane for cryosectioning. The tissues were stored at -80°C . and processed for biochemical analysis and histology assessment. Frozen cryosections (7 μm) were fixed with 4% paraformaldehyde for 15 minutes at room temperature. After washing with PBS, the slides were blocked with 3% BSA for 1 hour. The slides were incubated with primary antibodies against dystrophin (ab15277, 1:100, Abcam) and laminin- $\alpha 2$ (ALX-804-190-C100, 1:100, Enzo) at 4°C . for 1 hour. After that, the slides were washed extensively with PBS and incubated with secondary antibodies (Alexa Fluor 488 goat anti-rat IgG, Invitrogen, Carlsbad, CA or Alexa Fluor 568 donkey anti-rabbit IgG, Invitrogen) for 1 hour at room temperature. The slides were sealed with VECTASHIELD Antifade Mounting Medium with DAPI (Vector Laboratory, Burlingame, CA). All images were taken under a Nikon Ti-E fluorescence microscope (magnification 200 $\times$) (Nikon, Melville, NY). Laminin- $\alpha 2$ -positive and dystrophin-positive muscle fibers were counted using NIS-Elements AR version 4.50 (Nikon, Melville, NY). The amount of dystrophin positive muscle fibers is represented as a percentage of total laminin- $\alpha 2$ -positive muscle fibers.

(162) For trichrome staining, Masson's 2000 Trichrome Kit was used (American MasterTech, Lodi, CA). The muscle and heart sections were fixed with 4% paraformaldehyde for 1 hour at room temperature. After washing with PBS, the tissue sections were stained with Masson's trichrome reagent following the manufacturer's instruction.

(163) Western blot analysis. Mouse tissues from mdx.sup.4cv mice treated with or without AAV9-NG or AAV9-iNG were lysed with cold RIPA buffer supplemented with protease inhibitors and extracted protein samples were separated by SDS-PAGE (BioRad, 4-15%) and transferred onto Nitrocellulose membranes (0.45 μm). The rabbit polyclonal anti-dystrophin (E2660, 1:500, Spring Bioscience, Pleasanton, CA), rabbit polyclonal anti-Cas9 (C15310258-100, 1:1000, Diagenode, Denville, NJ) and rabbit monoclonal anti-Gapdh (#2118, 1:2000, Cell Signaling Technology, Danvers, MA) antibodies were used for immunoblotting analysis. HRP conjugated goat anti-mouse (1:4000) and goat anti-rabbit (1:4000) secondary antibodies were obtained from Cell Signaling Technology, Danvers, MA. The membranes were developed using ECL western blotting substrate (Pierce Biotechnology, Rockford, IL) and scanned by ChemiDoc XRS+ system (BioRad, Hercules, CA). Western blots were quantified using Image Lab 6.0.1 software (Bio-Rad Laboratories, Hercules, CA) according to the manufacturer's instruction.

(164) Extraction of genomic DNA and total RNA, PCR and Sanger sequencing. Genomic DNA from mouse tissues and cultured HEK293 cells were extracted using DNeasy Blood & Tissue Kit (Qiagen, Germantown, MD). Total RNA was extracted from mouse tissues and HEK293 cells using Quick-RNA MiniPrep Kit (ZYMO Research, Irvine, CA). Five μg of treated RNA was used as template for first-strand cDNA synthesis by using RevertAid RT Reverse Transcription Kit (Life Technologies, Carlsbad, CA). Aliquots of the RT product were used for RT-PCR analysis of dystrophin editing. PCR reactions were carried out with 100 ng genomic DNA or cDNA in the GoTaq Master Mix (Promega) according to the manufacturer's instruction. The primers used for RT-PCR of the reporter genes and PCR of endogenous loci were listed in Table 2. The PCR

products were purified using the Wizard SV Gel and PCR Clean-up System (Promega). Purified genomic DNA and RT PCR products (100 ng) were subjected to Sanger sequencing at the Genomics Shared Resource of the Ohio State University Comprehensive Cancer Center. The sequencing data were analyzed by BEAT program.

(165) Targeted deep sequencing. The on-target and off-target loci were first amplified by genomic DNA PCR and/or RT-PCR using gene-specific primers with Illumina adapters (primers are provided in Table 4). The first PCR products were purified using a commercial purification kit (Promega, Madison, WI, USA), diluted, pooled, and subjected to a second round PCR with primers including the index sequences. The final PCR products were electrophoresed on an agarose gel, showing a single sharp peak. The quality and quantity were assayed using an Agilent Bioanalyzer 2100 (Genomics Shared Resource, Ohio State University Comprehensive Cancer Center). The purified amplicons were pooled and sent for sequencing using a MiSeq nano-scale flow cell (paired-end 300 base-pair reads) at The Genomics Services Laboratory of Nationwide Children's Hospital. The FASTQ files were analyzed using CRISPResso2 with default parameters.

(166) RNA-seq experiments. RNA library preparation was performed using NEBNext® Ultra™ II Directional (stranded) RNA Kit for Illumina (NEB #E7760L New England Biolabs) with an initial input of 100 ng ng extracted RNA per sample, measured using Qubit RNA HS reagents (#Q32852 Invitrogen) for fragmentation, cDNA synthesis and amplification. Depletion of ribosomal RNA (rRNA) was carried out with NEBNext rRNA Depletion Kit (human, mouse, rat) from New England Biolabs (#E6310X). NEBNext Multiplex oligos indexes kits (E7335L, E7500L and E7710L) from New England Biolabs were used to barcode each library following the manufacturer protocol. RNA-seq libraries were examined using an Agilent 2100 Bioanalyzer and a High Sensitivity DNA kit (Agilent Technologies, Inc). RNA-seq libraries were sequenced on Novaseq SP Paired-End 150 bp format at The Genomics Services Laboratory of Nationwide Children's Hospital.

(167) RNA sequence variant calling and variant filtering. Illumina paired-end fastq sequencing reads were processed according to GATK Best Practices for RNA-seq variant calling. In brief, reads were aligned to the mouse mm10 reference genome using STAR version 1.5.2 in two-pass mode with the parameters implemented by the ENCODE project. Picard tools (version 2.19.0) was then applied to sort and mark duplicates of the mapped BAM files. The refined BAM files were subject to split reads that spanned splice junctions, local realignment, base recalibration and variant calling with SplitNCigarReads, IndelRealigner, BaseRecalibrator and HaplotypeCaller tools from GATK (version 4.1.2.0), respectively. Known variants in dbSNP version 142 were used during base quality recalibration. From all called variants, downstream analyses focused solely on single-nucleotide variants (SNVs) on canonical (1-22, X, Y and M) chromosomes. To identify variants with high confidence, clusters of at least five SNVs were filtered that were within a window of 35 bases and variants with Fisher strand values >30.0, qual by depth values <2.0 and sequencing depth <10. Base edits labelled as A-to-I comprise A-to-I edits called on the positive strand as well as T-to-C edits sourced from the negative strand, since the RNAs were converted into cDNA before sequencing, both the nucleotide and its complementary base can be sequenced. Results obtained with this pipeline can underestimate the actual number of RNA edits occurring in cells because of the high stringency of the variant calling pipeline and potential under-representation of intronic and intergenic RNA in our experiments.

(168) Any confident variants found in wild-type Neuro2a cells were considered to be SNPs and were filtered out from the base-editor-transfected groups for off-target analysis. Similarly, any confident variants found in control mdx.sup.4cv heart samples were filtered out from the AAV9-iNG group for off-target analysis. The editing rate was calculated as the number of mutated reads divided by the sequencing depth for each site.

(169) ClinVar database analysis. The ClinVar data was converted into a tab-delimited flat file. A python script (clinvar.py) was written to process the tab-delimited flat file of ClinVar data.

(170) Statistical analysis. The data were expressed as mean±S.E.M. and analyzed with GraphPad Prism 8.0.1 software (San Diego, USA). Statistical significance was determined using one-way ANOVA followed by Bonferroni post hoc-tests for multiple groups or student's t-test for two groups. A P value of less than 0.05 is regarded as significant.

(171) Data availability. The sequencing data have been deposited in the NCBI SRA under project accession numbers (PRJNA673243).

Example 7. Use of the System for Treatment of Spinal Muscular Atrophy

(172) The iABE-NGA system can be broadly used to rescue dystrophin expression in DMD and treat other conditions, in addition to correcting point mutations in DMD and dysferlinopathy. For example, by targeting the splice sites, iABE-NGA can mediate exon skipping and rescue dystrophin expression in a human induced pluripotent stem cell-derived cardiomyocytes with a large deletion from exon 48 through 54 in DMD gene (FIG. 30).

(173) Mutations in the survival motor neuron 1 (SMN1) gene cause spinal muscular atrophy (SMA). There is a paralogous gene SMN2 in human, present in almost all SMA patients. The SMN2 is different from SMN1 in exon 7, with position 6 converted from C to T in SMN2, which results in the skipping of exon 7 in SMN2 mRNA and non-functional SMN2 protein. Using iABE-NGA-mediated editing of A36 showed that the exon 7 of SMN2 was spliced in (FIG. 31), and thus this can be a therapy for SMA.

(174) Moreover, by targeting the splice sites, one can generate loss-of-functions in the therapeutic targets such as those involved in the regulation of plasma cholesterol levels. Previous studies identified a number of protective loss-of-functions in genes such as ANGPTL3, APOC3 and ASGR1, which reduce the plasma low cholesterol levels and renders cardioprotection. iABE-NGA can install such loss-of-function mutations in the aforementioned genes, which can be used as a one-time cure for dyslipidemia (FIG. 32).

(175) TABLE-US-00001 TABLE 1 List of ABE variants engineered in this study. Name

Description ABE-NG ABEmax with SpCas9-NG mutations R1335V/

L1111R/D1135V/G1218R/E1219F/A1322R/T1337R ABE-NGA ABE-NG with R1335Q mutation ABE-NGC ABE-NG with R1335E mutation ABE-NG-loop ABE-NG with the loop sequence from ScCas9 (amino acids 367-376) inserted ABE-NGX ABE-NG with A262T/R324L/S409I/E480K/E543D/M694I mutations ABE-NGX-NGA ABE-NGX with R1335Q ABE-NGX-NGC ABE-NGX with R1335E ABE-NGX-loop ABE-NGX with the loop sequence from ScCas9 (amino acids 367-376) inserted ABEmaxSc ABEmax with SpCas9 nickase replaced with ScCas9 nickase ABE-NGm ABE-NG with the dimeric TadA-TadA* replaced with monomeric TadA* containing two additional mutations A56G and V82G iABE-NGA ABE-NGA with the dimeric TadA-TadA* replaced with monomeric TadA* containing two additional mutations A56G and V82G

(176) TABLE-US-00002 TABLE 2 List of gRNA target sequences and primers for PCR in this study. Name Sequence SEQ ID NOs Mdx4cv-gRNA

GTTATCTCCTGTTCTGCAGC SEQ ID NO: 570 NT-gRNA

GTTTATGTCACCAGAGTAAC SEQ ID NO: 571 mDMD-i52-F

GAGGTAATAGAGCCAAGCCCT SEQ ID NO: 572 mDMD-i53-R

GCAAGAATTCCACTTTTCACTTCCT SEQ ID NO: 573 mDMD-E51-F

CTGTCATCTCCAACTAGAAATGC SEQ ID NO: 574 mDMD-E55-R

GCAGCCTCTTGCTCACTTACTC SEQ ID NO: 575 S1-gRNA

GATGACAGGCAGGGGCACCG SEQ ID NO: 576 S1-F TTCCAGTGGTTCAATGGTCA

SEQ ID NO: 577 S1-R CTTTCAACCCGAACGGAGAC SEQ ID NO: 578 VEGFA-

S5-gRNA GAGCGAGCAGCGTCTTCGAG SEQ ID NO: 579 VEGFA-S12-gRNA

GCAGACGGCAGTCACTAGGG SEQ ID NO: 580 VEGFA-S14-gRNA

GGGAAGCTGGGTGAATGGAG SEQ ID NO: 581 VEGFA-F

AGCTGTTTGGGAGGTCAGAA SEQ ID NO: 582 VEGFA-R

AGGGAGCAGGAAAGTGAGGT SEQ ID NO: 583 Site13-gRNA

GTCGCAGGACAGCTTTTCCT SEQ ID NO: 584 Site13-F
TGTAGCTACGCCTGTGATGG SEQ ID NO: 585 Site13-R TGCCCTGAGATCTTTTCCTC
SEQ ID NO: 586 FANCF-gRNA GATCCAGGTGCTGCAGAAGG SEQ ID NO: 587
FANCF-F CTCTTGCCTCCACTGGTTGT SEQ ID NO: 588 FANCF-R
TCGGTAGGATGCCCTACATC SEQ ID NO: 589 Q623X-gRNA
ATCCTACAGCATGGTGGCTG SEQ ID NO: 590 Puro-F
AGTGGTCTCCGGAACCTCCGCGCCCC SEQ ID NO: 591 GCAAC GFP-R
TCCTTGAAGAAGATGGTGCG SEQ ID NO: 592

pCMV_NG-ABE8.17 Expressing ABE8.17-NG; used in FIG. 4 pZC0106 pCMV_NG-ABE8.20 Expressing ABE8.20-NG; used in FIG. 4 pZC0111 pCMV_SpG-ABE8e(V106W) Expressing ABE8e-SpG(V106W); used in FIG. 4 pPW-0007 pLKO-puro-2A-Q623X-GFP Q623X reporter; used in FIG. 4b. pPW-0010 pLenti-Q623X-ogRNA Q623X targeting gRNA; used in FIG. 4b. pXL-0419 pCMV-ABE7.9 (Addgene Expressing ABE7.9; #102918) used in FIG. 8. pXL-0419 pCMV-ABE7.9 (Addgene Expressing ABE7.9; #102918) used in FIG. 8. pYG9010 pBac-rAAV-IntC-SpCas9n Expressing Npu_N-SpCas9 nickase; used in FIG. 8. pYG9011 pBac-rAAV-ABE7.10v3 Expressing GFP-TadA-TadA*-Npu_C; used in FIG. 8.

(178) TABLE-US-00004 TABLE 4 List of primers used for NGS in this study.

Name	Sequence	Mdx4cv-	ACACTCTTTCCCTACACGACGCTCTTCCGATCTGAACTCAT
SEQ ID E52-F	TACTGCTGCCCAGA	NO: 593	Mdx4cc-
GTGACTGGAGTTCAGACGTGTGCTCTTCCGATCGACCTGTT	SEQ ID E53-R	CGGCTTCTTCCTTA	NO: 594
Mdx4cv-	ACACTCTTTCCCTACACGACGCTCTTCCGATCTAAATTTCC	SEQ ID i52-F	ACTGTCTTCTCTTGAGT
NO: 595	Mdx4cv-	GTGACTGGAGTTCAGACGTGTGGTCTTCCGATCGCTTGCCT	SEQ ID i53-R
CTGACCTGTCCTAT	NO: 596	mChr16	ACACTCTTTCCCTACACGACGCTCTTCCGATCTGTGACTAG
SEQ ID OT-F	GGGCAAAGCAAGAT	NO: 597	mChr16
GTGACTGGAGTTCAGACGTGTGCTCTTCCGATCCTTCCAAA	SEQ ID OT-R	CTTTCTGCCCATT	NO: 598
mChr10	ACACTCTTTCCCTACACGACGCTCTTCCGATCTAACACAGC	SEQ ID T-F	GTGCTCTTTCCTTAC
NO: 599	mChr10	GTGACTGGAGTTCAGACGTGTGCTCTTCCGATCGTTCAGAA	SEQ ID T-R
GAACATCCCGTTGAC	NO: 600	NGS-	AATGATACGGCGACCACCGAGATCTACACTCTTTCCCTACA
SEQ ID final-F	CGAC	NO: 601	NGS- CAAGCAGAAGACGGCATAACGAGATCTTGTAGTGACTGGAG
SEQ ID final-R1	TTCAGACGT	NO: 602	NGS- CAAGCAGAAGACGGCATAACGAGATCGTGACTGGA
SEQ ID final-R2	GTTCAGACGT	NO: 603	NGS- CAAGCAGAAGACGGCATAACGAGATCCGTCCGTGACTGGAG
SEQ ID final-R3	TTCAGACGT	NO: 604	NGS- CAAGCAGAAGACGGCATAACGAGATATGTCAGTGACTGGAG
SEQ ID final-R4	TTCAGACGT	NO: 605	NGS- CAAGCAGAAGACGGCATAACGAGAT GTCCGC
SEQ ID final-R5	GTGACTGGAGTTCAGACGT	NO: 606	NGS- CAAGCAGAAGACGGCATAACGAGAT TTAGGC
SEQ ID final-R6	GTGACTGGAGTTCAGACGT	NO: 607	NGS- CAAGCAGAAGACGGCATAACGAGAT
CGATGT	SEQ ID final-R7	GTGACTGGAGTTCAGACGT	NO: 608
NGS- CAAGCAGAAGACGGCATAACGAGAT	TGACCA	SEQ ID final-R8	GTGACTGGAGTTCAGACGT
NO: 609	NGS- CAAGCAGAAGACGGCATAACGAGAT	AGTCAA	SEQ ID final-R9
GTGACTGGAGTTCAGACGT	NO: 610	NGS- CAAGCAGAAGACGGCATAACGAGAT	AGTTCC
SEQ ID final- GTGACTGGAGTTCAGACGT	NO: 611	R10	NGS- CAAGCAGAAGACGGCATAACGAGAT
GATCAG	SEQ ID final- GTGACTGGAGTTCAGACGT	NO: 612	R11
NGS- CAAGCAGAAGACGGCATAACGAGAT	ACAGTG	SEQ ID final- GTGACTGGAGTTCAGACGT	NO: 613
R12	NGS- CAAGCAGAAGACGGCATAACGAGAT	TATACT	SEQ ID final- GTGACTGGAGTTCAGACGT
NO: 614	R13	NGS- CAAGCAGAAGACGGCATAACGAGAT	CAACAA
SEQ ID final- GTGACTGGAGTTCAGACGT	NO: 615	R14	NGS- CAAGCAGAAGACGGCATAACGAGAT
GTTGTT	SEQ ID final- GTGACTGGAGTTCAGACGT	NO: 616	RIS
NGS-			

CAAGCAGAACGGCATACGAGAT TCGGTT SEQ ID final-
GTGACTGGAGTTCAGACGT NO: 617 R16 NGS- CAAGCAGAAGACGGCATAACGAGAT
AGTATT SEQ ID final- GTGACTGGAGTTCAGACGT NO: 618 R17 NGS-
CAAGCAGAAGACGGCATAACGAGAT TCTTGT SEQ ID final-
GTGACTGGAGTTCAGACGT NO: 619 R18
(179) TABLE-US-00005 TABLE 5 re- vari- vari- pair refer- sgRNA_re- chr pos hgvs_c hgvs_p
all_traits ants ant type editor ence_seq pair X 31173588 NM_004020.3:c.2843 +
NP_003997.1:p.Gln3427Ter Dilated G > A intron ABE SEQ ID SEQ ID 5081C > T cardiomy-
variant NO: 48 NO: 222; opathy SEQ ID 3B; Becker NO: 223 muscular dystrophy; Duchenne
muscular dystrophy X 32464674 NM_004006.2:c.3188G > NP_003997.1:p.Trp1063Ter Duchenne
C > T nonsense ABE SEQ ID SEQ ID A muscular NO: 49 NO: 224; dystrophy; SEQ ID
DUCHENNE NO: 225; MUSCULAR SEQ ID DYSTROPHY NO: 226; SEQ ID NO: 227 X
32217037 NM_004006.2:c.6317G > NP_003997.1:p.Trp2106Ter not provided; C > T nonsense
ABE SEQ ID SEQ ID A Not Provided NO: 50 NO: 228; SEQ ID NO: 229 X 32362826
NM_004006.2:c.5287C > NP_003997.1:p.Arg1763Ter Dilated G > A nonsense ABE SEQ ID SEQ
ID T cardiomy- NO: 51 NO: 230; opathy SEQ ID 3B; Becker NO: 231; muscular SEQ ID
dystrophy; NO: 232 Duchenne muscular dystrophy X 31178668 NM_004006.2:c.10223 +
NP_004010.1:p.Thr340= Dilated C > T synon- ABE SEQ ID SEQ ID 1G > A cardiomy- ymous
NO: 52 NO: 233; opathy variant SEQ ID 3B; Becker NO: 234; muscular SEQ ID dystrophy; NO:
235; Duchenne SEQ ID muscular NO: 236 dystrophy X 31180423 NM_004006.2:c.10033C >
NP_003997.1:p.Arg3345Ter Dilated G > A nonsense ABE SEQ ID SEQ ID T cardiomy- NO: 53
NO: 237; opathy SEQ ID 3B; Becker NO: 238 muscular dystrophy; Duchenne muscular dystrophy;
not provided X 31478163 NM_004006.2:c.8880G > NP_003997.1:p.Trp2960Ter not specified C >
T nonsense ABE SEQ ID SEQ ID A NO: 54 NO: 239 X 31479043 NM_004006.2:c.8608C >
NP_003997.1:p.Arg2870Ter Dilated G > A nonsense ABE SEQ ID SEQ ID T cardiomy- NO: 55
NO: 240 opathy 3B; Becker muscular dystrophy; Duchenne muscular dystrophy X 32697947
NM_004006.2:c.883C > NP_003997.1:p.Arg295Ter Becker G > A nonsense ABE SEQ ID SEQ ID
T muscular NO: 56 NO: 241; dystrophy; SEQ ID Duchenne NO: 242; muscular SEQ ID dystrophy;
NO: 243; Dilated SEQ ID cardiomy- NO: 244; opathy 3B SEQ ID NO: 245 X 32573766
NM_004006.2:c.1683G > NP_003997.1:p.Trp561Ter Duchenne C > T nonsense ABE SEQ ID
SEQ ID A muscular NO: 57 NO: 246; dystrophy; SEQ ID Duchenne NO: 247; muscular SEQ ID
dystrophy NO: 248; SEQ ID NO: 249 X 32484918 NM_004006.2:c.2803 + Dilated C > T splice
ABE SEQ ID SEQ ID 1G > A cardiomy- donor NO: 58 NO: 250 opathy variant 3B; Dilated
cardiomy- opathy 3B X 32573744 NM_004006.2:c.1704 + Becker C > T splice ABE SEQ ID SEQ
ID 1G > A muscular donor NO: 59 NO: 251; dystrophy; variant SEQ ID Duchenne NO: 252;
muscular SEQ ID dystrophy; NO: 253; not specified SEQ ID NO: 254 X 32816541
NM_004006.2:c.457C > NP_003997.1:p.Gln153Ter Duchenne G > A nonsense ABE SEQ ID SEQ
ID T muscular NO: 60 NO: 255; dystrophy SEQ ID NO: 256; SEQ ID NO: 257; SEQ ID NO: 258
X 32348501 NM_004006.2:c.5353C > NP_003997.1:p.Gln1785Ter Dilated G > A nonsense ABE
SEQ ID SEQ ID T cardiomy- NO: 61 NO: 259; opathy SEQ ID 3B; Dilated NO: 260; cardiomy-
SEQ ID opathy 3B NO: 261 X 32411811 NM_004006.2:c.4174C > NP_003997.1:p.Gln1392Ter
Duchenne G > A nonsense ABE SEQ ID SEQ ID T muscular NO: 62 NO: 262; dystrophy; SEQ ID
Becker NO: 263; muscular SEQ ID dystrophy NO: 264; SEQ ID NO: 265; SEQ ID NO: 266 X
31182861 NM_004006.2:c.9851G > NP_003997.1:p.Trp3284Ter Dilated C > T nonsense ABE
SEQ ID SEQ ID A cardiomy- NO: 63 NO: 267; opathy SEQ ID 3B; Dilated NO: 268 cardiomy-
opathy 3B X 32809577 NM_004006.2:c.565C > NP_003997.1:p.Gln189Ter Duchenne G > A
nonsense ABE SEQ ID SEQ ID T muscular NO: 64 NO: 269 dystrophy; Becker muscular
dystrophy X 32216981 NM_004006.2:c.6373C > NP_003997.1:p.Gln2125Ter Duchenne G > A
nonsense ABE SEQ ID SEQ ID T muscular NO: 65 NO: 270 dystrophy; DUCHENNE
MUSCULAR DYSTROPHY; Dilated cardiomy- opathy 3B; Becker muscular dystrophy X

32573812 NM_004006.2:c.1637G > NP_003997.1:p.Trp546Ter Motor delay; C > T nonsense ABE
SEQ ID SEQ ID A Muscle NO: 66 NO: 271; weakness; SEQ ID Muscle NO: 272; cramps; EMG
SEQ ID abnormality; NO: 273; EMG; SEQ ID myopathic NO: 274; abnormalities; SEQ ID Calf
muscle NO: 275 hypertrophy X 31507314 NM_004006.2:c.8357G > NP_003997.1:p.Trp2786Ter
Duchenne C > T nonsense ABE SEQ ID SEQ ID A muscular NO: 67 NO: 276; dystrophy; SEQ ID
Duchenne NO: 277; muscular SEQ ID dystrophy NO: 278; SEQ ID NO: 279; SEQ ID NO: 280 X
32463458 NM_004006.2:c.3413G > NP_003997.1:p.Trp1138Ter Becker C > T nonsense ABE
SEQ ID SEQ ID A muscular NO: 68 NO: 281; dystrophy SEQ ID NO: 282; SEQ ID NO: 283 X
31223071 NM_004006.2:c.9337C > NP_003997.1:p.Arg3113Ter Dilated G > A nonsense ABE
SEQ ID SEQ ID T cardiomy- NO: 69 NO: 284; opathy SEQ ID 3B; Becker NO: 285 muscular
dystrophy; Duchenne muscular dystrophy; not provided; Not Provided X 31929602
NM_004006.2:c.6906G > NP_003997.1:p.Trp2302Ter Dilated C > T 5 prime ABE SEQ ID SEQ
ID A cardiomy- UTR NO: 70 NO: 286 opathy variant 3B; Becker muscular dystrophy; Duchenne
muscular dystrophy X 32595765 NM_004006.2:c.1594C > NP_003997.1:p.Gln532Ter Duchenne
G > A nonsense ABE SEQ ID SEQ ID T muscular NO: 71 NO: 287; dystrophy SEQ ID NO: 288;
SEQ ID NO: 289; SEQ ID NO: 290 X 32573834 NM_004006.2:c.1615C >
NP_003997.1:p.Arg539Ter Dilated G > A nonsense ABE SEQ ID SEQ ID T cardiomy- NO: 72
NO: 291; opathy SEQ ID 3B; Becker NO: 292; muscular SEQ ID dystrophy; NO: 293; Duchenne
SEQ ID muscular NO: 294; dystrophy SEQ ID NO: 295 X 31679565 NM_004006.2:c.7682G >
NP_003997.1:p.Trp2561Ter Dilated C > T nonsense ABE SEQ ID SEQ ID A cardiomy- NO: 73
NO: 296; opathy SEQ ID 3B; Becker NO: 297 muscular dystrophy; Duchenne muscular dystrophy
X 31679564 NM_004006.2:c.7683G > NP_003997.1:p.Trp2561Ter Dilated C > T nonsense ABE
SEQ ID SEQ ID A cardiomy- NO: 74 NO: 298 opathy 3B; Becker muscular dystrophy; Duchenne
muscular dystrophy X 32485057 NM_004006.2:c.2665C > NP_003997.1:p.Arg889Ter Duchenne
G > A nonsense ABE SEQ ID SEQ ID T muscular NO: 75 NO: 299; dystrophy SEQ ID NO: 300;
SEQ ID NO: 301; SEQ ID NO: 302; SEQ ID NO: 303 X 32849781 NM_004006.2:c.133C >
NP_003997.1:p.Gln45Ter Duchenne G > A 5 prime ABE SEQ ID SEQ ID T muscular UTR NO:
76 NO: 304; dystrophy; variant SEQ ID Becker NO: 305; muscular SEQ ID dystrophy NO: 306;
SEQ ID NO: 307 X 32501767 NM_004006.2:c.2368C > NP_003997.1:p.Gln790Ter Duchenne G >
A nonsense ABE SEQ ID SEQ ID T muscular NO: 77 NO: 308; dystrophy; SEQ ID Duchenne NO:
309; muscular SEQ ID dystrophy NO: 310 X 32545250 NM_004006.2:c.2077C >
NP_003997.1:p.Gln693Ter Duchenne G > A nonsense ABE SEQ ID SEQ ID T muscular NO: 78
NO: 311; dystrophy SEQ ID NO: 312; SEQ ID NO: 313 X 31679429 NM_004006.2:c.7818G >
NP_003997.1:p.Trp2606Ter Dilated C > T nonsense ABE SEQ ID SEQ ID A cardiomy- NO: 79
NO: 314; opathy SEQ ID 3B; Dilated NO: 315 cardiomy- opathy 3B X 31679492
NM_004006.2:c.7755G > NP_003997.1:p.Trp2585Ter Duchenne C > T nonsense ABE SEQ ID
SEQ ID A muscular NO: 80 NO: 316; dystrophy; SEQ ID Duchenne NO: 317; muscular SEQ ID
dystrophy NO: 318; SEQ ID NO: 319; SEQ ID NO: 320 X 31875331 NM_004006.2:c.6955C >
NP_003997.1:p.Gln2319Ter Duchenne G > A 5 prime ABE SEQ ID SEQ ID T muscular UTR NO:
81 NO: 321 dystrophy; variant DUCHENNE MUSCULAR DYSTROPHY X 32342105
NM_004006.2:c.5917C > NP_003997.1:p.Gln1973Ter Duchenne G > A nonsense ABE SEQ ID
SEQ ID T muscular NO: 82 NO: 322 dystrophy; Duchenne muscular dystrophy X 31774193
NM_004006.2:c.7310 – Becker C > T splice ABE SEQ ID SEQ ID 1G > A muscular acceptor NO:
83 NO: 323; dystrophy; variant SEQ ID Duchenne NO: 324 muscular dystrophy X 31177932
NM_004006.2:c.10262C > NP_003997.1:p.Ala3421Val Becker G > A 500B ABE SEQ ID SEQ ID
T muscular down- NO: 84 NO: 325; dystrophy; stream SEQ ID BECKER variant NO: 326
MUSCULAR DYSTROPHY; not specified X 31658123 NM_004006.2:c.7894C >
NP_003997.1:p.Gln2632Ter Dilated G > A nonsense ABE SEQ ID SEQ ID T cardiomy- NO: 85
NO: 327 opathy 3B; Becker muscular dystrophy; Duchenne muscular dystrophy X 32468683
NM_004006.2:c.2977C > NP_003997.1:p.Gln993Ter Dilated G > A nonsense ABE SEQ ID SEQ

ID T cardiomy- NO: 86 NO: 328; opathy SEQ ID 3B; Dilated NO: 329 cardiomy- opathy 3B X 32699111 NM_004006.2:c.831 + Duchenne C > T splice ABE SEQ ID SEQ ID 1G > A muscular donor NO: 87 NO: 330; dystrophy variant SEQ ID NO: 331; SEQ ID NO: 332 X 32411772 NM_004006.2:c.4213C > NP_003997.1:p.Gln1405Ter Duchenne G > A nonsense ABE SEQ ID SEQ ID T muscular NO: 88 NO: 333; dystrophy; SEQ ID DUCHENNE NO: 334 MUSCULAR DYSTROPHY X 31178721 NM_004006.2:c.10171C > NP_003997.1:p.Arg3391Ter Dilated G > A nonsense ABE SEQ ID SEQ ID T cardiomy- NO: 89 NO: 335; opathy SEQ ID 3B; Becker NO: 336 muscular dystrophy; Duchenne muscular dystrophy; not provided; Not Provided X 31478983 NM_004006.2:c.8668G > NP_003997.1:p.Glu2890Lys Dilated C > T missense ABE SEQ ID SEQ ID A cardiomy- variant NO: 90 NO: 337; opathy SEQ ID 3B; Dilated NO: 338 cardiomy- opathy 3B X 31180369 NM_004006.2:c.10086 + Dilated C > T splice ABE SEQ ID SEQ ID 1G > A cardiomy- donor NO: 91 NO: 339; opathy variant SEQ ID 3B; Becker NO: 340; muscular SEQ ID dystrophy; NO: 341 Duchenne muscular dystrophy X 32364647 NM_004006.2:c.5089C > NP_003997.1:p.Gln1697Ter Duchenne G > A nonsense ABE SEQ ID SEQ ID T muscular NO: 92 NO: 342 dystrophy; Becker muscular dystrophy X 31496892 NM_004006.2:c.8443C > NP_003997.1:p.Gln2815Ter Dilated G > A nonsense ABE SEQ ID SEQ ID T cardiomy- NO: 93 NO: 343; opathy 3B; SEQ ID Duchenne NO: 344; muscular SEQ ID dystrophy NO: 345; SEQ ID NO: 346 X 32411868 NM_004006.2:c.4117C > NP_003997.1:p.Gln1373Ter Dilated G > A nonsense ABE SEQ ID SEQ ID T cardiomy- NO: 94 NO: 347; opathy SEQ ID 3B; Becker NO: 348 muscular dystrophy; Duchenne muscular dystrophy X 31819974 NM_004006.2:c.7309 + Dilated C > T splice ABE SEQ ID SEQ ID 1G > A cardiomy- donor NO: 95 NO: 349 opathy variant 3B; Becker muscular dystrophy; Duchenne muscular dystrophy X 31444621 NM_004006.2:c.8944C > NP_003997.1:p.Arg2982Ter Duchenne G > A nonsense ABE SEQ ID SEQ ID T muscular NO: 96 NO: 350; dystrophy; SEQ ID DUCHENNE NO: 351; MUSCULAR SEQ ID DYSTROPHY; NO: 352; Dilated SEQ ID cardiomy- NO: 353 opathy 3B; Becker muscular dystrophy X 31627852 NM_004006.2:c.8038C > NP_003997.1:p.Arg2680Ter Duchenne G > A nonsense ABE SEQ ID SEQ ID T muscular NO: 97 NO: 354; dystrophy; SEQ ID Becker NO: 355; muscular SEQ ID dystrophy NO: 356; SEQ ID NO: 357 X 32501833 NM_004006.2:c.2302C > NP_003997.1:p.Arg768Ter Duchenne G > A nonsense ABE SEQ ID SEQ ID T muscular NO: 98 NO: 358; dystrophy; SEQ ID DUCHENNE NO: 359; MUSCULAR SEQ ID DYSTROPHY; NO: 360 Dilated cardiomy- opathy 3B X 32365049 NM_004006.2:c.4996C > NP_003997.1:p.Arg1666Ter Dilated G > A nonsense ABE SEQ ID SEQ ID T cardiomy- NO: 99 NO: 361; opathy SEQ ID 3B; not NO: 362 provided; Duchenne muscular dystrophy X 32345999 NM_004006.2:c.5530C > NP_003997.1:p.Arg1844Ter Dilated G > A nonsense ABE SEQ ID SEQ ID T cardiomy- NO: 100 NO: 363; opathy SEQ ID 3B; Becker NO: 364; muscular SEQ ID dystrophy; NO: 365; Duchenne SEQ ID muscular NO: 366 dystrophy X 32468573 NM_004006.2:c.3087G > NP_003997.1:p.Trp1029Ter Dilated C > T nonsense ABE SEQ ID SEQ ID A cardiomy- NO: 101 NO: 367; opathy SEQ ID 3B; Dilated NO: 368 cardiomy- opathy 3B X 32518098 NM_004006.2:c.2202G > NP_003997.1:p.Trp734Ter Duchenne C > T nonsense ABE SEQ ID SEQ ID A muscular NO: 102 NO: 369; dystrophy SEQ ID NO: 370; SEQ ID NO: 371; SEQ ID NO: 372 X 31820055 NM_004006.2:c.7229G > NP_003997.1:p.Trp2410Ter Dilated C > T 5 prime ABE SEQ ID SEQ ID A cardiomy- UTR NO: 103 NO: 373; opathy variant SEQ ID 3B; Dilated NO: 374 cardiomy- opathy 3B X 32650983 NM_004006.2:c.961 – Dilated G > A intron ABE SEQ ID SEQ ID 5831C > T cardiomy- variant NO: 104 NO: 375; opathy 3B; SEQ ID Duchenne NO: 376; muscular SEQ ID dystrophy NO: 377; SEQ ID NO: 378; SEQ ID NO: 379 X 31478129 NM_004006.2:c.8914C > NP_003997.1:p.Gln2972Ter Duchenne G > A nonsense ABE SEQ ID SEQ ID T muscular NO: 105 NO: 380; dystrophy SEQ ID NO: 381; SEQ ID NO: 382; SEQ ID NO: 383; SEQ ID NO: 384; SEQ ID NO: 385 X 32389605 NM_004006.2:c.4414C > NP_003997.1:p.Gln1472Ter Duchenne G > A nonsense ABE SEQ ID SEQ ID T muscular NO: 106 NO: 386; dystrophy; SEQ ID DUCHENNE NO: 387; MUSCULAR SEQ ID DYSTROPHY NO:

388; SEQ ID NO: 389 X 32595855 NM_004006.2:c.1504C > NP_003997.1:p.Gln502Ter
 Duchenne G > A nonsense ABE SEQ ID SEQ ID T muscular NO: 107 NO: 390 dystrophy X
 32699219 NM_004006.2:c.724C > NP_003997.1:p.Gln242Ter Duchenne G > A nonsense ABE
 SEQ ID SEQ ID T muscular NO: 108 NO: 391; dystrophy; SEQ ID DUCHENNE NO: 392;
 MUSCULAR SEQ ID DYSTROPHY NO: 393; SEQ ID NO: 394 X 33211304 NM_004006.2:c.9G
 > NP_003997.1:p.Trp3Ter Becker C > T nonsense ABE SEQ ID SEQ ID A muscular NO: 109 NO:
 395; dystrophy; SEQ ID BECKER NO: 396; MUSCULAR SEQ ID DYSTROPHY; NO: 397
 Dilated cardiomy- opathy 3B; Duchenne muscular dystrophy; not provided; Not Provided X
 32454833 NM_004006.2:c.3433 – Dilated C > T splice ABE SEQ ID SEQ ID 1G > A cardiomy-
 acceptor NO: 110 NO: 398; opathy variant SEQ ID 3B; Dilated NO: 399; cardiomy- SEQ ID
 opathy 3B NO: 400; SEQ ID NO: 401; SEQ ID NO: 402 X 32472252 NM_004006.2:c.2861G >
 NP_003997.1:p.Trp954Ter Duchenne C > T nonsense ABE SEQ ID SEQ ID A muscular NO: 111
 NO: 403; dystrophy SEQ ID NO: 404; SEQ ID NO: 405 X 32389614 NM_004006.2:c.4405C >
 NP_003997.1:p.Gln1469Ter Dilated G > A nonsense ABE SEQ ID SEQ ID T cardiomy- NO: 112
 NO: 406; opathy SEQ ID 3B; Dilated NO: 407; cardiomy- SEQ ID opathy 3B NO: 408 X
 32438240 NM_004006.2:c.4071 + Duchenne C > T splice ABE SEQ ID SEQ ID 1G > A muscular
 donor NO: 113 NO: 409; dystrophy variant SEQ ID NO: 410; SEQ ID NO: 411; SEQ ID NO: 412;
 SEQ ID NO: 413; SEQ ID NO: 414 X 32491492 NM_004006.2:c.2407C >
 NP_003997.1:p.Gln803Ter Duchenne G > A nonsense ABE SEQ ID SEQ ID T muscular NO: 114
 NO: 415 dystrophy; Duchenne muscular dystrophy X 31206581 NM_004006.2:c.9649 + Duchenne
 C > T splice ABE SEQ ID SEQ ID 1G > A muscular donor NO: 115 NO: 416; dystrophy variant
 SEQ ID NO: 417; SEQ ID NO: 418 X 32390175 NM_004006.2:c.4240C >
 NP_003997.1:p.Gln1414Ter Duchenne G > A nonsense ABE SEQ ID SEQ ID T muscular NO: 116
 NO: 419; dystrophy; SEQ ID not specified NO: 420; SEQ ID NO: 421; SEQ ID NO: 422; SEQ ID
 NO: 423 X 32491387 NM_004006.2:c.2512C > NP_003997.1:p.Gln838Ter Duchenne G > A
 nonsense ABE SEQ ID SEQ ID T muscular NO: 117 NO: 424; dystrophy; SEQ ID Becker NO:
 425; muscular SEQ ID dystrophy NO: 426; SEQ ID NO: 427 X 32342154 NM.004006.2:c.5868G
 > NP_003997.1:p.Trp1956Ter Duchenne C > T nonsense ABE SEQ ID SEQ ID A muscular NO:
 118 NO: 428; dystrophy; SEQ ID Becker NO: 429; muscular SEQ ID dystrophy NO: 430; SEQ ID
 NO: 431 X 32816509 NM_004006.2:c.489G > NP_003997.1:p.Trp163Ter Dilated C > T nonsense
 ABE SEQ ID SEQ ID A cardiomy- NO: 119 NO: 432; opathy SEQ ID 3B; Dilated NO: 433
 cardiomy- opathy 3B X 32454685 NM_004006.2:c.3580C > NP_003997.1:p.Gln1194Ter Dilated
 G > A nonsense ABE SEQ ID SEQ ID T cardiomy- NO: 120 NO: 434; opathy SEQ ID 3B; Becker
 NO: 435 muscular dystrophy; Duchenne muscular dystrophy X 32573529 NM_004006.2:c.1812 +
 Becker C > T splice ABE SEQ ID SEQ ID 1G > A muscular donor NO: 121 NO: 436 dystrophy;
 variant Muscular dystrophy, Becker; Duchenne muscular dystrophy; not provided; Not Provided X
 31223046 NM_004006.2:c.9361 + Dilated C > T splice ABE SEQ ID SEQ ID 1G > A cardiomy-
 donor NO: 122 NO: 437 opathy variant 3B; Becker muscular dystrophy; Duchenne muscular
 dystrophy X 32545310 NM_004006.2:c.2017C > NP_003997.1:p.Gln673Ter Duchenne G > A
 nonsense ABE SEQ ID SEQ ID T muscular NO: 123 NO: 438; dystrophy; SEQ ID DUCHENNE
 NO: 439 MUSCULAR DYSTROPHY X 32343232 NM_004006.2:c.5641C >
 NP_003997.1:p.Gln1881Ter Duchenne G > A nonsense ABE SEQ ID SEQ ID T muscular NO: 124
 NO: 440; dystrophy; SEQ ID Duchenne NO: 441; muscular SEQ ID dystrophy NO: 442; SEQ ID
 NO: 443 X 31679430 NM_004006.2:c.7817G > NP_003997.1:p.Trp2606Ter Duchenne C > T
 nonsense ABE SEQ ID SEQ ID A muscular NO: 125 NO: 444 dystrophy; Duchenne muscular
 dystrophy X 32809559 NM_004006.2:c.583C > NP_003997.1:p.Arg195Ter Dilated G > A
 nonsense ABE SEQ ID SEQ ID T cardiomy- NO: 126 NO: 445; opathy SEQ ID 3B; Becker NO:
 446; muscular SEQ ID dystrophy; NO: 447; Duchenne SEQ ID muscular NO: 448 dystrophy; not
 provided; Not Provided X 32573786 NM_004006.2:c.1663C > NP 003997.1:p.Gln555Ter
 Duchenne G > A nonsense ABE SEQ ID SEQ ID T muscular NO: 127 NO: 449; dystrophy; SEQ

ID Duchenne NO: 450; muscular SEQ ID dystrophy NO: 451; SEQ ID NO: 452 X 31507313
NM_004006.2:c.8358G > NP_003997.1:p.Trp2786Ter Dilated C > T nonsense ABE SEQ ID SEQ
ID A cardiomy- NO: 128 NO: 453; opathy SEQ ID 3B; Dilated NO: 454; cardiomy- SEQ ID
opathy 3B NO: 455; SEQ ID NO: 456; SEQ ID NO: 457; SEQ ID NO: 458 X 32491463
NM_004006.2:c.2436G > NP_003997.1:p.Trp812Ter Dilated C > T nonsense ABE SEQ ID SEQ
ID A cardiomy- NO: 129 NO: 459; opathy SEQ ID 3B; Dilated NO: 460; cardiomy- SEQ ID
opathy 3B NO: 461 X 31658118 NM_004006.2:c.7899G > NP_003997.1:p.Trp2633Ter not
specified C > T nonsense ABE SEQ ID SEQ ID A NO: 130 NO: 462; SEQ ID NO: 463; SEQ ID
NO: 464; SEQ ID NO: 465; SEQ ID NO: 466 X 32448495 NM_004006.2:c.3747G >
NP_003997.1:p.Trp1249Ter Duchenne C > T nonsense ABE SEQ ID SEQ ID A muscular NO: 131
NO: 467; dystrophy SEQ ID NO: 468; SEQ ID NO: 469; SEQ ID NO: 470; SEQ ID NO: 471 X
31348571 NM_004006.2:c.9148C > NP_003997.1:p.Gln3050Ter not provided; G > A nonsense
ABE SEQ ID SEQ ID T Not Provided NO: 132 NO: 472; SEQ ID NO: 473; SEQ ID NO: 474;
SEQ ID NO: 475 X 32485072 NM_004006.2:c.2650C > NP_003997.1:p.Gln884Ter Dilated G > A
nonsense ABE SEQ ID SEQ ID T cardiomy- NO: 133 NO: 476; opathy SEQ ID 3B; Becker NO:
477; muscular SEQ ID dystrophy; NO: 478; Duchenne SEQ ID muscular NO: 479 dystrophy; not
provided X 32463444 NM_004006.2:c.3427C > NP_003997.1:p.Gln1143Ter Duchenne G > A
nonsense ABE SEQ ID SEQ ID T muscular NO: 134 NO: 480; dystrophy; SEQ ID Duchenne NO:
481; muscular SEQ ID dystrophy NO: 482 X 32565742 NM_004006.2:c.1952G > NP
003997.1:p.Trp651Ter Duchenne C > T nonsense ABE SEQ ID SEQ ID A muscular NO: 135 NO:
483; dystrophy; SEQ ID DUCHENNE NO: 484; MUSCULARSEQ ID DYSTROPHY; NO: 485;
Dilated SEQ ID cardiomy- NO: 486 opathy 3B X 31147421 NM_004006.2:c.10651C >
NP_003997.1p.Gln3551Ter not provided; G > A nonsense ABE SEQ ID T Not Provided NO: 136
X 31178700 NM_004006.2:c.10192C > NP_003997.1:p.Gln3398Ter Dilated G > A nonsense ABE
SEQ ID T cardiomy- NO: 137 opathy 3B; Dilated cardiomy- opathy 3B X 31178751
NM_004006.2:c.10141C > NP_003997.1:p.Arg3381Ter Duchenne G > A nonsense ABE SEQ ID T
muscular NO: 138 dystrophy; DUCHENNE MUSCULAR DYSTROPHY; Dilated cardiomy-
opathy 3B; not provided; Not Provided X 31180437 NM_004006.2:c.10019G >
NP_003997.1:p.Cys3340Tyr Duchenne C > T missense ABE SEQ ID A muscular variant NO: 139
dystrophy, mental retardation, and absence of erg b-wave; DUCHENNE MUSCULAR
DYSTROPHY, MENTAL RETARDA- TION, AND ABSENCE OF ERG B-WAVE X 31182784
NM_004006.2:c.9928C > NP_003997.1:p.Gln3310Ter Duchenne G > A nonsense ABE SEQ ID T
muscular NO: 140 dystrophy X 31209497 NM_004006.2:c.9563 + Duchenne C > T splice ABE
SEQ ID 1G > A muscular donor NO: 141 dystrophy variant X 31223062 NM_004006.2:c.9346C >
NP_003997.1:p.Gln3116Ter Dilated G > A nonsense ABE SEQ ID T cardiomy- NO: 142 opathy
3B; Dilated cardiomy- opathy 3B X 31478330 NM_004006.2:c.8713C >
NP_003997.1:p.Arg2905Ter Duchenne G > A nonsense ABE SEQ ID T muscular NO: 143
dystrophy; DUCHENNE MUSCULAR DYSTROPHY; Dilated cardiomy- opathy 3B; Becker
muscular dystrophy X 31478995 NM_004006.2:c.8656C > NP_003997.1:p.Gln2886Ter Dilated G
> A nonsense ABE SEQ ID T cardiomy- NO: 144 opathy 3B; Becker muscular dystrophy;
Duchenne muscular dystrophy X 31627681 NM_004006.2:c.82.09C > NP_003997.1:p.Gln2737Ter
Duchenne G > A nonsense ABE SEQ ID T muscular NO: 145 dystrophy X 31679575
NM_004006.2:c.7672C > NP_003997.1:p.Gln2558Ter not provided; G > A nonsense ABE SEQ ID
T Duchenne NO: 146 muscular dystrophy X 31836729 NM_004006.2:c.7189C >
NP_003997.1:p.Gln2.397Ter Dilated G > A 5 prime ABE SEQ ID T cardiomy- UTR NO: 147
opathy variant 3B; Becker muscular dystrophy; Duchenne muscular dystrophy X 31929718
NM_004006.2:c.6790C > NP_003997.1:p.Gln2264Ter Duchenne G > A 5 prime ABE SEQ ID T
muscular UTR NO: 148 dystrophy; variant DUCHENNE MUSCULAR DYSTROPHY X
32287536 NM_004006.2:c.6283C > NP_003997.1:p.Arg2095Ter Dilated G > A nonsense ABE
SEQ ID T cardiomy- NO: 149 opathy 3B; Dilated cardiomy- opathy 3B X 32287596

NM_004006.2:c.6223C > NP_003997.1:p.Gln2075Ter Duchenne G > A nonsense ABE SEQ ID T muscular NO: 150 dystrophy X 32287680 NM_004006.2:c.6139C > NP_003997.1:p.Gln2047Ter Elevated G > A nonsense ABE SEQ ID T serum NO: 151 creatine phosphokinase X 32287702 NM_004006.2:c.6118 – Duchenne C > T splice ABE SEQ ID 1G > A muscular acceptor NO: 152 dystrophy variant X 32342171 NM_004006.2:c.5851C > NP_003997.1:p.Gln1951Ter not provided; G > A nonsense ABE SEQ ID T not provided NO: 153 X 32345975 NM_004006.2:c.5554C > NP_003997.1:p.Gln1852Ter Dilated G > A nonsense ABE SEQ ID T cardiomy- NO: 154 opathy 3B; Dilated cardiomy- opathy 3B X 32345978 NM_004006.2:c.5551C > NP_003997.1:p.Gln1851Ter Duchenne G > A nonsense ABE SEQ ID T muscular NO: 155 dystrophy; DUCHENNE MUSCULAR DYSTROPHY; Duchenne muscular dystrophy X 32346023 NM_004006.2:c.5506C > NP_003997.1:p.Gln1836Ter Duchenne G > A nonsense ABE SEQ ID T muscular NO: 156 dystrophy X 32364602 NM_004006.2:c.5134C > NP_003997.1:p.Gln1712Ter Dilated G > A nonsense ABE SEQ ID T cardiomy- NO: 157 opathy 3B; Dilated cardiomy- opathy 3B X 32364605 NM_004006.2:c.5131C > XP_003997.1:p.Gln1711Ter Duchenne G > A nonsense ABE SEQ ID T muscular NO: 158 dystrophy; Duchenne muscular dystrophy X 32365175 NM_004006.2:c.4870C > NP_003997.1:p.Gln1624Ter Duchenne G > A nonsense ABE SEQ ID T muscular NO: 159 dystrophy X 32389536 NM_004006.2:c.4483C > NP_003997.1:p.Gln1495Ter Duchenne G > A nonsense ABE SEQ ID T muscular NO: 160 dystrophy X 32389644 NM_004006.2:c.4375C > NP_003997.1:p.Arg1459Ter Dilated G > A nonsense ABE SEQ ID T cardiomy- NO: 161 opathy 3B; Duchenne muscular dystrophy X 32454778 NM_004006.2:c.3487C > NP_003997.1:p.Gln1163Ter not provided; G > A nonsense ABE SEQ ID T Not Provided NO: 162 X 32463438 NM_004006.2:c.3432 + Dilated C > T splice ABE SEQ ID 1G > A cardiomy- donor NO: 163 opathy variant 3B; Becker muscular dystrophy; Duchenne muscular dystrophy X 32464585 NM_004006.2:c.3276 + Dilated C > T splice ABE SEQ ID 1G > A cardiomy- donor NO: 164 opathy variant 3B; Becker muscular dystrophy; Duchenne muscular dystrophy X 32464603 NM_004006.2:c.3259C > XP_003997.1:p.Gln1087Ter not provided; G > A nonsense ABE SEQ ID T Not Provided NO: 165 X 32468509 NM_004006.2:c.3151C > NP_003997.1:p.Arg1051Ter Dilated G > A nonsense ABE SEQ ID T cardiomy- NO: 166 opathy 3B; Duchenne muscular dystrophy; Becker muscular dystrophy X 32468704 NM_004006.2:c.2956C > XP_003997.1:p.Gln986Ter Duchenne G > A nonsense ABE SEQ ID T muscular NO: 167 dystrophy; Becker muscular dystrophy; Dilated cardiomy- opathy 3B; not provided; Not Provided X 32472310 NM_004006.2:c.2804 – Dilated C > T splice ABE SEQ ID 1G > A cardiomy- acceptor NO: 168 opathy variant 3B; Becker muscular dystrophy; Duchenne muscular dystrophy X 32484925 NM_004006.2:c.2797C > NP_003997.1:p.Gln933Ter Duchenne G > A nonsense ABE SEQ ID T muscular NO: 169 dystrophy; Duchenne muscular dystrophy X 32484964 NM_004006.2:c.2758C > NP_003997.1:p.Gln920Ter Dilated G > A nonsense ABE SEQ ID T cardiomy- NO: 170 opathy 3B; Dilated cardiomy- opathy 3B X 32491276 NM_004006.2:c.2622 + Dilated C > T splice ABE SEQ ID 1G > A cardiomy- donor NO: 171 opathy variant 3B; Dilated cardiomy- opathy 3B X 32491344 NM_004006.2:c.2555G > NP_003997.1:p.Trp852Ter not provided; C > T nonsense ABE SEQ ID A Not Provided NO: 172 X 32491480 NM_004006.2:c.2419C > NP_003997.1:p.Gln807Ter Dilated G > A nonsense ABE SEQ ID T cardiomy- NO: 173 opathy 3B; Becker muscular dystrophy; Duchenne muscular dystrophy X 32501803 NM_004006.2:c.2332C > NP_003997.1:p.Gln778Ter Dilated G > A nonsense ABE SEQ ID T cardiomy- NO: 174 opathy 3B; Becker muscular dystrophy; Duchenne muscular dystrophy X 32545158 NM_004006.2:c.2168 + not provided; C > T splice ABE SEQ ID 1G > A Not Provided; donor NO: 175 Duchenne variant muscular dystrophy X 32545190 NM_004006.2:c.2137C > NP_003997.1:p.Gln713Ter Dilated G > A nonsense ABE SEQ ID T cardiomy- NO: 176 opathy 3B; Dilated cardiomy- opathy 3B X 32565704 NM_004006.2:c.1990C > NP_003997.1:p.Gln664Ter Dilated G > A nonsense ABE SEQ ID T cardiomy- NO: 177 opathy 3B; Becker muscular

dystrophy; Duchenne muscular dystrophy X 32565782 NM_004006.2:c.1912C >
NP_003997.1:p.Gln638Ter Duchenne G > A nonsense ABE SEQ ID T muscular NO: 178
dystrophy; Becker muscular dystrophy X 32573796 NM_004006.2:c.1653G >
NP_003997.1:p.Trp551Ter Duchenne C > T nonsense ABE SEQ ID A muscular NO: 179
dystrophy; not provided X 32595870 NM_004006.2:c.1489C > NP_003997.1:p.Gln497Ter
Duchenne G > A nonsense ABE SEQ ID T muscular NO: 180 dystrophy; DUCHENNE
MUSCULAR DYSTROPHY; Duchenne muscular dystrophy X 32614320 NM_004006.2:c.1465C
> NP_003997.1:p.Gln489Ter Dilated G > A nonsense ABE SEQ ID T cardiomy- NO: 181 opathy
3B; Becker muscular dystrophy; Duchenne muscular dystrophy X 32614397
NM_004006.2:c.1388G > NP_003997.1:p.Trp463Ter Duchenne C > T nonsense ABE SEQ ID A
muscular NO: 182 dystrophy; Duchenne muscular dystrophy X 32644131 NM_004006.2:c.1331 +
Duchenne C > T splice ABE SEQ ID 1G > A muscular donor NO: 183 dystrophy; variant
Duchenne muscular dystrophy X 32644139 NM_004006.2:c.1324C > NP_003997.1:p.Gln442Ter
Duchenne G > A nonsense ABE SEQ ID T muscular NO: 184 dystrophy; Duchenne muscular
dystrophy X 32644202 NM_004006.2:c.1261C > NP_003997.1:p.Gln421Ter Dilated G > A
nonsense ABE SEQ ID T cardiomy- NO: 185 opathy 3B; Becker muscular dystrophy; Duchenne
muscular dystrophy X 32645020 NM_004006.2:c.1093C > NP_003997.1:p.Gln365Ter Duchenne
G > A nonsense ABE SEQ ID T muscular NO: 186 dystrophy; Becker muscular dystrophy
32809492 NM_004006.2:c.649 + Dilated C > T splice ABE SEQ ID 1G > A cardiomy- donor NO:
187 opathy variant 3B; Becker muscular dystrophy; Duchenne muscular dystrophy X 32816641
NM.004006.2:c.358 – Becker C > T splice ABE SEQ ID 1G > A muscular acceptor NO: 188
dystrophy variant X 32823297 NM_004006.2:c.355C > NP_003997.1:p.Gln119Ter Duchenne G >
A 5 prime ABE SEQ ID T muscular UTR NO: 189 dystrophy; variant Duchenne muscular
dystrophy X 32849727 NM_004006.2:c.186 + Duchenne C > T splice ABE SEQ ID 1G > A
muscular donor NO: 190 dystrophy variant X 32849736 NM_004006.2:c.178C >
NP_003997.1:p.Gln60Ter Duchenne G > A 5 prime ABE SEQ ID T muscular UTR NO: 191
dystrophy; variant DUCHENNE MUSCULAR DYSTROPHY; not provided X 33020138
NM_004006.2:c.93 + Becker C > T splice ABE SEQ ID 1G > A muscular donor NO: 192
dystrophy; variant Duchenne muscular dystrophy X 32595756 NM_004006.2:c.1602 + not
provided; C > T splice ABE SEQ ID SEQ ID 1G > A Not Provided donor NO: 193 NO: 487 variant
X 31206668 NM_004006.2:c.9564 – Dilated C > T splice ABE SEQ ID SEQ ID 1G > A cardiomy-
acceptor NO: 194 NO: 488; opathy variant SEQ ID 3B; Becker NO: 489; muscular SEQ ID
dystrophy; NO: 490; Duchenne SEQ ID muscular NO: 491 dystrophy X 32217062
NM_004006.2:c.6292C > NP_003997.1:p.Arg2098Ter Duchenne G > A nonsense ABE SEQ ID
SEQ ID T muscular NO: 195 NO: 492 dystrophy; DUCHENNE MUSCULAR DYSTROPHY;
Dilated cardiomy- opathy 3B X 32644314 NM_004006.2:c.1150 – Duchenne C > T splice ABE
SEQ ID SEQ ID 1G > A muscular acceptor NO: 196 NO: 493; dystrophy; not variant SEQ ID
provided; Not NO: 494; Provided SEQ ID NO: 495 X 32849821 NM_004006.2:c.94 – Duchenne
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(180) TABLE-US-00006 SEQUENCES SEQ ID NO: 1 (pCMV-ABEmaxNG)

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LSQLGGD SEQ ID NO: 14, SpCas9(D10A)-NGX: Protein sequence:

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GETAEATRLKRTARRRYTRRKNRICYLQEIFS NEMAKVDDSFHRLEESFLVEEDKKHE
RHPIFGNIVDEVAYHEKYPTIYHLRKKLVDSTDKADLR LIYLALAHMIKIRGHFLIEGDL
NPDNSDVKLFIQLVQTYNQLFEENPINASGVDAKAILSARLSKSRLENLIAQLPGEKK
NGLFGNLIASLGLTPNFKS NFDLAEDTKLQLSKDTYDDDLDNLLAQIGDQYADLFLAA
KNLSDAILLSDILRVNTEITKAPLSASMIKLYDEHHQDLTLLKALVRQQLPEKYKEIFFD
QSKNGYAGYIDGGASQEEFYKFIKPILEKMDGTEELLVKLNREDLLRKQRTFDNGIIPHQ
IHLGELHAILRRQEDFYFPFLKDNREKIEKILTRIPYYVGPLARGNSRFAWMTRKSEETIT
PWNFEKVVDKGASAQSFIERMTNFDKNLPNEKVLPKHSLLEYFTVYNELTKVKYVTE
GMRKPAFLSGDQKKAIVDLLFKTNRKVTVKQLKEDYFKKIECFDSVEISGVEDRFNASL
GTYHDLKIIKDKDFLDNEENEDILEDIVLTTLTFEDREMIEERLKTYAHLFDDKVMKQL
KRRRYTGWGRLSRKLINGIRDKQSGKTILDFLKSDGFANRNFIQLIHDDSLTFKEDIQKA
QVSGQGDSLHEHIANLAGSPAIIKKGILQTVKVVDELVKVMGRHKPENIVIAMARENQTT
QKGQKNSRERMKRIIEGKELGSQILKEHPVENTQLQNEKLYLYYLQNGRDMYVDQEL
DINRLSDYDVDHIVPQSFLKDDSIDNKVLTRSDKNRGKSDNVPSEEVVKKMKNYWRQL
LNAKLITQRKFDNLTAKAERGGLSELDKAGFIKRQLVETRQITKHVAQILDSRMNTKYDE
NDKLIREVKVITLKSCLVSDFRKDFQFYKVRINNYHHAHDAYLNAVVG TALIKKYPKL
ESEFVYGDYKVYDVRKMIKSEQEIGKATAKYFFYSNIMNFFKTEITLANGEIRKRPLIE
TNGETGEIVWDKGRDFATVRKVLSMPQVNIVKKTEVQTGGFSKESIRPKRNSDKLIARK
KDWDPPKKYGGFVSPTVAYSVLVVAKEVGKSKKLKSVKELLGITIMERSSSFENPIDFL
EAKGYKEVKKDLIIKLPKYSLFELENGRKRMLASARFLQKGNELALPSKYVNFLYLASH
YEKLKGSPEDEQKQLFVEQHKHYLDEIIIEQISEFSKRVLADANLDKVL SAYNKH RD
PIREQAENIIHLFTLTNLGAPRAFKYFDTTIDRKVYRSTKEVL DATLIHQ SITGLYETR ID
SQLGGD SEQ ID NO: 15, SpCas9(D10A)-NGA: Protein sequence:

MDKKYSIGLAIGTNSVGWAVITDEYKVPSKKFKVLGNTDRHSIKKNLIGALLFDS
GETAEATRLKRTARRRYTRRKNRICYLQEIFS NEMAKVDDSFHRLEESFLVEEDKKHE
RHPIFGNIVDEVAYHEKYPTIYHLRKKLVDSTDKADLR LIYLALAHMIKFRGHFLIEGDL
NPDNSDVKLFIQLVQTYNQLFEENPINASGVDAKAILSARLSKSRLENLIAQLPGEKK
NGLFGNLIASLGLTPNFKS NFDLAEDA KLQLSKDTYDDDLDNLLAQIGDQYADLFLAA
KNLSDAILLSDILRVNTEITKAPLSASMIKRYDEHHQDLTLLKALVRQQLPEKYKEIEFD
QSKNGYAGYIDGGASQEEFYKFIKPILEKMDGTEELLVKLNREDLLRKQRTFDNGSIPH
QIHLGELHAILRRQEDFYFPFLKDNREKIEKILTRIPYYVGPLARGNSRFAWMTRKSEETI
TPWNFEKVVDKGASAQSFIERMTNFDKNLPNEKVLPKHSLLEYFTVYNELTKVKYVI

EGMRKPAFLSGEQKKAIVDLLFKTNRKVTVKQLKEDYFKKIECFDSVEISGVEDRFNAS
LGTYHDLLKIIKDKDFLDNEENEDILEDIVLTTLTLFEDREMIEERLKTYYAHLFDDKVMKQ
LKRRRYTGWGRLSRKLINGIRDKQSGKTILDFLKSDGFANRNFQMQLIHDDSLTFKEDIQ
KAQVSGQGDSLHEHIANLAGSPAIIKKGILQTVKVVDDELVKVMGRHKPENIVIAMAREN
QTTQKGQKNSRERMKRIEEGIKELGSQILKEHPVENTQLQNEKLYLYYLQNGRDMYVD
QELDINRLSDYDVDHIVPQSFLKDDSIDNKVLTRSDKNRGKSDNVPSEEVVKMKMKNYW
RQLLNAKLITQRKFDNLTKAERGGLSELDKAGFIKRQLVETRQITKHVAQILDSRMNTK
YDENDKLIREVKVITLKSCLVSDFRKDFQFYKVVREINNYHHAHDAYLNAVVGITALIKK
YPKLESEFVYGDYKVYDVRKMIKSESEQEIGKATAKYFFYSNIMNFFKTEITLANGEIRK
RPLIETNGETGEIVWDKGRDFATVRKVL SMPQVNIVKKTEVQTGGFSKESIRPKRNSDK
LIARKKDWDPKKYGGFVSPTVAYSVLVVAKEKVGKSKKLKSVKELLGITIMERSSEFEK
NPIDFLEAKGYKEVKKDLIIKLPKYSLFELENGRKRMLASARFLQKGNELALPSKYVNF
LYLASHYEKLKGSPEDEQKQLFVEQHKHYLDEIIEQISEFSKRVLADANLDKVL SAYN
KHRDKPIREQAENIIHLFTLTNLGAPRAFKYFDTTIDRKQYRSTKEVL DATLIHQ SITGLY

ETRIDLSQLGGD SEQ ID NO: 16, SpCas9(D10A)-NGC: Protein sequence:
MDKKYSIGLAIGTNSVGWAVITDEYKVPSKKFKVLGNTDRHSIKKNLIGALLFDS
GETAEATRLKRIARRRYTRRKNRICYLQEIFSNEMAKVDDDSFFHRLEESFLVEEDKKHE
RHPIFGNIVDEVAYHEKYPTIYHLRKKLVDSTDKADLRLIYLALAHMIKFRGHFLIEGDL
NPDNSDVKLFIQLV'QTYNQLFEENPINASGVDAKAILSARLSKSRLENLIAQLPGEKK
NGLFGNLIASLGLTPNFKSNFDLAEDAKLQLSKDTYDDDLDNLLAQIGDQYADLFLAA
KNLSDAILLSDILRVNTEITKAPLSASMIKRYDEHHQDLTLLKALVRQQQLPEKYKEIFFD
QSKNGYAGYIDGGASQEEFYKFIKPILEKMDGTEELLVKLNREDLLRKQRTFDNGSIPH
QIHLGELHAILRRQEDFYFPLKDNREKIEKILTRIPYYVGPLARGNSRFAWMTRKSEETI
TPWNFEEVVDKGASAQSFIERMTNFDKNLPNEKVLPKHSLLYEYFTVYNELTKVKYVT
EGMRKPAFLSGEQKKAIVDLLFKTNRKVTVKQLKEDYFKKIECFDSVEISGVEDRFNAS
LGTYHDLLKIIKDKDFLDNEENEDILEDIVLTTLTLFEDREMIEERLKTYYAHLFDDKVMKQ
LKRRRYTGWGRLSRKLINGIRDKQSGKTILDFLKSDGFANRNFQMQLIHDDSLTFKEDIQ
KAQVSGQGDSLHEHIANLAGSPAIIKKGILQTVKVVDDELVKVMGRHKPENIVIAMAREN
QTTQKGQKNSRERMKRIEEGIKELGSQILKEHPVENTQLQNEKLYLYYLQNGRDMYVD
QELDINRLSDYDVDHTVPQSFLKDDSIDNKVLTRSDKNRGKSDNVPSEEVVKMKMKNYW
RQLLNAKIITQRKFDNLTKAERGGLSELDKAGFIKRQLVETRQITKHVAQILDSRMNTK
YDENDKLIREVKVITLKSCLVSDFRKDFQFYKVVREINNYHHAHDAYLNAVVGITALIKK
YPKLESEFVYGDYKVYDVRKMIKSESEQEIGKATAKYFFYSNIMNFFKTEITLANGEIRK
RPLIETNGETGEIVWDKGRDFATVRKVL SMPQVNIVKKTEVQTGGFSKESIRPKRNSDK
LIARKKDWDPKKYGGFVSPTVAYSVLVVAKEKVGKSKKLKSVKELLGITIMERSSEFEK
NPIDFLEAKGYKEVKKDLIIKLPKYSLFELENGRKRMLASARFLQKGNELALPSKYVNF
LYLASHYEKLKGSPEDEQKQLFVEQHKHYLDEIIEQISEFSKRVLADANLDKVL SAYN
KHRDKPIREQAENIIHLFTLTNLGAPRAFKYFDTTIDRKEYRSTKEVL DATLIHQ SITGLY

ETRIDLSQLGGD SEQ ID NO: 17, SpCas9(D10A)-NGX-NGA: Protein sequence:
MDKKYSIGLAIGTNSVGWAVITDEYKVPSKKFKVLGNTDRHSIKKNLIGALLFDS
GETAEATRLKRTARRRYTRRKNRICYLQEIFSNEMAKVDDDSFFHRLEESFLVEEDKKHE
RHPIFGNIVDEVAYHEKYPTIYHLRKKLVDSTDKADLRLIYLALAHMIKFRGHFLIEGDL
NPDNSDVKLFIQLVQTYNQLFEENPINASGVDAKAILSARLSKSRLENLIAQLPGEKK
NGLFGNLIASLGLTPNFKSNFDLAEDTKLQLSKDTYDDDLDNLLAQIGDQYADLFLAA
KNLSDAILLSDILRVNTEITKAPLSASMIKLYDEHHQDLTLLKALVRQQQLPEKYKEIFFD
QSKNGYAGYIDGGASQEEFYKFIKPILEKMDGTEELLVKLNREDLLRKQRTFDNGIIPHQ
IHLGELHAILRRQEDFYFPLKDNREKIEKILTRIPYYVGPLARGNSRFAWMTRKSEETIT
PWNFEKVVDKGASAQSFIERMTNFDKNLPNEKVLPKHSLLYEYFTVYNELTKVKYVTE
GMRKPAFLSGDQKKAIVDLLFKTNRKVTVKQLKEDYFKKIECFDSVEISGVEDRFNASL
GTYHDLLKIIKDKDFLDNEENEDILEDIVLTTLTLFEDREMIEERLKTYYAHLFDDKVMKQL

KRRRYTGWGRLSRKLINGIRDKQSGKTILDFLKSDGFANRNFIQLIHDDSLTFKEDIQKA
QVSGQGDSLHEHIANLAGSPAIAKKGILQTVKVVDELVKVMGRHKPENIVIAMARENQTT
QKGQKNSRERMKRIIEGIELGSQILKEHPVENTQLQNEKLYLYYLQNGRDMYVDQEL
DINRLSDYDVDHIVPQSFLKDDSIDNKVLTRSDKNRGKSDNVPSEEVVKKMKNYWRQL
LNAKLITQRKFDNLTAKAERGGLSELDKAGFIKRQLVETRQITKHVAQILDSRMNTKYDE
NDKLIREVKVITLKSCLVSDFRKDFQFYKVREINNYHHAHDAYLNAVVG TALIKKYPKL
ESEFVYGDYKVYDVRKMIKSEQEIGKATAKYFFYSNIMNFFKTEITLANGEIRKRPLIE
TNGETGEIVWDKGRDFATVRKVLSMPQVNIVKKTEVQTGGFSKESIRPKRNSDKLIARK
KDWDPPKKYGGFVSPTVAYSVLVVAKEKVGKSKKLKSVKELLGITIMERSSEFEKNPIDFL
EAKGYKEVKKDLIIKLPKYSLFELENGRKRMLASARFLQKGNELALPSKYVNFLYLASH
YEKLKGSPEDNEQKQLFVEQHKHYLDEIEQISEFSKRVLADANLDKVL SAYNKHARDK
PIREQAENIIHLFTLTNLGAPRAFKYFDTTIDRKQYRSTKEVLDATLIHQ SITGLYETRIDL
SQLGGD SEQ ID NO: 18, SpCas9(D10A)-NGX-NGC: Protein sequence:

MDKKYSIGLAIGTNSVGWAVITDEYKVPSKKFKVLGNTDRHSIKKNLIGALLFDS
GETAEATRLKRTARRRYTRRKNRICYLQEIFS NEMAKVDDSFHRLEESFLVEEDKKHE
RHPIFGNIVDEVAYHEKYPTIYHLRKKLVDSTDKADLR LIYLALAHMIKFRGHFLIEGDL
NPDNSDVKLF IQLVQTYNQLFEENPINASGVDAKAILSARLSKSRLENLIAQLPGEKK
NGLFGNLI ALSLGLTPNFKS NFDLAEDTKLQLSKDTYDDDLDNLLAQIGDQYADLFLAA
KNLSDAILLSDILRVNTEITKAPLSASMIKLYDEHHQDLTLLKALVRQQLPEKYKEIFFD
QSKNGYAGYIDGGASQEEFYKFIKPILEKMDGTEELLVKLNREDLLRKQRTFDNGIIPHQ
IHLGELHAILRRQEDFY PFLKDNREKIEKILTRIPYYVGPLARGNSRFAWMTRKSEETIT
PWNFEKVVDKGASAQSFIERMTNFDKNLPNEKVL PKHSLLYEYFTVYNELTKVKYVTE
GMRKPAFLSGDQKKAIVDLLFKTNRKVTVKQLKEDYFKKIECFDSVEISGVEDRFNASL
GTYHDLLKIIKDKDFLDNEENEDILEDIVLTLT LFEDREMIEERLKT YAHLFDDKVMKQL
KRRRYTGWGRLSRKLINGIRDKQSGKTILDFLKSDGFANRNFIQLIHDDSLTFKEDIQKA
QVSGQGDSLHEHIANLAGSPAIAKKGILQTVKVVDELVKVMGRHKPENIVIAMARENQTT
QKGQKNSRERMKRIIEGIELGSQILKEHPVENTQLQNEKLYLYYLQNGRDMYVDQEL
DINRLSDYDVDHIVPQSFLKDDSIDNKVLTRSDKNRGKSDNVPSEEVVKKMKNYWRQL
LNAKLITQRKFDNLTAKAERGGLSELDKAGFIKRQLVETRQITKHVAQILDSRMNTKYDE
NDKLIREVKVITLKSCLVSDFRKDFQFYKVREINNYHHAHDAYLNAVVG TALIKKYPKL
ESEFVYGDYKVYDVRKMIKSEQEIGKATAKYFFYSNIMNFFKTEITLANGEIRKRPLIE
TNGETGEIVWDKGRDFATVRKVLSMPQVNIVKKTEVQTGGFSKESIRPKRNSDKLIARK
KDWDPPKKYGGFVSPTVAYSVLVVAKEKVGKSKKLKSVKELLGITIMERSSEFEKNPIDFL
EAKGYKEVKKDLIIKLPKYSLFELENGRKRMLASARFLQKGNELALPSKYVNFLYLASH
YEKLKGSPEDNEQKQLFVEQHKHYLDEIEQISEFSKRVLADANLDKVL SAYNKHARDK
PIREQAENIIHLFTLTNLGAPRAFKYFDTTIDRKE YRSTKEVLDATLIHQ SITGLYETRIDL
SQLGGD SEQ ID: 19, SpCas9(D10A)-NG-loop: Protein sequence:

MDKKYSIGLAIGTNSVGWAVITDEYKVPSKKFKVLGNTDRHSIKKNLIGALLFDS
GETAEATRLKRTARRRYTRRKNRICYLQEIFS NEMAKVDDSFHRLEESFLVEEDKKHE
RHPIFGNIVDEVAYHEKYPTIYHLRKKLVDSTDKADLR LIYLALAHMIKFRGHFLIEGDL
NPDNSDVKLF IQLVOTYNQLFEENPINASGVDAKAILSARLSKSRLENLIAQLPGEKK
NGLFGNLI ALSLGLTPNFKS NFDLAEDAKLQLSKDTYDDDLDNLLAQIGDQYADLFLAA
KNLSDAILLSDILRVNTEITKAPLSASMIKRYDEHHQDLTLLKALVRQQLPEKYKEIFFD
QSKNGYAGYVGADKKLRKRSSKLATEEEFYKFIKPILEKMDGTEELLVKLNREDLLRK
QRTFDNGSIPHQIHLGELHAILRRQEDFY PFLKDNREKIEKILTRIPYYVGPLARGNSRF
AWMTRKSEETITPWNFEKVVDKGASAQSFIERMTNFDKNLPNEKVL PKHSLLYEYFTV
YNELTKVKYVTEGMRKPAFLSGEQKKAIVDLLFKTNRKVTVKQLKEDYFKKIECFDSV
EISGVEDRFNASLGTYHDLLKIIKDKDFLDNEENEDILEDIVLTLT LFEDREMIEERLKT Y
AHLFDDKVMKQLKRRRYTGWGRLSRKLINGIRDKQSGKTILDFLKSDGFANRNFMQLI
HDDSLTFKEDIQKAQVSGQGDSLHEHIANLAGSPAIAKKGILQTVKVVDELVKVMGRHK

PENVIEMARENQTTQKGQKNSRERMKRIEELGSQILKEHPVENTQLQNEKLYLYLQNGRDMYVDQELDINRLSDYDVDHIVPQSFLKDDSIDNKVLTRSDKNRGKSDNVPS
EEVVKMKKNYWRQLLNAKLITQRKFDNLTKAERGGLSELDKAGFIKRQLVETRQITKH
VAQILDSRMNTKYDENDKLIREVVKVITLKSCLVSDFRKDFQFYKVVREINNYHHAHDAY
LNAVVGTAIIKKYPKLESEFVYGDYKVYDVRKMIKSEQEIGKATAKYFFYSNIMNFF
KTEITLANGEIRKRPLIETNGETGEIVWDKGRDFATVRKVLSPQVNIKKTEVQTGGFS
KESIRPKRNSDKLIARKKDWDPKKYGGFVSPTVAYSVLVAKVEKGKSKKLKSVKELL
GITIMERSSEKKNPIDFLEAKGYKEVKKDLIIKLPKYSLFELENGRKRMLASARFLQKGN
ELALPSKYVNFYLYLASHYEKLKGSPEDENEQKQLFVEQHKHYLDEHIEQISEFSKRVLAD
ANLDKVL SAYNKHDKPIREQAENIIHLFTLTNLGAPRAFKYFDTTIDRKVYRSTKEVLD
ATLIHQ SITGLYETRIDLSQLGGD SEQ ID NO: 20, SpCas9(D10A)-NGX-loop:

Protein sequence:

MDKKYSIGLAIGTNSVGWAVITDEYKVPSKKFKVLGNTDRHSIKKNLIGALLFDS
GETAEATRLKRTARRRYTRRKNRICYLQEIFSNEMAKVDDSFHRLEESFLVEEDKKHE
RHPIFGNIVDEVAYHEKYPTIYHLRKKLVDSTDKADLRILIYLA AHMIKFRGHFLIEGDL
NPDNSDVKLFIQLVQTYNQLFEENPINASGVDAKAILSARLSKSRLENLIAQLPGEKK
NGLFGNLIASLGLTPNFKSNFDLAEDTKLQLSKDTYDDDLDNLLAQIGDQYADLFLAA
KNLSDAILLSDILRVNTEITKAPLSASMIKLYDEHHQDLTLLKALVRQQLP EKYKEIFFD
QSKNGYAGYVGADKKLRKRSSKLATEEEFYKFIKPILEKMDGTEELLVKLNREDLLRK
QRTFDNGIIPHQIHLGELHAILRRQEDFY PFLKDNREKIEKILTFRIPYYVGPLARGNSRFA
WMTRKSEETITPWNFEKVVDKGASAQSFIERMTNFDKNLPNEKVLPKHSLLYEYFTVY
NELTKVKYVTEGMRKPAFLSGDQKKAIVDLLFKTNRKVTVKQLKEDYFKKIECFDSVEI
SGVEDRFNASLGTYHDLLKIIKDKDFLDNEENEDILEDIVLTTLTFEDREMIEERLKTYA
HLFDDKVMKQLKRRRYTGWGRLSRKLINGIRDKQSGKTILDFLKSDGFANRNFIQLIHD
DSLTFKEDIQKAQVSGQGDSLHEHIANLAGSPAIIKGILQTVKVVDDELVKVMGRHKPEN
IVIEMARENQTTQKGQKNSRERMKRIEELGSQILKEHPVENTQLQNEKLYLYYLQ
NGRDMYVDQELDINRLSDYDVDHIVPQSFLKDDSIDNKVLTRSDKNRGKSDNVPS
EEVVKMKKNYWRQLLNAKLITQRKFDNLTKAERGGLSELDKAGFIKRQLVETRQITKHVAQ
ILDSRMNTKYDENDKLIREVVKVITLKSCLVSDFRKDFQFYKVVREINNYHHAHDAYLNA
VVGTAIIKKYPKLESEFVYGDYKVYDVRKMIKSEQEIGKATAKYFFYSNIMNFFKTEI
TLANGEIRKRPLIETNGETGEIVWDKGRDFATVRKVLSPQVNIKKTEVQTGGFSKESI
RPKRNSDKLIARKKDWDPKKYGGFVSPTVAYSVLVAKVEKGKSKKLKSVKELLGITI
MERSSEKKNPIDFLEAKGYKEVKKDLIIKLPKYSLFELENGRKRMLASARFLQKGNELA
LPSKYVNFYLYLASHYEKLKGSPEDENEQKQLFVEQHKHYLDEHIEQISEFSKRVLADANL
DKVLSAYNKHDKPIREQAENIIHLFTLTNLGAPRAFKYFDTTIDRKVYRSTKEVLDATL
IHQSITGLYETRIDLSQLGGD SEQ ID NO: 21, ScCas9(D10A): Protein sequence:

MEKKYSIGLAIGTNSVGWAVITDDYKVPSKKFKVLGNTNRKSIKKNLMGALLFD
SGETAETRLKRTARRRYTRRKNRIRYLQEIFANEMAKLDDSFQRLEESFLVEEDKKN
ERHPIFGNLADEVAYHRNYPTIYHLRKKLADSPEKADLRILIYLA AHIIKFRGHFLIEGKL
NAENSDVAKLFYQLIQTYNQLFEESPLDEIEVDAKGILSARLSKSKRLEKLI AVFPNEKK
NGLFGNIIALALGLTPNFKSNFDLTEDAKLQLSKDTYDDDLDLDELGQIGDQYADLFSAA
KNLSDAILLSDILRSNSEVTKAPLSASMVKRYDEHHQDLALLKTLVRQQFP EKYAEIFK
DDTKNGYAGYVGIGIKHRKRTTKLATQEEFYKFIKPILEKMDGAEELLAKLNRDDLLRK
QRTFDNGSIPHQIHLKELHAILRRQEEFY PFLKENREKIEKILTFRIPYYVGPLARGNSRFA
WLTRKSEEAITPWNFEKVVDKGASAQSFIERMTNFDQELPNKKVLPHSLLYEYFTVYN
ELTKVKYVTERMRKPEFLSGEQKKAIVDLLFKTNRKVTVKQLKEDYFKKIECFDSVEIIG
VEDRFNASLGTYHDLLKIIKDKDFLDNEENEDILEDIVLTTLTFEDREMIEERLKTYAHLF
DDKVMKQLKRRHYTGWGRLSRKMINGIRDKQSGKTILDFLKSDGFSNRNFMQLIHDDS
LTFKEEIEKAQVSGQGDSLHEQIADLAGSPAIIKGILQTVKIVDELVKVMGHKPENIVIE
MARENQTTTGLQQSRERKKRIEELGKELSKILKENPVENTQLQNEKLYLYYLQNGRD

MYVDQELDSDYDNLVDSQFSDVNRGKSDNVPSEEVVKKM
KNYWRQLLNAKLITQRKFDNLTKAERGGLSEADKAGFIKRQLVETRQITKHVARILDSR
MNTKRDKNDKPIREVKVITLKSCLVSDFRKDFQLYKVRDINNYHHAHDAYLNAVVG
ALIKKYPKLESEFVYGDYKVYDVRKMIKSEQEIGKATAKRFFYSNIMNFFKTEVKLAN
GEIRKRPLIETNGETGEVVWNKEKDFATVRKVLAMPQVNIVKKTEVQTGGFSKESILSK
RESAKLIPRKKGWDTRKYGGFGSPTVAYSILVVAKVEKGKAKKLKSVKVLVGITIMEK
GSYEKDPIGFLEAKGYKDIKKELIFKLPKYSLFELENGRRRMLASATELQKANELVLPQH
LVRLLYYTQNISATTGSNNLGYIEQHREEFKEIFEKIIDFSEKYILKNKVNSNLKSSFDEQF
AVSDSILLSNSFVSLLKYTSFGASGGFTFLDLVDVKQGRRLRYQTVTEVLDATLIYQSITGL
YETRDTLSQLGGD SEQ ID NO: 22, TadA-TadA*: Protein sequence:
SEVEFSHEYWMRHALTLAKRAWDEREVPVGAVLVHNNRVIGEGWNRPIGRHD
PTAHAEIMALRQGGLVMQNYRLIDATLYVTLEPCVMCAGAMIHSRIGRVVFGARDAKT
GAAGSLMDVLHHPGMNHRVEITEGILADECAALLSDFFRMRRQEIKAKKKAQSSTDSG
GSSGGSSGSETPGTSESATPESSGGSSGSSGSEVEFSHEYWMRHALTLAKRARDEREVPV
GAVLVLNNRVIGEGWNRAIGLHDPTAHAEIMALRQGGLVMQNYRLIDATLYVTLEPCV
MCAGAMIHSRIGRVVFGVRNAKTGAAGSLMDVLHYPGMNHRVEITEGILADECAALL
CYFFRMPRQVFNAQKKAQSSTD SEQ ID: 23, TadA*(A56G_V82G): Protein
sequence: SEVEFSHEYWMRHALTLAKRARDEREVPVGAVLVLNNRVIGEGWNRAIGLHDP
TGHAEIMALRQGGLVMQNYRLIDATLYGTLEPCVMCAGAMIHSRIGRVVFGVRNAKT
GAAGSLMDVLHYPGMNHRVEITEGILADECAALLCYFFRMPRQVFNAQKKAQSSTD
SEQ ID NO: 24, Gp41-1-N: Protein sequence: CLDLK
TQVQTPQGMKEISNIQVGDVLVSLNTGYNEVLNVEPKSKKKSYKITLEDG
KEIICSEHLFPTQTGEMNISGGLKEGMCLYVKE SEQ ID NO: 25, Gp41-1-C:
Protein sequence: MMLKKILKIEELDERELIDIEVSGNHLFYANDILTHNS SEQ ID NO:
26, Cfa-N: Protein sequence:
CLSYDTEILTVEYGFLPIGKIVEERIECTVYTVDKNGFVYTQPIAQWHNRGEQEV
FEYCLEDGSIIRATKDHKFMTTDQGMLPIDEIFERGLDLKQVDGLPN SEQ ID NO:
27, Cfa-C: Protein sequence: MVKIISRKSLGTQNVYDIGVEKDHNFLLKNGLVASN
SEQ ID NO: 28, meCMV:
tegagCGCGTGATGAGAGCAGCCACTACGGGTCTAGGCTGCCCCATGTAAGGAG
GCAAGGCCTGGGGACACCCGAGATGCCTGGTTATAATTAACCCAGACATGTGGCTG
CCCCCCCCCCCCCAACACCTGCTGCCTGCTAAAAATAACCCTGTCCCTGGTGGCctgc
atgccACTCACGGGGATTTCCTCAAGTCTCCACCCCATGACGTCAATGGGAGTTTGT
TGGCACCAAAATCAACGGGACTTTCCAAAATGTCGTAACAACCTCCGCCCCATTGAC
GCAAATGGGCGGTAGGCGTGTACGGTGGGAGGTCTATATAAGCAGAGCTGGTTTAG
TGAACCGTCAGATC SEQ ID NO: 29, SEMHP1:
CCCTTCAGATTAAAAATAACTGAGGTAAGGGCCTGGGTAGGGGAGGTGGTG
TGAGACGCTCCTGTCTCTCCTCTATCTGCCCATCGGCCCTTTGGGGAGGAGGAATGT
GCCCAAGGACTAAAAAAGGCCATGGAGCCAGAGGGGCGAGGGCAACAGACCTTT
CATGGGCAAACCTTGGGGCCCTGCTGactgtaGATGAGAGCAGCCACTACGGGTCTAG
GCTGCCCATGTAAGGAGGCAAGGCCTGGGGACACCCGAGATGCCTGGTTATAATTA
ACCCAGACATGTGGCTGCCCCCCCCCCCCCAACACCTGCTGCCTGCTAAAAATAACC
CTGTCCCTGGTGGccctgcatgCCCTCCCTGGGGACAGCCCCCTCCTGGCTAGTCACACCC
TGTAGGCTCCTCTATATAACCCAGGGGACAGGGGGCTGCCctcattctACCACCACCTCC
ACAGCACAGACAGACACTCAGGAGCCAGC SEQ ID NO: 30, Wild Type
SpCas9(D10A): Nucleotide sequence:
ATGGACAAGAAGTACAGCATCGGCCTGGCCATCGGCACCAACTCTGTGGGCT
GGGCCGTGATCACCGACGAGTACAAGGTGCCAGCAAGAAATTCAAGGTGCTGGGC
AACACCGACCGGCACAGCATCAAGAAGAACCTGATCGGAGCCCTGCTGTTTCGACAG
CGGCGAAACAGCCGAGGCCACCCGGCTGAAGAGAACCGCCAGAAGAAGATACACC

AGACGGAAGAACCGGATCTGCTATCTTACGTAAGATCTTACGCAAGATGCGCA
GGTGGACGACAGCTTCTTCCACAGACTGGAAGAGTCCTTCCTGGTGGAAAGAGGATA
AGAAGCACGAGCGGCACCCCATCTTCGGCAACATCGTGGACGAGGTGGCCTACCAC
GAGAAGTACCCCAACCATCTACCACCTGAGAAAGAACTGGTGGACAGCACCGACA
AGGCCGACCTGCGGCTGATCTATCTGGCCCTGGCCCACATGATCAAGTTCCGGGGC
CACTTCCTGATCGAGGGCGACCTGAACCCCGACAACAGCGACGTGGACAAGCTGTT
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ATAAGGGCCGGGATTTTGCCACCGTGCGGAAAGTGCTGAGCATGCCCCAAGTGAAT
ATCGTGAAAAAGACCGAGGTGCAGACAGGCGGCTTCAGCAAAGAGTCTATCCgGCC
CAAGAGGAACAGCGATAAGCTGATCGCCAGAAAGAAGGACTGGGACCCTAAGAAG
TACGGCGGCTTCGtCAGCCCCACCGTGGCCTATTCTGTGCTGGTGGTGGCCAAAGTG
GAAAAGGGCAAGTCCAAGAACTGAAGAGTGTGAAAGAGCTGCTGGGGATCACCA
TCATGGAAAGAAGCAGCTTCGAGAAGAATCCCATCGACTTTCTGGAAGCCAAGGGC
TACAAAGAAGTGAAAAAGGACCTGATCATCAAGCTGCCTAAGTACTCCCTGTTCGA
GCTGGAAAACGGCCCGGAAGAGAATGCTGGCCTCTGCCcGCtttCTGCAGAAGGGAAA
CGAACTGGCCCTGCCCTCCAAATATGTGAACTTCCTGTACCTGGCCAGCCACTATGA
GAAGCTGAAGGGCTCCCCCGAGGATAATGAGCAGAAACAGCTGTTTGTGGAACAGC
ACAAGCACTACCTGGACGAGATCATCGAGCAGATCAGCGAGTTCTCCAAGAGAGTG
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TAAGCCCATCAGAGAGCAGGCCGAGAATATCATCCACCTGTTTACCCTGACCAATC
TGGGAGCCCCTcggGCCTTCAAGTACTTTGACACCACCATCGACCGGAAGgtGTACcgg
AGCACCAAAGAGGTGCTGGACGCCACCCTGATCCACCAGAGCATCACCGGCCTGTA
CGAGACACGGATCGACCTGTCTCAGCTGGGAGGTGAC SEQ ID NO: 39,

SpCas9(D10A)-NGX-loop: Nucleotide sequence:

ATGGACAAGAAGTACAGCATCGGCCTGGCCATCGGCACCAACTCTGTGGGCT
GGGCCGTGATCACCGACGAGTACAAGGTGCCCAGCAAGAAATTCAAGGTGCTGGGC
AACACCGACCGGCACAGCATCAAGAAGAACCTGATCGGAGCCCTGCTGTTTCGACAG
CGGCGAAACAGCCGAGGCCACCCGGCTGAAGAGAACCGCCAGAAGAAGATACACC
AGACGGAAGAACCGGATCTGCTATCTGCAAGAGATCTTCAGCAACGAGATGGCCAA
GGTGGACGACAGCTTCTTCCACAGACTGGAAGAGTCCTTCCTGGTGGAAAGAGGATA
AGAAGCACGAGCGGCACCCCATCTTCGGCAACATCGTGGACGAGGTGGCCTACCAC
GAGAAGTACCCCACCATCTACCACCTGAGAAAGAACTGGTGGACAGCACCGACA
AGGCCGACCTGCGGCTGATCTATCTGGCCCTGGCCCACATGATCAAGTTCCGGGGC
CACTTCCTGATCGAGGGCGACCTGAACCCCGACAACAGCGACGTGGACAAGCTGTT
CATCCAGCTGGTGCAGACCTACAACCAGCTGTTCGAGGAAAACCCCATCAACGCCA
GCGGCGTGGACGCCAAGGCCATCCTGTCTGCCAGACTGAGCAAGAGCAGACGGCTG
GAAAATCTGATCGCCCAGCTGCCCCGGCGAGAAGAAGAATGGCCTGTTTCGGAAACCT
GATTGCCCTGAGCCTGGGCCTGACCCCCAACTTCAAGAGCAACTTCGACCTGGCCG
AGGATACCAAACCTGCAGCTGAGCAAGGACACCTACGACGACGACCTGGACAACCT
GCTGGCCCAGATCGGCGACCAGTACGCCGACCTGTTTCTGGCCGCCAAGAACCTGT
CCGACGCCATCCTGCTGAGCGACATCCTGAGAGTGAACACCGAGATACCAAGGCC
CCCCTGAGCGCCTCTATGATCAAGCTGTACGACGAGCACCAACAGGACCTGACCCT
GCTGAAAGCTCTCGTGCGGCAGCAGCTGCCTGAGAAGTACAAAGAGATTTTCTTCG
ACCAGAGCAAGAACGGCTACGCCGGCTACgtggggcgccgacaagaagctgcgcaagcgcagctctaaactg
gccacagagGAAGAGTTCTACAAGTTCATCAAGCCCATCCTGGAAAAGATGGACGGCAC
CGAGGAACTGCTCGTGAAGCTGAACAGAGAGGACCTGCTGCGGAAGCAGCGGACC
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GCGGCGGCAGGAAGATTTTACCATTCTGAAGGACAACCGGGAAAAGATCGAGA
AGATCCTGACCTTCCGCATCCCCTACTACGTGGGCCCTCTGGCCAGGGGAAACAGC
AGATTGCCTGGATGACCAGAAAGAGCGAGGAAACCATCACCCCTGGAACCTTCGA
GAAGGTGGTGGACAAGGGCGCTTCCGCCCAGAGCTTCATCGAGCGGATGACCAACT
TCGATAAGAACCTGCCCAACGAGAAGGTGCTGCCCAAGCACAGCCTGCTGTACGAG
TACTTCACCGTGTATAACGAGCTGACCAAAGTGAAATACGTGACCGAGGGAATGAG

CGCGCAAGCAAGCAACCGCATCCGCTACCTGCAGGAGAGATATTGCGCCAAACGAGATGGGCCAA
GCTGGACGACAGCTTCTTCCAGCGCCTGGAGGAGAGCTTCCTGGTGGAGGAGGACA
AGAAGAACGAGCGCCACCCCATCTTCGGCAACCTGGCCGACGAGGTGGCCTACCAC
CGCAACTACCCCAACCATCTACCACCTGCGCAAGAAGCTGGCCGACAGCCCCGAGAA
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CTTCTGATCGAGGGCAAGCTGAACGCCGAGAACAGCGACGTGGCCAAGCTGTTCT
ACCAGCTGATCCAGACCTACAACCAGCTGTTTCGAGGAGAGCCCCCTGGACGAGATC
GAGGTGGACGCCAAGGGCATCCTGAGCGCCCCGCCTGAGCAAGAGCAAGCGCCTGG
AGAAGCTGATCGCCGTGTTCCCCAACGAGAAGAAGAACGGCCTGTTTCGGCAACATC
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GGACGCCAAGCTGCAGCTGAGCAAGGACACCTACGACGACGACCTGGACGAGCTG
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CGACGCCATCCTGCTGAGCGACATCCTGCGCAGCAACAGCGAGGTGACCAAGGCC
CCCTGAGCGCCAGCATGGTGAAGCGCTACGACGAGCACCACCAGGACCTGGCCCTG
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GAGAAGATGGACGGCGCCGAGGAGCTGCTGGCCAAGCTGAACCGCGACGACCTGC
TGCGCAAGCAGCGCACCTTCGACAACGGCAGCATCCCCCACCAGATCCACCTGAAG
GAGCTGCACGCCATCCTGCGCCGCCAGGAGGAGTTCTACCCCTTCCTGAAGGAGAA
CCGCGAGAAGATCGAGAAGATCCTGACCTTCCGCATCCCCTACTACGTGGGGCCCC
TGGCCCCGCGGCAACAGCCGCTTCGCCTGGCTGACCCGCAAGAGCGAGGAGGCCATC
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TCGTGGACCTGCTGTTCAAGACCAACCGCAAGGTGACCGTGAAGCAGCTGAAGGAG
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CCTGTTTCGACGACAAGGTGATGAAGCAGCTGAAGCGCCGCCACTACACCGGCTGGG
GCCGCTGAGCCGCAAGATGATCAACGGCATCCGCGACAAGCAGAGCGGCAAGAC
CATCCTGGACTTCCTGAAGAGCGACGGCTTCAGCAACCGCAACTTCATGCAGCTGA
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AGAAGGGCATCCTGCAGACCGTGAAGATCGTGGACGAGCTGGTGAAGGTGATGGG
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AAGGGCCTGCAGCAGAGCCGCGAGCGCAAGAAGCGCATCGAGGAGGGCATCAAGG
AGCTGGAGAGCCAGATCCTGAAGGAGAACCCCGTGGAGAACACCCAGCTGCAGAA
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CAAGGCCGAGCGCGGCGGCTGAGCGAGGCCGACAAGGCCGGCTTCATCAAGCGC
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CATGAACACCAAGCGCGACAAGAACGACAAGCCCATCCGCGAGGTGAAGGTGATC
ACCCTGAAGAGCAAGCTGGTGAAGCGACTTCCGCAAGGACTTCCAGCTGTACAAGGT
GCGCGACATCAACAACCTACCACCACGCCCACGACGCCTACCTGAACGCCGTGGTGG

GCACCGCTGACCTGACCTCAAGCTGAGAGCGAGTTTCGTGTACGGCGGAC
TACAAGGTGTACGACGTGCGCAAGATGATCGCCAAGAGCGAGCAGGAGATCGGCA
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GTGAAGCTGGCCAACGGCGAGATCCGCAAGCGCCCCCTGATCGAGACCAACGGCG
AGACCGGCGAGGTGGTGTGGAACAAGGAGAAGGACTTCGCCACCGTGCGCAAGGT
GCTGGCCATGCCCCAGGTGAACATCGTGAAGAAGACCGAGGTGCAGACCGGCGGGCT
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GAAGGGCTGGGACACCCGCAAGTACGGCGGGCTTCGGCAGCCCCACCGTGGCCTACA
GCATCCTGGTGGTGGCCAAGGTGGAGAAGGGCAAGGCCAAGAAGCTGAAGAGCGT
GAAGGTGCTGGTGGGCATCACCATCATGGAGAAGGGCAGCTACGAGAAGGACCCC
ATCGGCTTCTCTGGAGGCCAAGGGCTACAAGGACATCAAGAAGGAGCTGATCTTCAA
GCTGCCCAAGTACAGCCTGTTCGAGCTGGAGAACGGCCGCGCCGCGCATGCTGGCCA
GCGCCACCGAGCTGCAGAAGGCCAACGAGCTGGTGTGCTGCCCCAGCACCTGGTGCGC
CTGCTGTACTACACCCAGAACATCAGCGCCACCACCGGCAGCAACAACCTGGGGCTA
CATCGAGCAGCACCGCGAGGAGTTCAAGGAGATATTCGAGAAGATCATCGACTTCA
GCGAGAAGTACATCCTGAAGAACAAGGTGAACAGCAACCTGAAGAGCAGCTTCGA
CGAGCAGTTCGCCGTGAGCGACAGCATCCTGCTGAGCAACAGCTTCGTGAGCCTGC
TGAAGTACACCAGCTTCGGCGCCAGCGGGCGGCTTCACCTTCCTGGACCTGGACGTG
AAGCAGGGCCGCGCTGCGCTACCAGACCGTGACCGAGGTGCTGGACGCCACCCTGAT
CTACCAGAGCATCACCGGCCTGTACGAGACCCGACCCGACCTGAGCCAGCTGGGCG
GCGAC

SEQ ID NO: 41, TadA-TadA* Nucleotide sequence:

TCTGAAGTCGAGTTTAGGCCACGAGTATTGGATGAGGCACGCACTGACCCTGG
CAAAGCGAGCATGGGATGAAAGAGAAGTCCCCGTGGGCGCCGTGCTGGTGCACAA
CAATAGAGTGATCGGAGAGGGATGGAACAGGCCAATCGGCCGCCACGACCCTACC
GCACACGCAGAGATCATGGCACTGAGGCAGGGAGGCCTGGTCATGCAGAATTACCG
CCTGATCGATGCCACCCTGTATGTGACACTGGAGCCATGCGTGATGTGCGCAGGAG
CAATGATCCACAGCAGGATCGGAAGAGTGGTGTTCGGAGCACGGGACGCCAAGAC
CGGCGCAGCAGGCTCCCTGATGGATGTGCTGCACCACCCCGGCATGAACCACCGGG
TGGAGATCACAGAGGGAATCCTGGCAGACGAGTGCGCCGCCCCTGCTGAGCGATTTC
TTTAGAATGCGGAGACAGGAGATCAAGGCCCAGAAGAAGGCACAGAGCTCCACCG
ACTCTGGAGGATCTAGCGGAGGATCCTCTGGAAGCGAGACACCAGGCACAAGCGA
GTCCGCCACACCAGAGAGCTCCGGCGGGCTCCTCCGGAGGATCCTCTGAGGTGGAGT
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GAGAGGGAGGTGCCTGTGGGAGCCGTGCTGGTGTGCTGAACAATAGAGTGATCGGCG
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GCCCTGAGACAGGGCGGCCTGGTCATGCAGAACTACAGACTGATTGACGCCACCCT
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CTGGCAGATGAATGTGCCGCCCTGCTGTGCTATTTCTTTCGGATGCCTAGACAGGTG
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SEQ ID: 42, TadA*

(A56G_V82G) Nucleotide sequence:

TCTGAGGTGGAGTTTTTCCCACGAGTACTGGATGAGACATGCCCTGACCCTGG
CCAAGAGGGCACGCGATGAGAGGGAGGTGCCTGTGGGAGCCGTGCTGGTGTGCTGAA
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GgCCATGCCGAAATTATGGCCCTGAGgCAGGGCGGCCTGGTCATGCAGAACTACAG
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CATGATCCACTCTAGGATCGGCCGCGTGGTGTGTTGGCGTGAGGAACGCAAAAACCG
GCGCCGCGAGGCTCCCTGATGGACGTGCTGCACTACCCCGGCATGAATCACCGCGTC
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CGGATGCCATAGACAGGTGTTCAATGCTCAGAAGGCCAGAGCTCCACCGAC
SEQ ID NO: 43, Gp41-1-N: Nucleotide sequence:
TGTTTGGATCTGAAAACGCAAGTTCAAACGCCACAGGGTATGAAAGAAATAT
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ATGTATTTCCAAAAAGCAAGAAAAAAGTTACAAGATAACTCTGGAAGATGGAAA
AGAAATTATCTGTTCTGAGGAGCATCTGTTTCCGACCCAAACAGGGGAGATGAATA
TCAGTGGCGGTCTCAAAGAGGGTATGTGTTTGTATGTCAAGGaataa SEQ ID NO:
44, Gp41-1-C: Nucleotide sequence:
ATGATGCTCAAGAAGATCCTCAAGATTGAAGAGTTGGACGAGCGCGAGCTT
ATAGACATAGAAGTCAGTGGTAATCACCTTTTCTACGCAAATGACATTTTGACTCAC
AACTCC SEQ ID NO: 45, Cfa-N: Nucleotide sequence:
TGTCTCAGTTATGAcACCGAAATCCTGACAGTCGAGTATGGATtTtCTGCCGATC
GGCAAGATTGTGGAGgAGAGAATTGAATGTACGGTCTATAcgGTCGACAAgAATGGTt
tCgTCTACACCCAACCAaTTGCTCAATGGCATaATCGAGGGGAGCAGGAGGTGTTTGA
GTATTGCCTGGAGGACGGGTCAaTCATTAGAGCTACAAAGGACCATAAGTTTATGAC
AacCGATGGTCAAATGCTGCCGATAGATGAAATATTGAAAGGGgACTGGATCTTAa
GCaAGTCGATggCCTTCCAaac SEQ ID NO: 46, Cfa-C, Nucleotide sequence:
ATGgTcAAgATTatCAgcCGCAAATcCTTGGGGAcACAGAATGTATATGACATCG
GCGTGGAaAaGGATCACAATTTTctgCTGAAGAATGGTcTTgTTGCTtccAAt SEQ ID
NO: 47, Tad A,
SEVEFSHEYWMRHALTLAKRARDEREVPVGAVLVLNNRVIGEGWNRAIGLHDP
TAHAEIMALRQGGLVMQNYRLIDATLYVTFEPCVMCAGAMIHSRIGRVVFGVRNAKT
GAAGSLMDVLHYPGMNHRVEITEGILADECAALLCYFFRMPRQVFNAQKKAQSSTD
SEQ ID NO: 48 TGAGTATCATCGTGTGAAAGCTgAGGGGACGAGGCAGGCCTATAA
SEQ ID NO: 49 AAAAACATCAACTTCAGCCATCCATTTCTTCAGGGTTTGTATGTG
SEQ ID NO: 50 TATATCATAATGAAAACGCCGCCATTTCTCAACAGATCTGTCAA
SEQ ID NO: 51 CTGTGTGAAATGGCTGCAAATCgATGGTTGAGCTCTGAGATTTGG
SEQ ID NO: 52 GTTCTGCTTTTGTACTACTCACGTTTCCATGTTGTCCCCCTCTA
SEQ ID NO: 53 ATTTTATGGCCTTTTGTCAACTCgACCAGAAAAAAGCAGCTTTGG
SEQ ID NO: 54 TGAGGAGATCGCCCACGGGCTGCCAGGATCCCTTGATCACCTCAG
SEQ ID NO: 55 GGCTGCTCTGTGAGAAATATTCgTACAGTCTCAAGAGTACTCATG
SEQ ID NO: 56 GTGTAGGCATAGCTCTTGAATCgAGGCTTAGGGGAAGAAGTTCTC
SEQ ID NO: 57 CCTGTTCTTCAGTAAGACGTTGCCATTTGAGAAGGATGTCTTGTA
SEQ ID NO: 58 GCCATTTTAGGCTTTTTACTTACTTGTCTGTAGCTCTTTCTCTCT
SEQ ID NO: 59 CTAGTTTCTCACACATGACACACCTGTTCTTCAGTAAGACGTTGC
SEQ ID NO: 60 GTGAAGTTGATTACATTAACCTgTGGATAATTACGAGTTGATTGT
SEQ ID NO: 61 tttgtatATCTGAGTTAAACTgCTCCAATTCCTTCAAAGGAATG SEQ
ID NO: 62 TCTGCAATATAAGCTGCCAACTgCTTGTCAATGAATGTGAGGGAC SEQ
ID NO: 63 GGAAGTTGATTACATTAACCTgTGGATAATTACGAGTTGATTGT
ID NO: 64 AGTCGTTGTGTGGCTGACTGCTgGCAAACCACACTATTCCAGTCA SEQ
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TCTAGGAGGGCGCCTCCCATCCTgTAGGTCACCTGAAGAGGTTCTCA SEQ ID NO: 77
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126 TACTGTATAGGAGGACCCTCCTTCcATGACTCAAGGCTTGGCTCTGGC SEQ ID NO: 126
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 134 AAAGAGATTGTCTATACCTGTTgGCACATGTGATCCCACTGAGTG SEQ ID NO:
 135 AAGTTTTTTGGACTAAATTATCCcAACACCGGGGCAAAGTTATCCAG SEQ ID NO:
 136 AGCTCAGCATCCCGGGGACTCTgGGGAGAGGTGGGCATCATTTCa SEQ ID NO:
 137 TTGTCCCCCTCTAAGACAGTCTgCACTGGCAGGTAGCCCATTCGG SEQ ID NO:
 138 TTCGCAAATAACCTTTTTGGTTcGAAATTTGTTTTTTAGTACCTTG SEQ ID NO:
 139 TGCAACTCGACCAGAAAAAAAGcAGCTTTGGCAGATGTCATAATT SEQ ID NO:
 140 TTGCAGATGTTACATTTGGCCTgATGCTTGGCAGTTTCTGCAGCA SEQ ID NO:
 141 AAATAAAAACATGCCATACGTAcGTATCATAAACATTcAGCAGCC SEQ ID NO:
 142 TACTTACAGCAAAGGGCCTTCTgCAGTCTTCGGAGTTTCATGGCA SEQ ID NO:
 143 TTGACCTCCTCAGCCTGCTTTcGtAGAACCCGAGTGACATTCTGG SEQ ID NO:
 144 ATTCAATTACCTCTGGGCTCCTgGTAGAGTTTCTCTAGTCCTTCC SEQ ID NO:
 145 AATGCCTGACTTACTTGCCATTgTTTCATCAGCTCTTTTACTCCC SEQ ID NO:
 146 TGTACTTCATCCCACTGATTCTgAATTCTTTCAactagaataaaa SEQ ID NO: 147
 TGCTTCATTACCTTCACTGGCTgAGTGGCTGGTTTTTCCTTGTAC SEQ ID NO: 148
 TTtAATTGTTTGAGAATTCCCTgGCGCAGGGGGCAACTCTTCCACC SEQ ID NO: 149
 atatgtgtaCCTACCCTTGTCgGTCCTTGtacttttgtaact SEQ ID NO: 150
 TCAAGCTGGGAGAGAGCTTCCTgTAGCTTCACCCTTTCCACAGGC SEQ ID NO: 151
 ATGTCAATCCGACCTGAGCTTTgTTGTAGACTATCTTTTATATTc SEQ ID NO: 152
 GTTGTAGACTATCTTTTATATTctgtaataaaaaattttaaaac SEQ ID NO: 153
 TCCCGCCAGCGCTTGCTGAGCTgGATCTGAGTTGGCTCCACTGCC SEQ ID NO: 154
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 TGTTTTGTCTGTAAACAGCTgctgtttatctttatttctctctCGC SEQ ID NO: 156
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 aCCTTAAGCACGTCTTCTTTTTgCtgggggtttcttttctctgat SEQ ID NO: 158
 TTAAGCACGTCTTCTTTTTGcTgggggtttcttttctctgattca SEQ ID NO: 159
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 AGCTGTGACTGTACTACTTCCTgTTCCACACTCTTTGTTTCCAAT SEQ ID NO: 161
 TTGGCTGGTTTCTGGAATAATCgAAACTTCATGGAGACATCTTGT SEQ ID NO: 162
 TGCATCTCTGATAGATCTTTCTgGAGGCTTACAGTTTTCTCCAAA SEQ ID NO: 163
 ACAGTGAAAGAGATTGTCTATAcCTGTTGGCACATGTGATCCCAC SEQ ID NO: 164
 AAGGCATCATATAAAAATCTTAcTCTGCACTGTTTCAGCTGCTTT SEQ ID NO: 165
 CTTACTCTGCACTGTTTCAGCTgCTTTTTTAGAATTTCTGAATCC SEQ ID NO: 166
 TCTTGAATTACCTGAATTTTTcGAGTTTATTCATTTGCTCCTCT SEQ ID NO: 167
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TGT SEQ ID NO: 517 TCTaTGTTAGTGATGGCTGA GTG SEQ ID NO: 518
GTCTaTGTTAGTGATGGCTG AGT SEQ ID NO: 519
TGTCTaTGTTAGTGATGGCT GAG SEQ ID NO: 520
TTGTCTaTGTTAGTGATGGC TGA SEQ ID NO: 521
GTTGTCTaTGTTAGTGATGG CTG SEQ ID NO: 522
TGTCTaTGTTAGTGATGGCTG AGTGGT SEQ ID NO: 523
AGTTGTCTaTGTTAGTGATGG CTGAGT SEQ ID NO: 524
TCTCaAACATCTTCTCCTGA TGT SEQ ID NO: 525
GTCTCaAACATCTTCTCCTG ATG SEQ ID NO: 526
AAGTCTCaAACATCTTCTCC TGA SEQ ID NO: 527
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ATGCAGaATTTGGAACAGAG GCG SEQ ID NO: 532
AATGCAGaATTTGGAACAGA GGC SEQ ID NO: 533
TGCaGCTTCTCGAGTTCATA GGA SEQ ID NO: 534
TTGCaGCTTCTCGAGTTCAT AGG SEQ ID NO: 535
TGCTaGACCCATGTCCTGAT GGC SEQ ID NO: 536
CTGCCTaGACCCATGTCCTGA TGG SEQ ID NO: 537
ACTGCTaGACCCATGTCCTG ATG SEQ ID NO: 538
TTCTGACTGCTaGACCCATGT CCTGAT SEQ ID NO: 539
TGGTATaTATTTCTGGTGGC AAA SEQ ID NO: 540 GTGGTATaTATTTCTGGTGG

CAA SEQ ID NO: 541 GTGTGGTATaTATTCTGGTG GCAAAT SEQ ID NO: 542 TGTCaCATCAAATTTTCAAG TGA SEQ ID NO: 543 ATGTCaCATCAAATTTTCAA GTG SEQ ID NO: 544 TGTTCaTCCCCTATTATGAA GAA SEQ ID NO: 545 CTGTTCaTCCCCTATTATGA AGA SEQ ID NO: 546 CCTGTTCaTCCCCTATTATG AAG SEQ ID NO: 547 CCCTGTTCaTCCCCTATTATG AAGAAT SEQ ID NO: 548 TTACTTCaATCCGTAATGAT TGT SEQ ID NO: 549 TTAGaTAGGAAAAGATGTGG AGC SEQ ID NO: 550 ATTAGaTAGGAAAAGATGTG GAG SEQ ID NO: 551 CATTAGaTAGGAAAAGATGT GGA SEQ ID NO: 552 ACATTAGaTAGGAAAAGATG TGG SEQ ID NO: 553 TTGaAATATAATCCTCCACT GGC SEQ ID NO: 554 TTTGaAATATAATCCTCCAC TGG SEQ ID NO: 555 ATTTGaAATATAATCCTCCA CTG SEQ ID NO: 556 TTGTCaGACCCAGCTCAGGA GAA SEQ ID NO: 557 ATTGTCaGACCCAGCTCAGG AGA SEQ ID NO: 558 GATTGTCaGACCCAGCTCAG GAG SEQ ID NO: 559 TGATTGTCaGACCCAGCTCAG GAGAAT SEQ ID NO: 560 TTGTCTaAATATCACTGACT AAA SEQ ID NO: 561 ATTGTCTaAATATCACTGAC TAA SEQ ID NO: 562 TTGTTCTaCAAAACCCGCAG TGC SEQ ID NO: 563 TTTaACAATGCTCAACCAGC TGG SEQ ID NO: 564 TTTTaACAATGCTCAACCAG CTG SEQ ID NO: 565 GCTTTTaACAATGCTCAACC AGC SEQ ID NO: 566 AGCTTTTaACAATGCTCAAC CAG SEQ ID NO: 567 TTTaGGCAGCCTCCTTCCCC TGA SEQ ID NO: 568 CTTTaGGCAGCCTCCTTCCC CTG SEQ ID NO: 569 tttcttctctacaaTTCCCG ATCAAT SEQ ID NO: 570 GTTATCTCCTGTTCTGCAGC SEQ ID NO: 571 GTTTATGTCACCAGAGTAAC SEQ ID NO: 572 GAGGTAATAGAGCCAAGCCCT SEQ ID NO: 573 GCAAGAATTCCACTTTTCACTTCCT SEQ ID NO: 574 CTGTCATCTCCAAACTAGAAATGC SEQ ID NO: 575 GCAGCCTCTTGCTCACTTACTC SEQ ID NO: 576 GATGACAGGCAGGGGCACCG SEQ ID NO: 577 TTCCAGTGGTTCAATGGTCA SEQ ID NO: 578 CTTTCAACCCGAACGGAGAC SEQ ID NO: 579 GAGCGAGCAGCGTCTTCGAG SEQ ID NO: 580 GCAGACGGCAGTCACTAGGG SEQ ID NO: 581 GGGAAGCTGGGTGAATGGAG SEQ ID NO: 582 AGCTGTTTGGGAGGTCAGAA SEQ ID NO: 583 AGGGAGCAGGAAAGTGAGGT SEQ ID NO: 584 GTCGCAGGACAGCTTTTCCT SEQ ID NO: 585 TGTAGCTACGCCTGTGATGG SEQ ID NO: 586 TGCCCTGAGATCTTTTCCTC SEQ ID NO: 587 GATCCAGGTGCTGCAGAAGG SEQ ID NO: 588 CTCTTGCCTCCACTGGTTGT SEQ ID NO: 589 TCGGTAGGATGCCCTACATC SEQ ID NO: 590 ATCCTACAGCATGGTGGCTG SEQ ID NO: 591 AGTGGTCTCCGGAAACCTCCGCGCCCCGCAAC SEQ ID NO: 592 TCCTTGAAGAAGATGGTGCG SEQ ID NO: 593 ACACTCTTTCCCTACACGACGCTCTTCCGATCTGAACTCATTACTGCTGCCCA GA SEQ ID NO: 594 GTGACTGGAGTTCAGACGTGTGCTCTTCCGATCGACCTGTTCCGGCTTCTTCCT TA SEQ ID NO: 595 ACACTCTTTCCCTACACGACGCTCTTCCGATCTAAATTTCCACTGTCTTCTCTT GAGT SEQ ID NO: 596 GTGACTGGAGTTCAGACGTGTGCTCTTCCGATCGCTTGCCTCTGACCTGTCCT AT SEQ ID NO: 597

ACACTCTTTCCCTACACGACGCTCTTCCGATCTGTGACTAGGGGCAAAGCAA GAT
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ID NO: 599
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SEQ ID NO: 600
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SEQ ID NO: 601 AATGATACGGCGACCACCGAGATCTACACTCTTTCCCTACACGAC
SEQ ID NO: 602
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SEQ ID NO: 604
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NO: 605 CAAGCAGAAGACGGCATAACGAGATATGTCAGTGACTGGAGTTCAGACGT
SEQ ID NO: 606 CAAGCAGAAGACGGCATAACGAGAT GTCCGC
GTGACTGGAGTTCAGACGT SEQ ID NO: 607
CAAGCAGAAGACGGCATAACGAGAT TTAGGC GTGACTGGAGTTCAGACGT SEQ
ID NO: 608 CAAGCAGAAGACGGCATAACGAGAT CGATGT
GIGACIGGAGTTCAGACGT SEQ ID NO: 609 CAAGCAGAAGACGGCATAACGAGAT
TGACCA GTGACTGGAGTTCAGACGT SEQ ID NO: 610
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GATATCTCCTGTTCTGCAGGAGA SEQ ID NO: 630 GGATTTCOAAGTCTCCACCC
SEQ ID NO: 631 TCCCACCGTACACGCCTAC SEQ ID NO: 632
GGTTTCAGACAAAATCAAAAAGAAGGAAGGTGCTCACATTCCTTAAATTAAG GA
SEQ ID NO: 633
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ID NO: 634 GTGCTCACATTCCTTAAATTAAGG SEQ ID NO: 635

GGCTCACATTCCCTTAAATTAAGGA

(181) Unless defined otherwise, all technical and scientific terms used herein have the same meanings as commonly understood by one of skill in the art to which the disclosed invention belongs. Publications cited herein and the materials for which they are cited are specifically incorporated by reference.

(182) Those skilled in the art will appreciate that numerous changes and modifications can be made to the preferred embodiments of the invention and that such changes and modifications can be made without departing from the spirit of the invention. It is, therefore, intended that the appended claims cover all such equivalent variations as fall within the true spirit and scope of the invention.

Claims

1. A fusion protein comprising a Cas9 nickase and a monomeric adenine deaminase, wherein the Cas9 nickase comprises amino acid substitutions L1111R, D1135V, G1218R, E1219F, A1322R, and T1337R, and at least one additional mutation selected from R1335V, R1335Q or R1335E when compared to SEQ ID NO: 11, and wherein the monomeric adenine deaminase comprises amino acid substitutions A56G and V82G when compared to SEQ ID NO: 47.
2. The fusion protein of claim 1, wherein the Cas9 nickase comprises the amino acid sequence of SEQ ID NO: 15.
3. The fusion protein of claim 1, wherein the Cas9 nickase when in conjunction with a bound guide RNA (gRNA) specifically binds to a target nucleic acid sequence.
4. The fusion protein of claim 1, wherein the Cas9 nickase recognizes a NG protospacer adjacent motif (PAM) sequence.
5. The fusion protein of claim 1, wherein the adenine deaminase comprises the amino acid sequence set forth in SEQ ID NO: 23.
6. A system for base editing comprising: a first nucleotide sequence encoding a N-terminal portion of a Cas9 nickase fused at its C-terminus to an intein-N; and a second nucleotide sequence encoding an intein-C fused to the N-terminus of a C-terminal portion of the Cas9 nickase; wherein the first nucleotide sequence further comprises a nucleotide sequence encoding a monomeric adenine deaminase fused to the N-terminus of the N-terminal portion of the Cas9 nickase, wherein the Cas9 nickase comprises amino acid substitutions L1111R, D1135V, G1218R, E1219F, A1322R, and T1337R, and at least one additional mutation selected from R1335V, R1335Q or R1335E when compared to SEQ ID NO: 11, and wherein the monomeric adenine deaminase comprises amino acid substitutions A56G and V82G when compared to SEQ ID NO: 47.
7. The system of claim 6, wherein the N-terminal portion of the Cas9 nickase comprises the amino acid positions 2-573 of the amino acid sequence of SEQ ID NO: 15 and the C-terminal portion of the Cas9 nickase comprises the amino acid positions 574-1368 of the amino acid sequence of SEQ ID NO: 15.
8. The system of claim 6, wherein the intein-N is a Cfa intein-N or a gp41-1 intein-N.
9. The system of claim 6, wherein the intein-C is a Cfa intein-C or a gp41-1 intein-C.
10. The system of claim 6, wherein the first nucleotide sequence or the second nucleotide sequence further comprises a nucleotide encoding a guide RNA (gRNA).
11. A composition comprising: a first recombinant adeno-associated virus (AAV) particle comprising a first nucleotide sequence encoding a N-terminal portion of a Cas9 nickase fused at its C-terminus to an intein-N; and a second recombinant AAV particle comprising a second nucleotide sequence encoding an intein-C fused to the N-terminus of a C-terminal portion of the Cas9 nickase; wherein first nucleotide sequence further comprises a nucleotide sequence encoding a monomeric adenine deaminase fused to the N-terminus of the N-terminal portion of the Cas9 nickase, wherein the Cas9 nickase comprises amino acid substitutions L1111R, D1135V, G1218R, E1219F, A1322R, and T1337R, and at least one additional mutation selected from R1335V, R1335Q or R1335E when

compared to SEQ ID NO: 11, and wherein the monomeric adenine deaminase comprises amino acid substitutions A56G and V82G when compared to SEQ ID NO: 47.

12. A method of treating a genetic disease in a subject, comprising administering a therapeutically effective amount of the composition of claim 11.

13. The method of claim 12, wherein the genetic disease is a muscular dystrophy or spinal muscular atrophy.

14. The method of claim 13, wherein the muscular dystrophy is Duchenne muscular dystrophy.

15. The method of claim 13, wherein the muscular dystrophy is dysferlinopathy.
